

EVENTS AND PROBABILITIES

Experiment – an activity with an uncertain outcome

Event – one or more possible outcomes of an experiment

Experiment: Rolling two dices

Event A – rolling a total of 7

Event B – rolling a total greater than 5

Experiment: Taking an exam

Event A = pass

Event B = fail

Probability of an event is $P(A)$, $0 \leq P(A) \leq 1$

EVENTS AND PROBABILITIES

One definition (frequentist approach):

$P(A) = m/n$ m - experiments in which A occurs

n – total number of experiments (all equally likely)

A - number of happy days in a week $\Rightarrow P(A) = 5/7 = 71.42\%$

Another definition:

Bayesian probability(subjective probability)

“The probability that your laptop will break in the next 6 months”

CONTINGENCY TABLES

AGE(YEARS)				
	< 21	21-65	> 65	
Sex	Y -young	A -adult	S -senior	TOTAL
Male(M)	60	20	40	120
Female(F)	40	30	10	80
TOTAL	100	50	50	200

Marginal Probabilities:

$$P(M) = 120/200 = .6 ; P(Y) = 100/200 = .5 ; \text{ IF } P(M)=.6 \Rightarrow P(\bar{A}) = 1 - P(A) \Rightarrow 1 - .6 = .4$$

Conditional Probabilities

Joint Probabilities:

$$P(F \text{ and } Y) = 40/200 = .2 ;$$

$$P(M \text{ or } Y) = (100 + 120)/200 = 220/200 \text{ NOT GOOD We counted Young males twice}$$

$$P(M \text{ or } Y) = (40 + 120)/200 = .8$$

$$P(A|B) = P(A \text{ and } B)/P(B)$$

For independent events

$$P(A|B) = P(A) * P(B)$$

Conditional Probabilities:

$$P(M|Y) = 60/100 = .6$$

Events A and B are independent if and only if:

$$1. P(A|B) = P(A) \text{ with } P(B) \neq 0; 2. P(B|A) = P(B) \text{ with } P(A) \neq 0; P(A \text{ and } B) = P(A) * P(B)$$

Two events A and B are mutually exclusive if they cannot occur simultaneously or $P(A \text{ and } B) = 0$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

DISCRETE RANDOM VARIABLES

Intuitive definition =>

What is a Random Variable

A random variable is a variable which represents the outcome of a trial, an experiment or event. It is a specific number which is different each time the trial, experiment or event is repeated.

when throwing 2 dice:



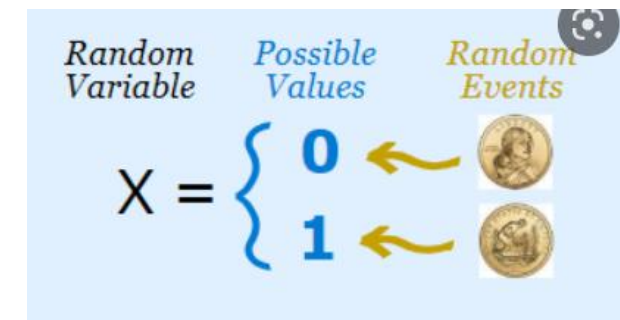
sample space:

$$S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

let x be the random variable
equal to the sum:

$$P(x=12) = \frac{1}{36}$$

A discrete random variable is a type of random variable that has a countable number of distinct values that can be assigned to it, such as in a coin toss.



More consistent definition =>

A function that assigns a numerical value to each outcome of an experiment

DISCRETE RANDOM VARIABLES

Flip a coin three times. What are the possible outcomes?

How many ways can get 1 head out of 3 flips?

H –heads; T-tails

TTT, TTH, THT, HTT, THH, HHT, HTH, HHH

A – event of observing 0 Heads in 3 flips (TTT) = 1/8

B – event of observing 1 Head in 3 flips (TTH, THT, HTT) = 3/8

C – event of observing 2 Heads in 3 flips (HHT, HTH, THH) = 3/8

D - event of observing 3 Heads in 3 flips (HHH) = 1/8

$X = \left\{ \begin{array}{l} 0 \text{ with probability } 1/8 \\ 1 \text{ with probability } 3/8 \\ 2 \text{ with probability } 3/8 \\ 3 \text{ with probability } 1/8 \end{array} \right.$

The function that assigns a probability to each value of X is called a probability mass function (PMF). $P(X=x)=P(x)=x/6$ for $x=1,2,3$

$P(x)$ should be between 0 and 1 inclusive for each x
 $\sum P(x) = 1$

$$P(x) = \binom{n}{x} p^x q^{n-x} = \frac{n!}{(n-x)!x!} p^x q^{n-x}$$

where

n = the number of trials (or the number being sampled)

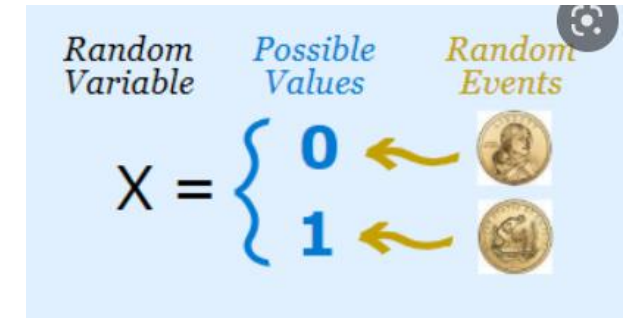
x = the number of successes desired

p = probability of getting a success in one trial

$q = 1 - p$ = the probability of getting a failure in one trial

DISCRETE RANDOM VARIABLES

$$X = \begin{cases} 0 & \text{with probability } 1/8 \\ 1 & \text{with probability } 3/8 \\ 2 & \text{with probability } 3/8 \\ 3 & \text{with probability } 1/8 \end{cases}$$



Mean $\mu = \sum x \cdot P(x) = 0 \cdot 1/8 + 1 \cdot 3/8 + 2 \cdot 3/8 + 3 \cdot 1/8 = 1.5$ This is the value of X on the average.

Variance

Variance of a Random Variable

$$\sigma^2 = \sum (x - \mu)^2 P(x)$$

Variance of a Random Variable

$$\sigma^2 = \sum x^2 P(x) - \mu^2$$

X	P(X)	X*X	X^2*P(X)	μ^2	Variance
0	1/8	0	0	2.25	
1	3/8	1	3/8	2.25	
2	3/8	4	12/8	2.25	
3	1/8	9	9/8	2.25	
		Sum	24/8=3	2.25	3-2.25=0.75

A probability mass function (pmf) is a function over the sample space of a discrete random variable X which gives the probability that X is equal to a certain value. $f(x) = P[X=x]$.

BINOMIAL DISTRIBUTION

DESCRIPTION	EXAMPLE
Experiment consisting of n repetitions (trials)	$n=3$ (flips a coin)
Each trial has two mutually exclusive possible outcomes referred as success and failure	Success = head, Failure = tail
The n trials are independent	The results of one coin flip do not affect the results of another flip
The probability of a success for each trial is denoted p and remains the same for each individual trial	p =the probability of flipping a head on a particular trial
The random variable X is the number of successes out of n trials	X =the number of heads out of three flips

BINOMIAL DISTRIBUTION

Mean $\mu = np$; Variance $\sigma^2 = np(1-p)$

Sometimes, instead of an exact number of successes, you want to know the probability of getting r or more successes or r or less successes. To calculate the probability of getting any range of successes:

Use the binomial probability formula to calculate the probability of success (P) for all possible values of r you are interested in. Sum the values of P for all r within the range of interest.

The probability of getting two or fewer successes when flipping a coin four times ($p = 0.5$ and $n = 4$) would be:

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$P(X \leq 2) = 37.5\% + 25\% + 6.25\%$$

$$P(X \leq 2) = 68.75\%$$

This calculation is made easy using the options available on the binomial distribution calculator. You can change the settings to calculate the probability of getting:

Exactly r successes: $P(X = r)$

r or more successes: $P(X \geq r)$

r or fewer successes: $P(X \leq r)$

Between r_0 and r_1 successes $P(r_0 \leq X \leq r_1)$

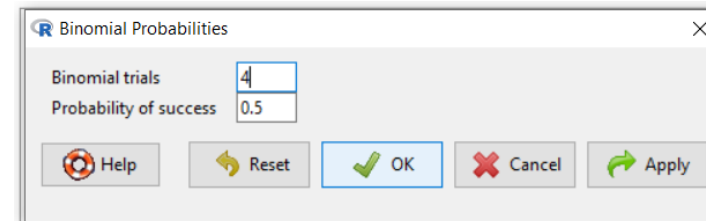
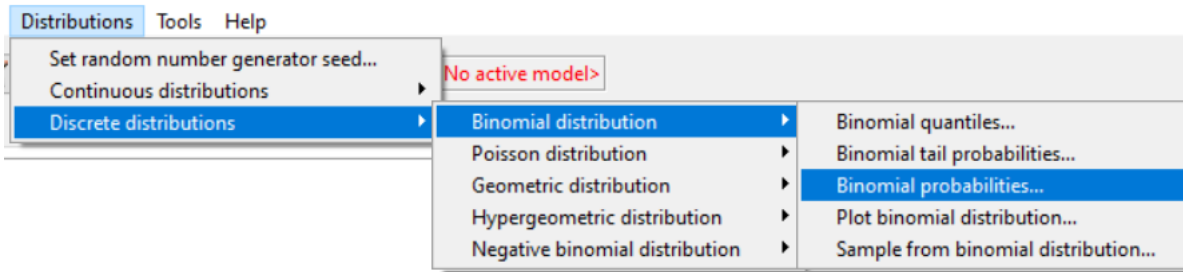
BINOMIAL DISTRIBUTION

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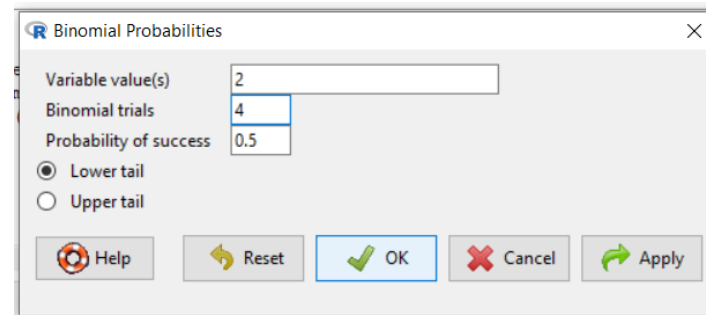
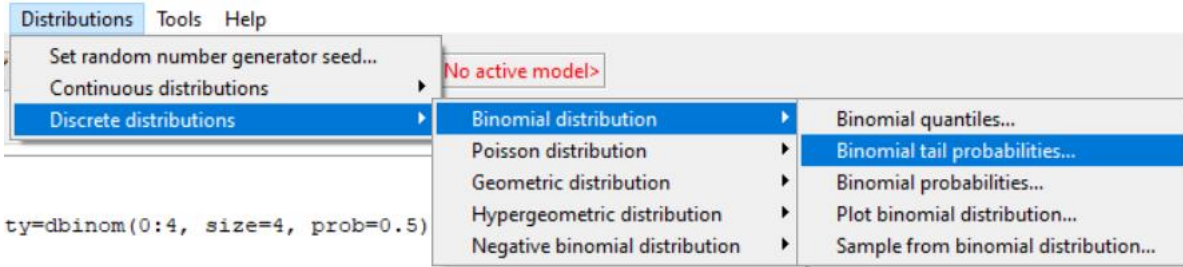
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Probability	
0	0.0625
1	0.2500
2	0.3750
3	0.2500
4	0.0625



```
> pbinom(c(2), size=4, prob=0.5, lower.tail=TRUE)
[1] 0.6875
```

BINOMIAL DISTRIBUTION

Example: A lot of 2500 dishes arrives at a store. The contract says that the store will accept the dishes if a sample of 100 dishes from the lot has no more than one cracked dish. What is the probability of accepting the lot if in fact 5% of the lot (125 dishes) have cracks? **Approximately binomial distribution**

- *n – number of trials = 100;

- *each trial has success=cracked dish and failure=good dish

- *probability of success $p=0.05$

- * X = number of success out of n trials = number of cracked dishes out of 100

The cumulative probability $P(\text{accept}) = P(X \leq 1) = P(0) + P(1) = 0.03708121$

BERNOULLI DISTRIBUTION

A random variable follows a Bernoulli distribution if it only has two possible outcomes: 0 or 1. For example, **suppose we flip a coin only one time**. Let the probability that it lands on heads be p . This means the probability that it lands on tails is $1-p$.

Thus, we could write:
$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$

If we flip a coin multiple times, then the sum of the Bernoulli random variables will follow a Binomial distribution.

1. A random variables that follows a Bernoulli distribution can only take on two possible values, but a random variable that follows a Binomial distribution can take on several values.

For example, in a single coin flip we will either have 0 or 1 heads. However, in a series of 5-coin flips we could have 0, 1, 2, 3, 4, or 5 heads.

2. In order for a random variable to follow a Binomial distribution, the probability of “success” in each Bernoulli trial must be equal and independent.

For example, if we define “success” as landing on heads, then the probability of success on each coin flip is equal to 0.5 and each flip is independent – the outcome of one coin flip does not affect the outcome of another.

POISSON DISTRIBUTION

A Poisson distribution is a tool that helps to predict the probability of certain events happening when you know how often the event has occurred. It gives us the probability of a given number of events happening in a fixed interval of time.

In [probability theory](#) and [statistics](#), the **Poisson distribution** is a [discrete probability distribution](#) that expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and [independently](#) of the time since the last event.

For instance, a call center receives an average of 180 calls per hour, 24 hours a day. The calls are independent; receiving one does not change the probability of when the next one will arrive. The number of calls received during any minute has a Poisson probability distribution with mean 3: the most likely numbers are 2 and 3 but 1 and 4 are also likely and there is a small probability of it being as low as zero and a very small probability it could be 10. Another example is the number of decay events that occur from a radioactive source during a defined observation period.

Bacteria are assumed to be distributed at random in some medium with a uniform density of λ (lambda) per unit area. The number of bacteria colonies found in a sample area of size A follows the Poisson distribution with a parameter μ equal to the product of λ and A .

POISSON DISTRIBUTION

Conditions for a Poisson distribution(over time or a specific area or volume):

1.The number of occurrences in one interval of time is unaffected by the number of occurrences in any other non-overlapping time interval. Variables are statistically independent. What happens between 10:00-11:00 is not affected by what happened between 7:00-8:00;

2.The expected(average) number of occurrences over any time period is proportional to the size of the time interval. We expect half as many occurrences between 3:00-3:30 as between 3:00-4:00. Caution to restaurants where during some hours the restaurants are more busy than other hours.

3. Events cannot occur exactly at the same time. The unit of time is that small that two occurrences do not happen at the same time.(Use seconds)

Situations meeting the above conditions:

The number of arrivals at the local bank over a five-minutes interval

The number of accidents reported along a 10 mile stretch of an intercity road

POISSON DISTRIBUTION

Poisson probability mass function

X = number of occurrences over time period T

μ = expected number of occurrences over T and $e=2.71828$

$$P(x) = (\mu^x * e^{-\mu}) / x! \quad \text{for } x=0,1,2,3\dots$$

Recall: A probability mass function (pmf) is a function over the sample space of a discrete random variable X which gives the probability that X is equal to a certain value. $f(x)=P[X=x]$.

For Poisson distribution: mean of $X = \sum x * P(x) = \mu$ and $\sigma^2 = \sum x^2 * P(x) - \mu^2 = \mu$

POISSON DISTRIBUTION

A new house builder decides to open an information kiosk inside one of the famous malls in Austin. Let's assume X = the number of customers arriving at the kiosk over a 5-minutes period. Supposing the Poisson requirements are met and:

$\mu=4$ customers over a 5 minutes period

1. What is the probability that over any 5 minutes interval exactly 4 people arrive at the kiosk?
2. What is the probability that more than 1 person arrive at the kiosk?
3. What is the probability that exactly 6 persons arrive at the kiosk over 10 minutes interval?

Answers:

1. $P(X=4)=(4^4 * e^{-4})/4! = 0.1954$ If you stand at the kiosk for many 5 minutes, 19.5% of the time you will see four people arriving at the kiosk.

2. $P(X>1) = P(X\geq 2)=P(X=2)+P(X=3) + \dots = 0.1465 + 0.1954 + \dots$ **Difficult and long**

Alternative: $P(X\geq 2) = 1-P(X<2)=1-P(X\leq 1)=1-P(0)+P(1)=0.9084$

3. For a 10 minutes interval $\mu=8$, therefore using formula for $\mu=8$ we get .1221

POISSON DISTRIBUTION

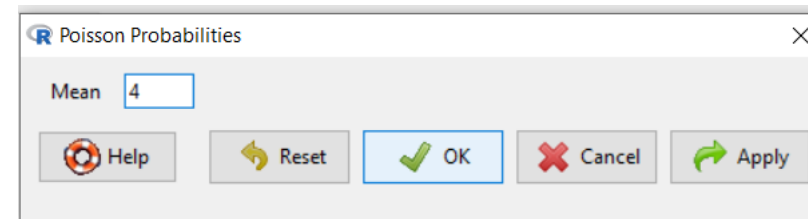
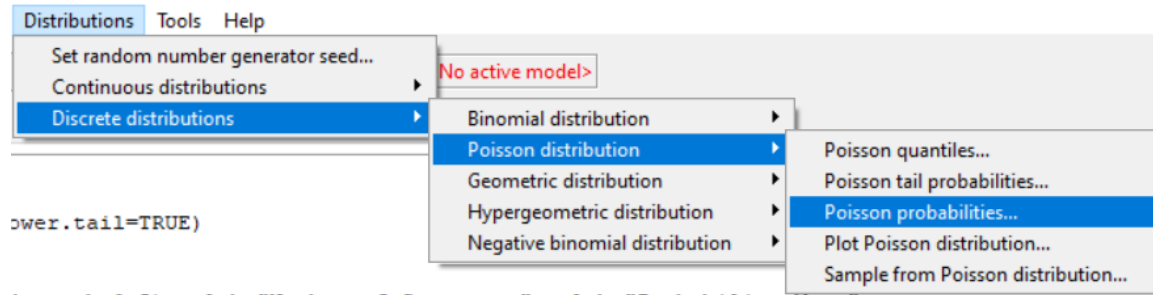
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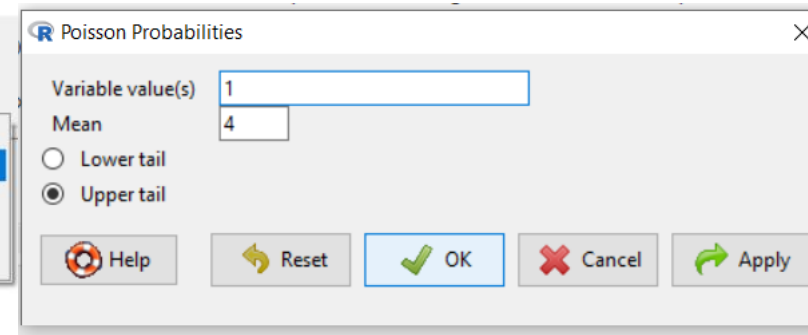
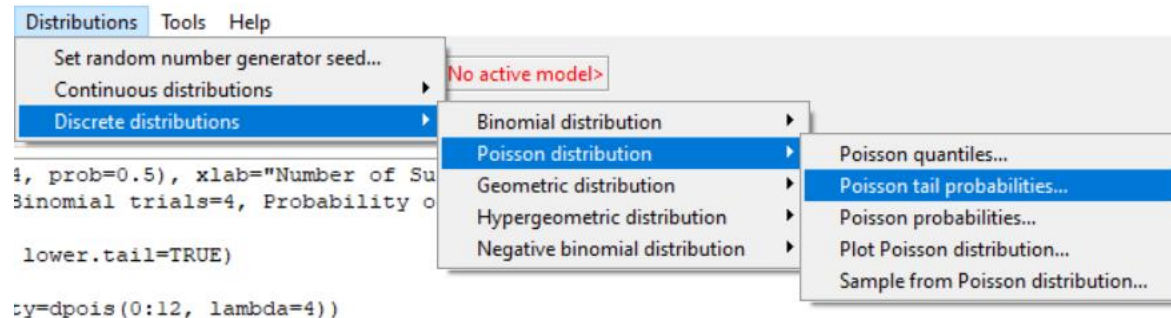
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	Probability
0	0.0183156389
1	0.0732625556
2	0.1465251111
3	0.1953668148
4	0.1953668148
5	0.1562934519
6	0.1041956346
7	0.0595403626
8	0.0297701813
9	0.0132311917
10	0.0052924767
11	0.0019245370
12	0.0006415123



```
> ppois(c(1), lambda=4, lower.tail=FALSE)
[1] 0.9084218
```

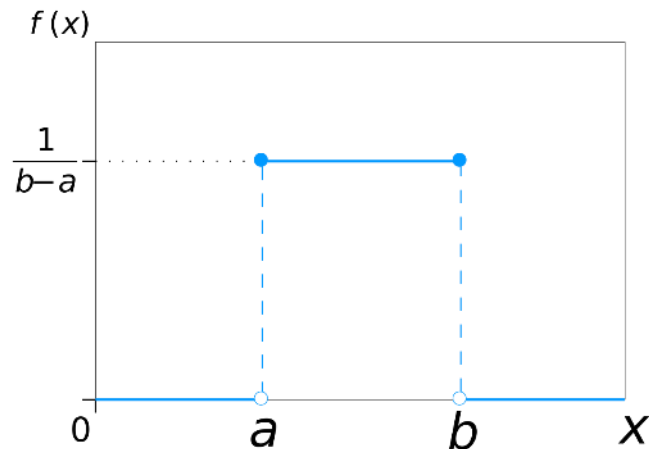
CONTINUOUS RANDOM VARIABLES

A continuous random variable is one which takes an infinite number of possible values. Continuous random variables are usually measurements. Examples include height, weight, the amount of sugar in an orange, the time required to run a mile. A continuous random variable is not defined at specific values.

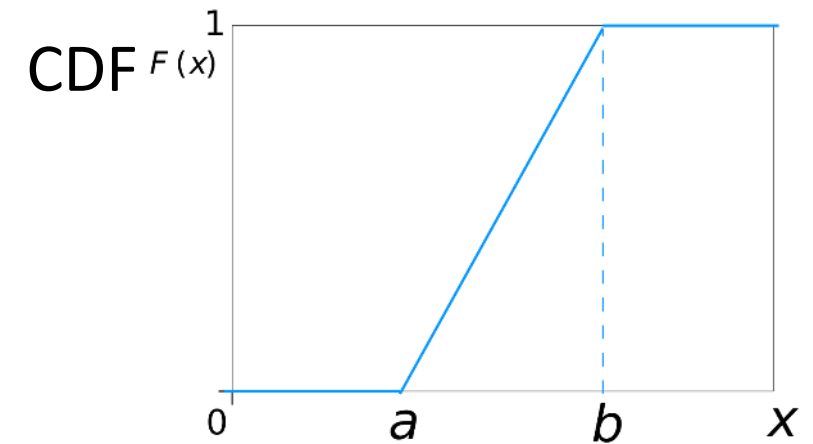
UNIFORM DISTRIBUTION

In probability theory and statistics, the continuous uniform distribution or rectangular distribution is a family of symmetric probability distributions. The distribution describes an experiment where there is an arbitrary outcome that lies between certain bounds. The bounds are defined by the parameters, a and b , which are the minimum and maximum values. The interval can either be closed (e.g. $[a, b]$) or open (e.g. (a, b)). [2] Therefore, the distribution is often abbreviated $U(a, b)$, where U stands for uniform distribution. The difference between the bounds defines the interval length; all intervals of the same length on the distribution's support are equally probable.

PMF



Area of Rectangle = (Length)(Height) = 1



UNIFORM DISTRIBUTION

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{for all other values} \end{cases}$$
$$\mu = \frac{a+b}{2}$$
$$\sigma = \frac{b-a}{\sqrt{12}}$$

According to the National Association of Insurance Commissioners, the average annual cost for automobile insurance in the United States in a recent year was \$691. Suppose automobile insurance costs are uniformly distributed in the United States with a range of from \$200 to \$1,182. What is the standard deviation of this uniform distribution? What is the height of the distribution? What is the probability that a person's annual cost for automobile insurance in the United States is between \$410 and \$825?

Solution

The mean is given as \$691. The value of a is \$200 and b is \$1,182.

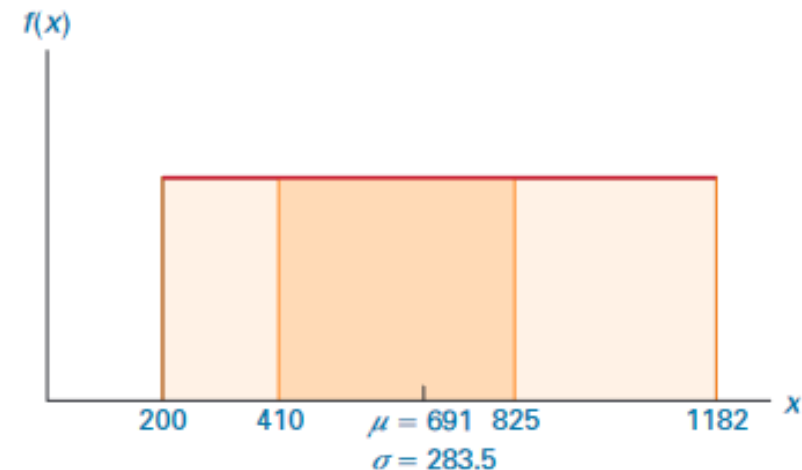
$$\sigma = \frac{b-a}{\sqrt{12}} = \frac{1,182-200}{\sqrt{12}} = 283.5$$

The height of the distribution is $\frac{1}{1,182-200} = \frac{1}{982} = .001$

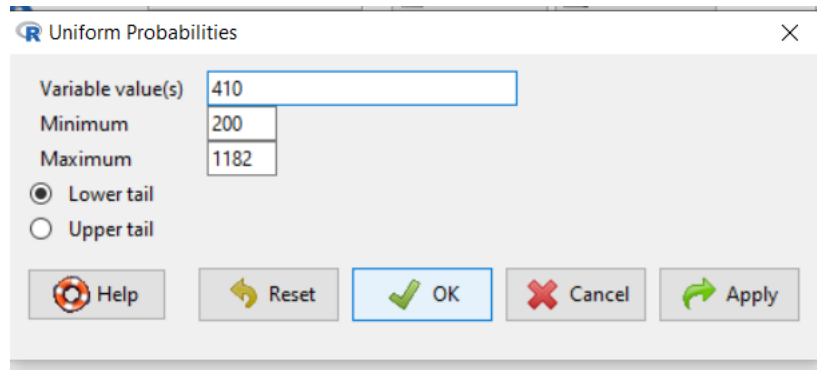
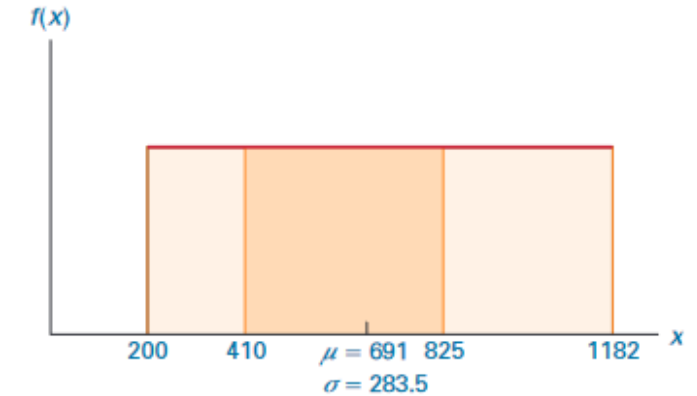
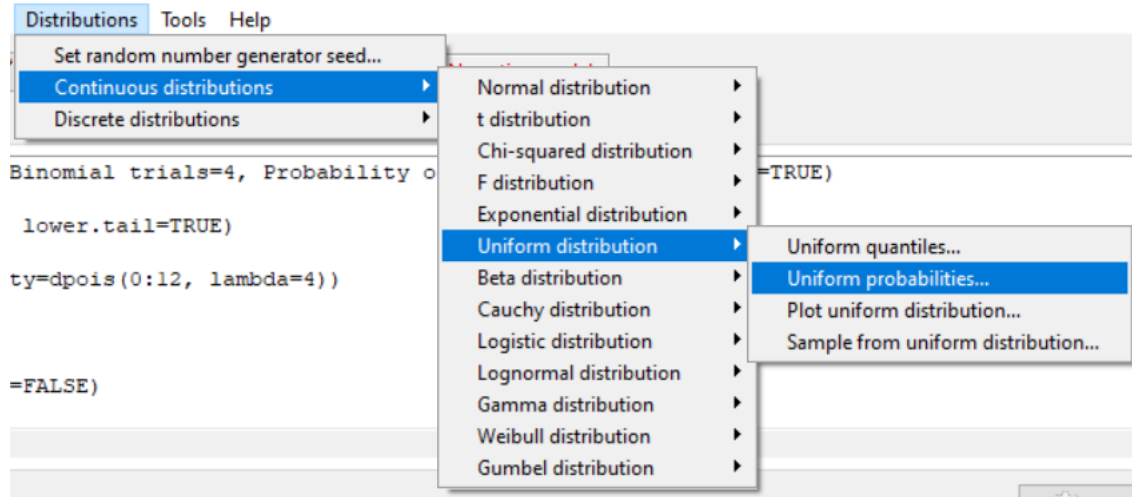
$$x_1 = 410 \text{ and } x_2 = 825$$

$$P(410 \leq x \leq 825) = \frac{825-410}{1,182-200} = \frac{415}{982} = .4226$$

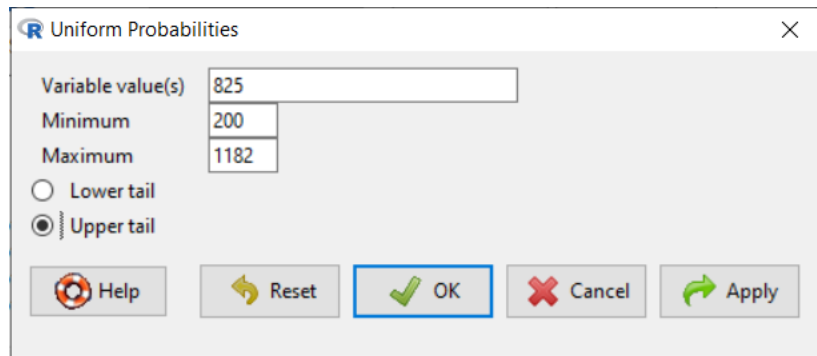
The probability that a randomly selected person pays between \$410 and \$825 annually for automobile insurance in the United States is .4226. That is, about 42.26% of all people in the United States pay in that range.



UNIFORM DISTRIBUTION



```
> punif(c(410), min=200, max=1182, lower.tail=TRUE)
[1] 0.2138493
```

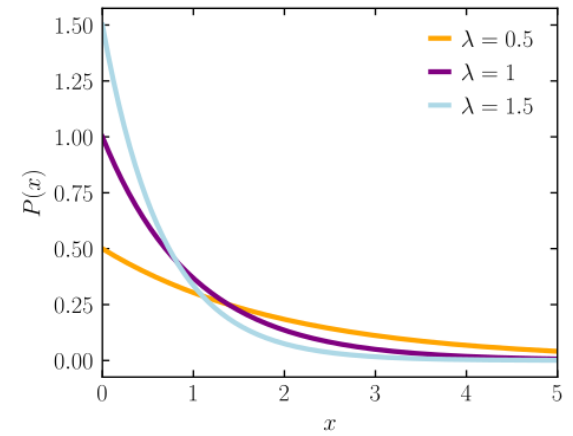


```
> punif(c(825), min=200, max=1182, lower.tail=FALSE)
[1] 0.3635438
```

$$=1-(0.2138+0.3635)=0.4226$$

EXPONENTIAL DISTRIBUTION

In probability theory and statistics, the exponential distribution is the probability distribution of the **time between events** in a Poisson point process, i.e., a process in which events occur continuously and independently at a constant average rate. It is a particular case of the gamma distribution.



Applications:

Observing the time **between** successive arrivals at a bank (**Not the number of arrivals over a specific period like in the Poisson distribution**)

Observing the lifetime of components of a car.

Observing the wait in line (queue) of people, cars, or telephone calls.

EXPONENTIAL DISTRIBUTION

Probability density function

The [probability density function](#) (pdf) of an exponential distribution is

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

Here $\lambda > 0$ is the parameter of the distribution, often called the *rate parameter*. The distribution is supported on the interval $[0, \infty)$. If a [random variable](#) X has this distribution, we write $X \sim \text{Exp}(\lambda)$.

The exponential distribution exhibits [infinite divisibility](#).

Cumulative distribution function

The [cumulative distribution function](#) is given by

$$F(x; \lambda) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

Parameters	$\lambda > 0$, rate, or inverse scale
Support	$x \in [0, \infty)$
PDF	$\lambda e^{-\lambda x}$
CDF	$1 - e^{-\lambda x}$
Quantile	$-\frac{\ln(1 - p)}{\lambda}$
Mean	$\frac{1}{\lambda}$
Median	$\frac{\ln 2}{\lambda}$
Mode	0
Variance	$\frac{1}{\lambda^2}$
Skewness	2
Ex. kurtosis	6

EXPONENTIAL DISTRIBUTION

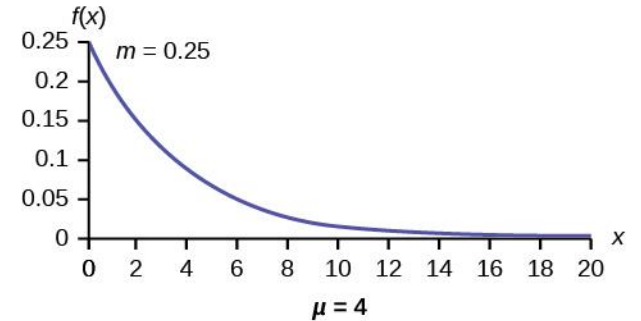
Let X = amount of time (in minutes) a postal clerk spends with his or her customer.

The time is known to have an exponential distribution with the average amount of time equal to four minutes. X is a continuous random variable since time is measured. It is given that $\mu = 4$ minutes. To do any calculations, you must know m , the decay parameter.

$m = 1/\mu$. Therefore, $m = 1/4 = 0.25$. The standard deviation, σ , is the same as the mean. $\mu = \sigma$. The distribution notation is $X \sim \text{Exp}(m)$. Therefore, $X \sim \text{Exp}(0.25)$.

The probability density function is $f(x) = me^{-mx}$. The number $e = 2.71828182846$. The curve is:

$f(x) = 0.25e^{-0.25x}$ where x is at least zero and $m = 0.25$.



For example, $f(5) = 0.25e^{-(0.25)(5)} = 0.072$. The postal clerk spends five minutes with the customers. The graph is as above.

EXPONENTIAL DISTRIBUTION

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Using the information in the example above, find the probability that a clerk spends four to five minutes with a randomly selected customer.

The curve is:

$$X \sim \text{Exp}(0.125); f(x) = 0.125e^{-0.125x}$$

a) Find $P(4 < x < 5)$.

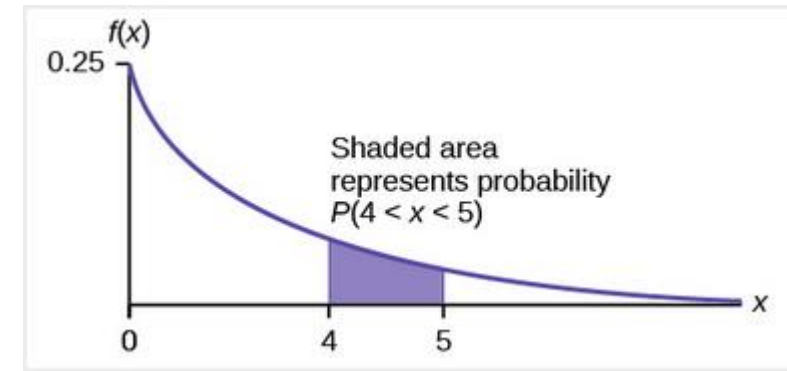
Solution:

The cumulative distribution function (CDF) gives the area to the left.

$$P(X < x) = 1 - e^{-mx}$$

$$P(X < 5) = 1 - e^{(-0.25)(5)} = 0.7135 \text{ and } P(x < 4) = 1 - e^{(-0.25)(4)} = 0.6321$$

$$P(4 < x < 5) = P(x < 5) - P(x < 4) = 0.7135 - 0.6321 = 0.0814.$$



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$$X \sim \text{Exp}(0.125); f(x) = 0.125e^{-0.125x}$$

b) Half of all customers are finished within how long? (Find the 50th percentile)

Solution:

Find the 50th percentile.

$P(X < k) = 0.50$, $k = 2.8$ minutes (calculator or computer) Half of all customers are finished within 2.8 minutes.

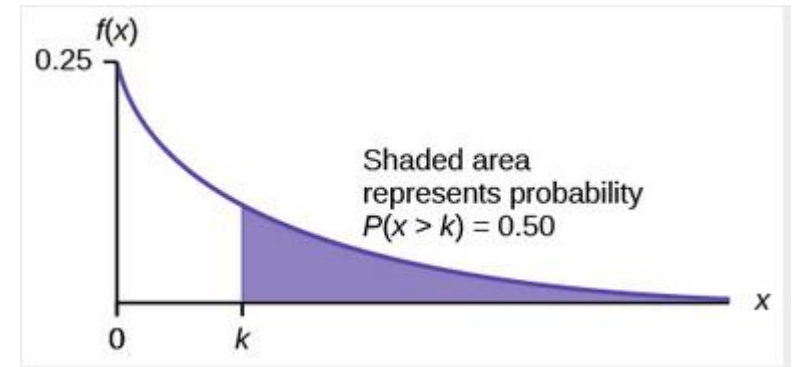
You can also do the calculation as follows:

$$P(X < k) = 0.50 \text{ and } P(X < k) = 1 - e^{-0.25k} \text{ Therefore, } 0.50 = 1 - e^{-0.25k} \text{ and } e^{-0.25k} = 1 - 0.50 = 0.5$$

Take natural logs: $\ln(e^{-0.25k}) = \ln(0.50)$. So, $-0.25k = \ln(0.50)$

Solve for k :

$$k = \ln(0.50)/(-0.25) = 2.8 \text{ minutes}$$



EXPONENTIAL DISTRIBUTION PROBLEMS

The number of days ahead travelers purchase their airline tickets can be modeled by an exponential distribution with the average amount of time equal to 15 days. Find the probability that a traveler will purchase a ticket fewer than ten days in advance. How many days do half of all travelers wait?

Solution: $P(x < 10) = 0.4866$ and the 50th percentile = 10.40

On the average, a certain computer part lasts ten years. The length of time the computer part lasts is exponentially distributed.

a) What is the probability that a computer part lasts more than 7 years?

Solution: Let x = the amount of time (in years) a computer part lasts.

$\mu=10$; $m = 1/\mu=0.10$

$P(x > 7)$. Draw the graph.

$P(x > 7) = 1 - P(x < 7)$.

Since $P(X < x) = 1 - e^{-mx}$ then $P(X > x) = 1 - (1 - e^{-mx}) = e^{-mx}$

$P(x > 7) = e^{(-0.1)(7)} = 0.4966$. The probability that a computer part lasts more than seven years is 0.4966.

b) On the average, how long would five computer parts last if they are used one after another?

Solution: On the average, one computer part lasts ten years. Therefore, five computer parts, if they are used one right after the other would last, on the average, $(5)(10) = 50$ years.

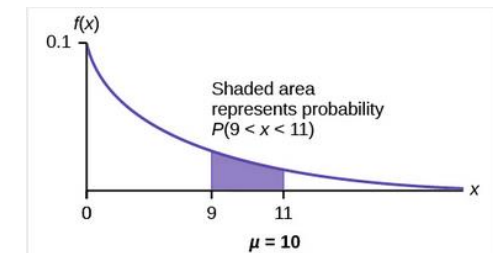
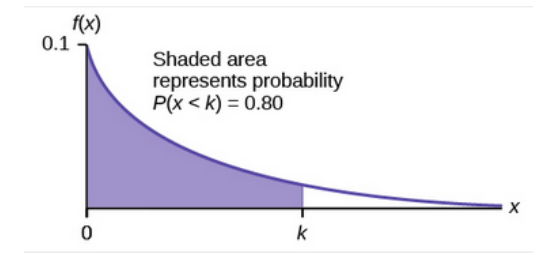
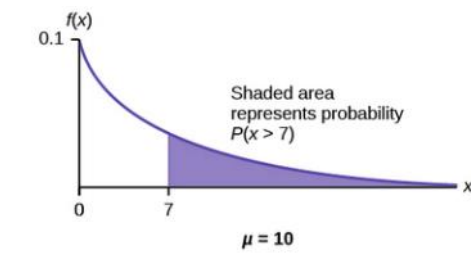
c) Eighty percent of computer parts last at most how long?

Solution: Find the 80th percentile. Draw the graph. Let k = the 80th percentile.

Solve for k : $k = \ln(1 - 0.80) / -0.1 = 16.1$ Eighty percent of the computer parts last at most 16.1 years.

d) What is the probability that a computer part lasts between nine and 11 years?

$P(9 < x < 11) = P(x < 11) - P(x < 9) = (1 - e^{(-0.1)(11)}) - (1 - e^{(-0.1)(9)}) = 0.6671 - 0.5934 = 0.0737$. The probability that a computer part lasts between nine and 11 years is 0.0737.



EXPONENTIAL DISTRIBUTION PROBLEMS

On the average, a certain computer part lasts ten years. The length of time the computer part lasts is exponentially distributed.

a) What is the probability that a computer part lasts more than 7 years?

Solution: Let x = the amount of time (in years) a computer part lasts.

$\mu=10$; $m = 1/\mu=0.10$; $P(x > 7)$; Draw the graph; $P(x > 7) = 1 - P(x < 7)$; Since $P(X < x) = 1 - e^{-mx}$ then $P(X > x) = 1 - (1 - e^{-mx}) = e^{-mx}$ $P(x > 7) = e^{(-0.1)(7)} = 0.4966$.

The probability that a computer part lasts more than seven years is 0.4966.

b) On the average, how long would five computer parts last if they are used one after another?

Solution: On the average, one computer part lasts ten years. Therefore, five computer parts, if they are used one right after the other would last, on the average, $(5)(10) = 50$ years.

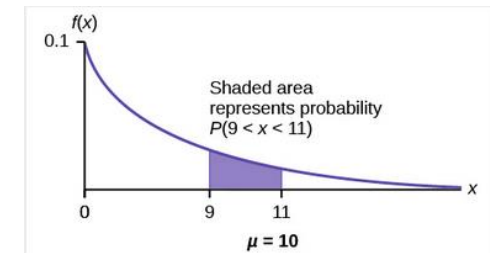
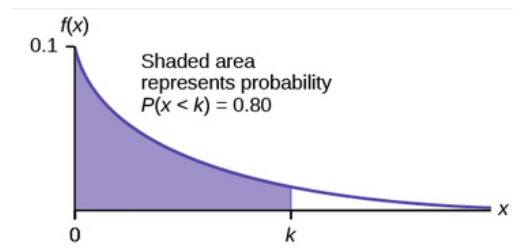
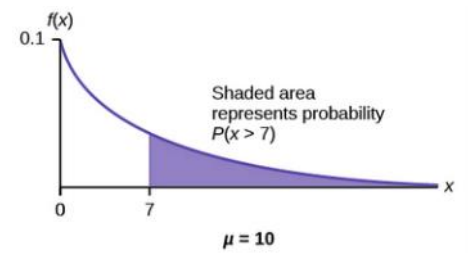
c) Eighty percent of computer parts last at most how long?

Solution: Find the 80th percentile. Draw the graph. Let k = the 80th percentile.

Solve for k : $k = \ln(1-0.80)/-0.1=16.1$ Eighty percent of the computer parts last at most 16.1 years.

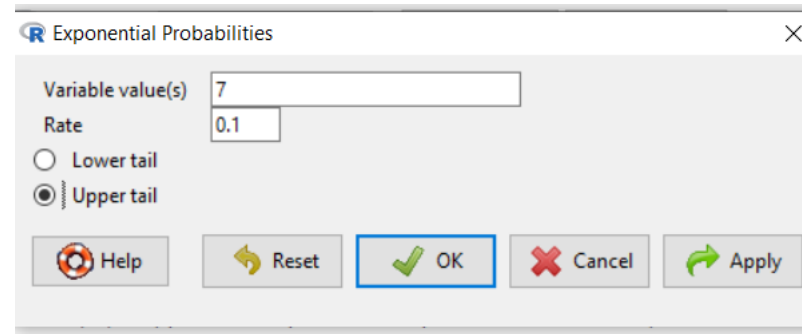
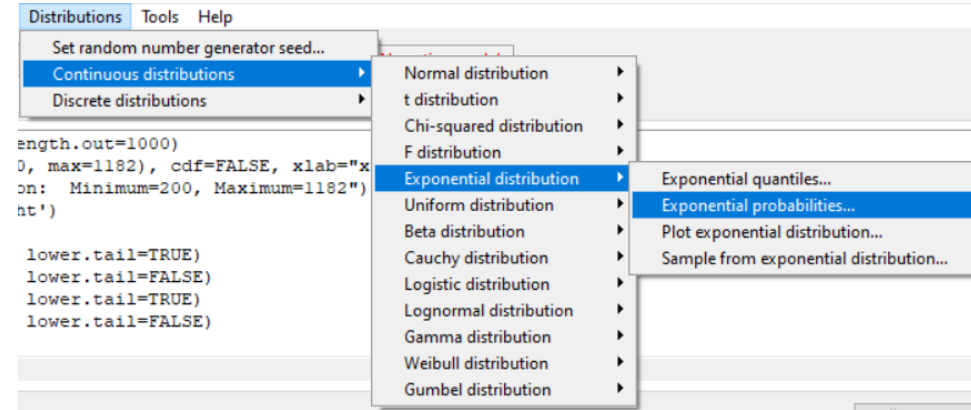
d) What is the probability that a computer part lasts between nine and 11 years?

$P(9 < x < 11) = P(x < 11) - P(x < 9) = (1 - e^{(-0.1)(11)}) - (1 - e^{(-0.1)(9)}) = 0.6671 - 0.5934 = 0.0737$. The probability that a computer part lasts between nine and 11 years is 0.0737.



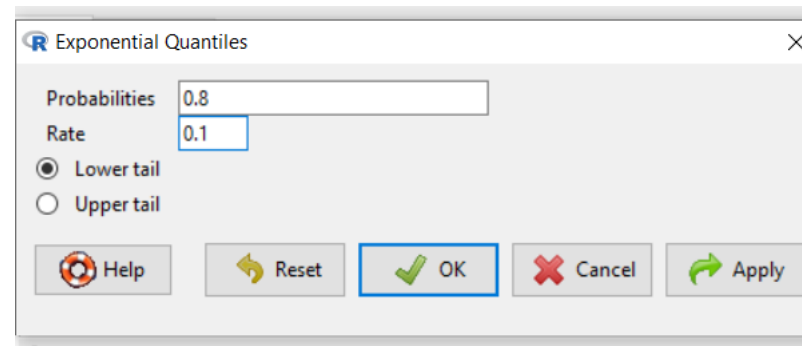
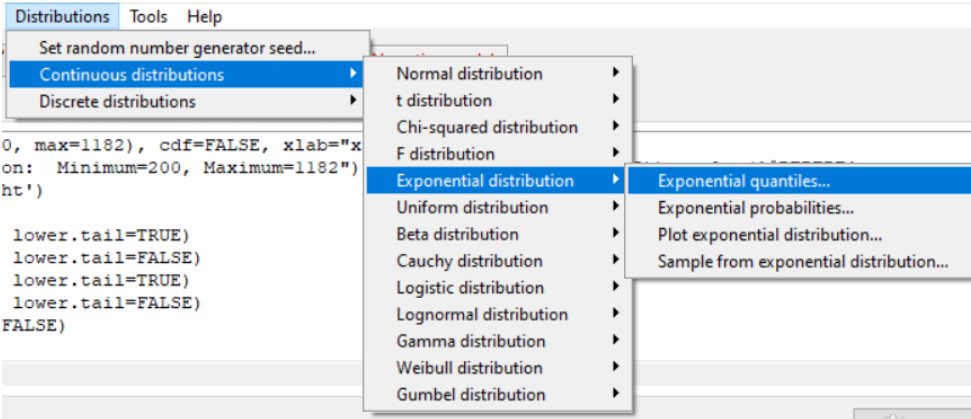
EXPONENTIAL DISTRIBUTION PROBLEMS

a)

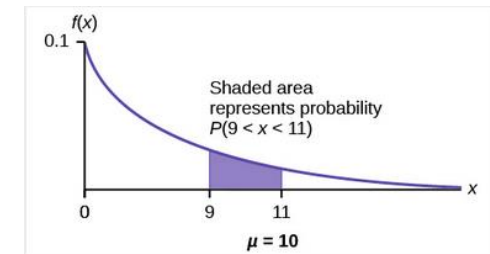
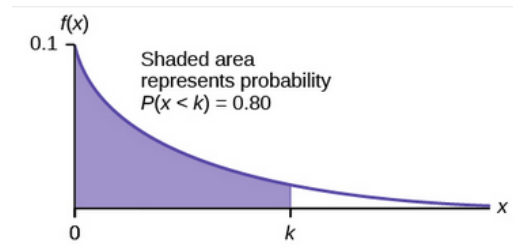
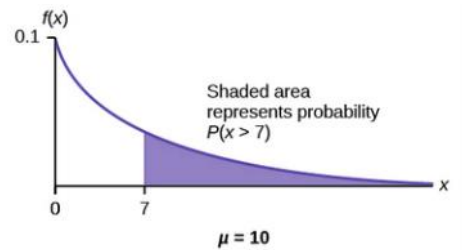


```
> pexp(c(7), rate=0.1, lower.tail=FALSE)
[1] 0.4965853
```

b)

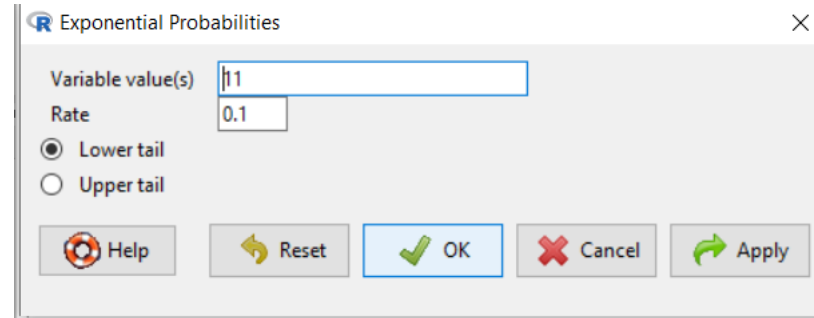
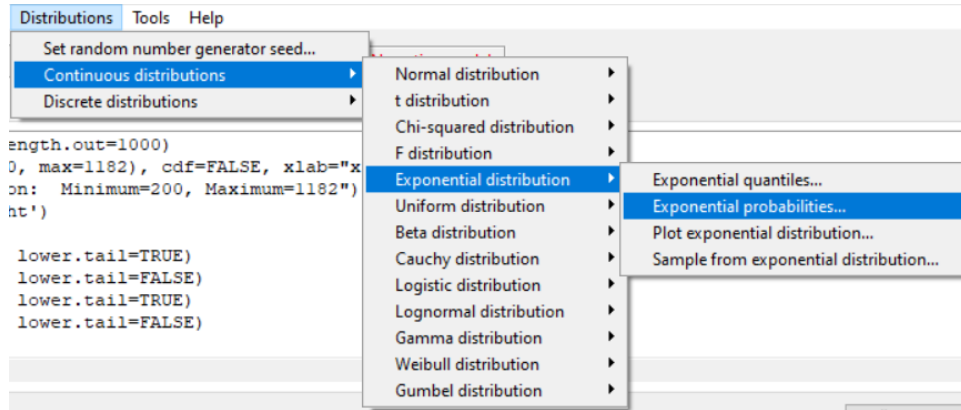


```
> qexp(c(0.8), rate=0.1, lower.tail=TRUE)
[1] 16.09438
```



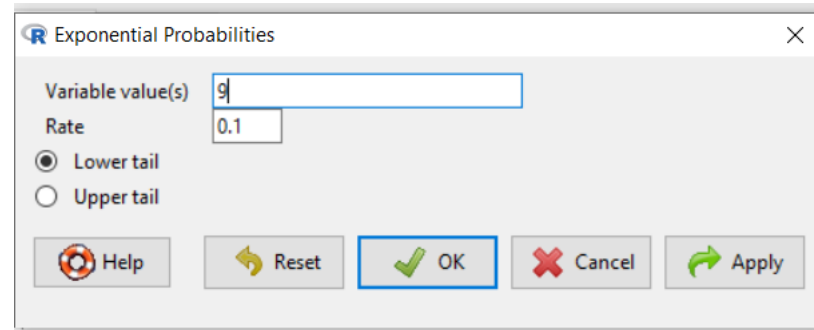
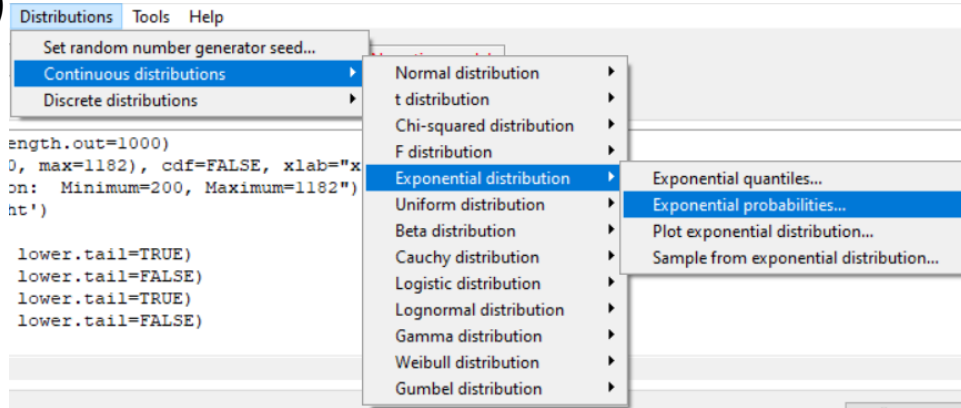
EXPONENTIAL DISTRIBUTION PROBLEMS

c1)



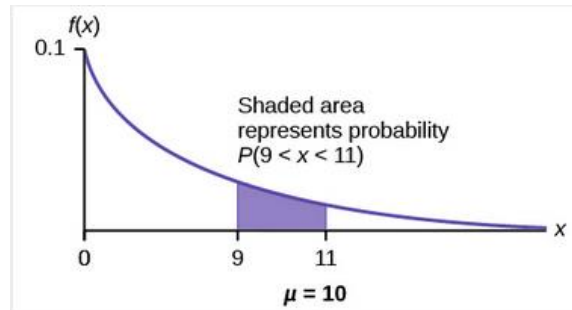
```
> pexp(c(11), rate=0.1, lower.tail=TRUE)
[1] 0.6671289
```

c2)



```
> pexp(c(9), rate=0.1, lower.tail=TRUE)
[1] 0.5934303
```

$$= 0.6671 - 0.5934 = 0.0737$$



NORMAL DISTRIBUTION

In statistics, a normal distribution (also known as Gaussian, Gauss, or Laplace–Gauss distribution) is a type of continuous probability distribution for a real-valued random variable. The general form of its probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Normal distributions are important in statistics and are often used in the natural and social sciences to represent real-valued random variables whose distributions are not known.[2][3] Their importance is partly due to the central limit theorem.

It states that, under some conditions, the average of many samples (observations) of a random variable with finite mean and variance is itself a random variable—whose distribution converges to a normal distribution as the number of samples increases.

Therefore, physical quantities that are expected to be the sum of many independent processes, such as measurement errors, often have distributions that are nearly normal.[4]

NORMAL DISTRIBUTION

The normal distribution exhibits the following characteristics.

It is a continuous distribution.

It is a symmetrical distribution about its mean.

It is asymptotic to the horizontal axis.

It is unimodal.

It is a family of curves.

Area under the curve is 1.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2[(x-\mu)/\sigma]^2}$$

where

μ = mean of x

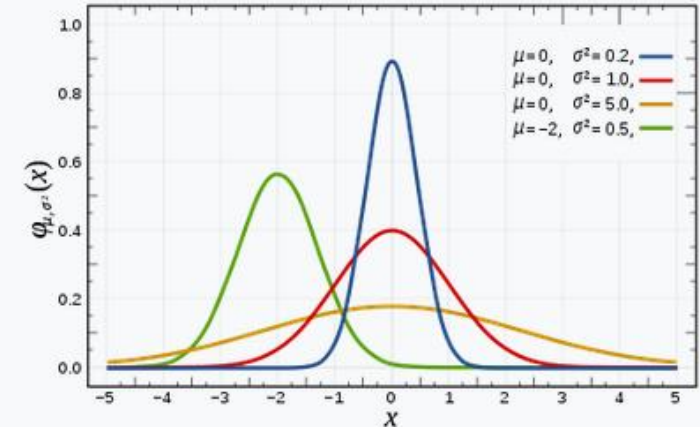
σ = standard deviation of x

$\pi = 3.14159 \dots$, and

$e = 2.71828 \dots$

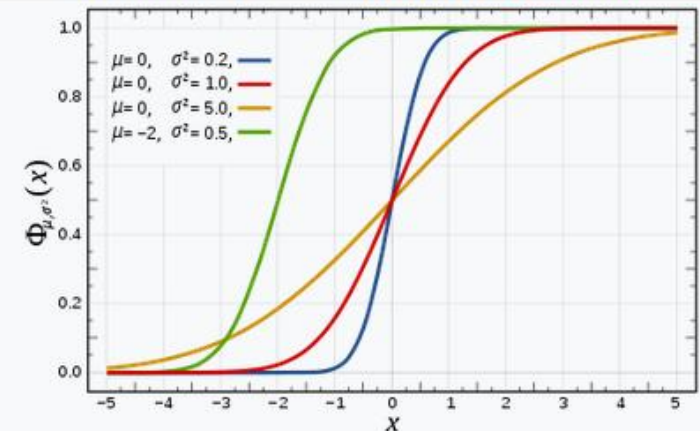
Normal distribution

Probability density function



The red curve is the *standard normal distribution*

Cumulative distribution function

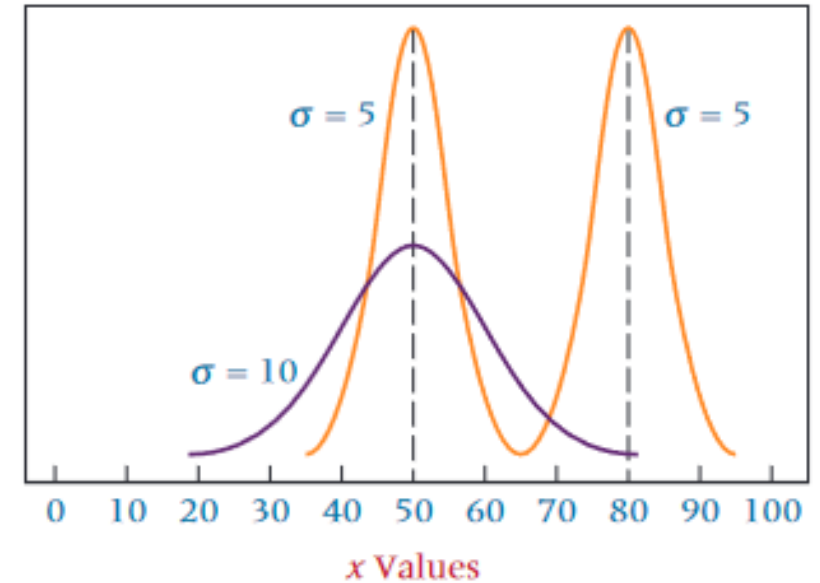


NORMAL DISTRIBUTION

Every unique pair of μ and σ values defines a different normal distribution. Figure 6.5 shows the Minitab graphs of normal distributions for the following three pairs of parameters.

1. $\mu = 50$ and $\sigma = 5$
2. $\mu = 80$ and $\sigma = 5$
3. $\mu = 50$ and $\sigma = 10$

Note that every change in a parameter (μ or σ) determines a different normal distribution. This characteristic of the normal curve (a family of curves) could make analysis by the normal distribution tedious because volumes of normal curve tables—one for each different combination of μ and σ —would be required. Fortunately, a mechanism was developed by which all normal distributions can be converted into a single distribution: the z distribution. This process yields the **standardized normal distribution** (or curve). The conversion formula for any x value of a given normal distribution follows.



NORMAL DISTRIBUTION

A **z score** is *the number of standard deviations that a value, x , is above or below the mean*. If the value of x is less than the mean, the z score is negative; if the value of x is more than the mean, the z score is positive; and if the value of x equals the mean, the associated z score is zero. This formula allows conversion of the distance of any x value from its mean into standard deviation units. A standard z score table can be used to find probabilities for any normal curve problem that has been converted to z scores. The **z distribution** is *a normal distribution with a mean of 0 and a standard deviation of 1*. Any value of x at the mean of a normal curve is zero standard deviations from the mean. Any value of x that is one standard deviation above the mean has a z value of 1. The empirical rule, introduced in Chapter 3, is based on the normal distribution in which about 68% of all values are within one standard deviation of the mean regardless of the values of μ and σ . In a z distribution, about 68% of the z values are between $z = -1$ and $z = +1$.

The z distribution probability values are given in Table A.5. Because it is so frequently used, the z distribution is also printed inside the cover of this text. For discussion purposes, a list of z distribution values is presented in Table 6.2.

Table A.5 gives the total area under the z curve between 0 and any point on the positive z -axis. Since the curve is symmetric, the area under the curve between z and 0 is the same whether z is positive or negative (the sign on the z value designates whether the z score is above or below the mean). The table areas or probabilities are always positive.

$$z = \frac{x - \mu}{\sigma}, \quad \sigma \neq 0$$



Diagram of a normal distribution curve. The horizontal axis is labeled with 0 at the center and z at a point to the right. The area under the curve between 0 and z is shaded in light orange.

SECOND DECIMAL PLACE IN z										
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224

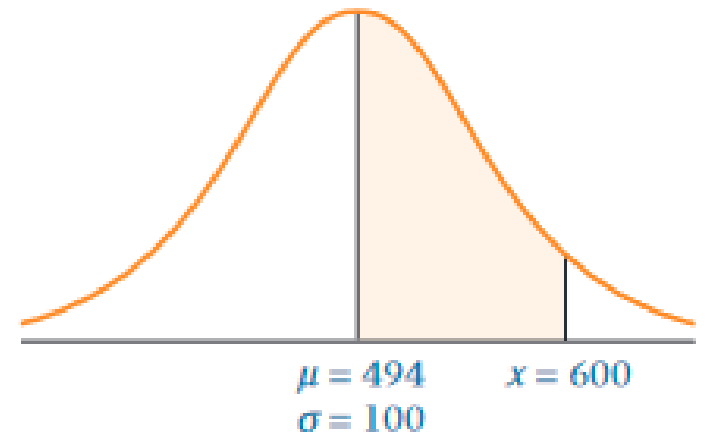
NORMAL DISTRIBUTION

Example: The Graduate Management Aptitude Test (GMAT), produced by the Educational Testing Service in Princeton, New Jersey, is widely used by graduate schools of business in the United States as an entrance requirement. Assuming that the scores are normally distributed, probabilities of achieving scores over various ranges of the GMAT can be determined. In a recent year, the mean GMAT score was 494 and the standard deviation was about 100. What is the probability that a randomly selected score from this administration of the GMAT is between 600 and the mean? That is:

$$P(494 \leq x \leq 600 | \mu = 494 \text{ and } \sigma = 100) = ?$$

The z formula yields the number of standard deviations away from the mean.

$$z = \frac{x - \mu}{\sigma} = \frac{600 - 494}{100} = \frac{106}{100} = 1.06$$



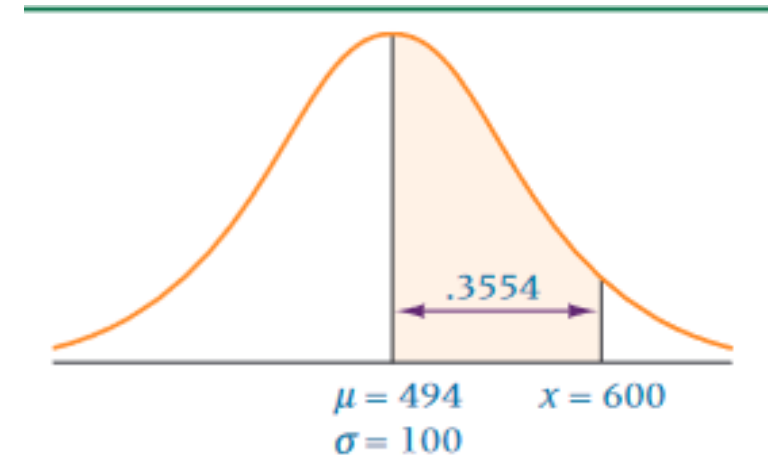
NORMAL DISTRIBUTION

The z value of 1.06 reveals that the GMAT score of 600 is 1.06 standard deviations more than the mean.

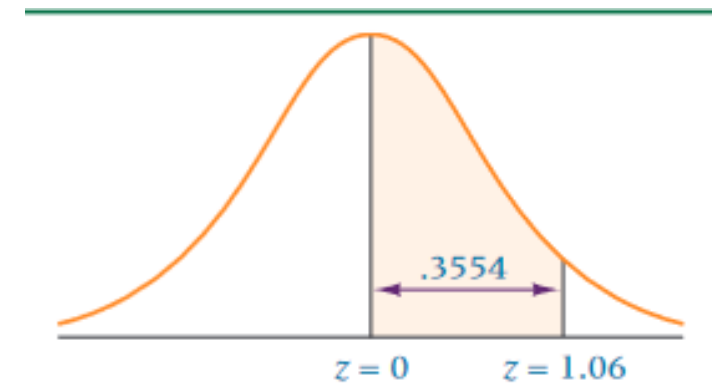
The z distribution values in z-Table give the probability of a value being between this value of x and the mean. The whole-number and tenths-place portion of the z score appear in the first column of table (the 1.0 portion of this z score).

Across the top of the table are the values of the hundredths-place portion of the z score. For this z score, the hundredths-place value is 6. The probability value in table for $z = 1.06$ is .3554. The shaded portion of the curve at the top of the table indicates that the probability value given always is the probability or area between an x value and the mean.

In this particular example, that is the desired area. Thus, the answer is that .3554 of the scores on the GMAT are between a score of 600 and the mean of 494. Figure 6.7(a) depicts graphically the solution in terms of x values. Figure 6.7(b) shows the solution in terms of z values



(a)



(b)

NORMAL DISTRIBUTION EXAMPLE

We believe that the lifetime of a bulb light is a normal random variable with $\mu=400$ and $\sigma=50$.

What percentage of the time will the burnout time, X , be less than 360?

$P(X < 360) = ?$; $z = (x - \mu) / \sigma \Rightarrow z = (360 - 400) / 50 = -0.8 \Rightarrow$ go to Z-Table and obtain the probability = 0.2119 that means that 21.19% of the time the bulb light will burn out in less than 360 hours.

a) Well: $P(X < 360) = P((X - 400) / 50 < (360 - 400) / 50) = P(Z < -0.8) \Rightarrow$ from Z-tables or using Rcmdr we get $P(X < 360) = 0.2119$

Find areas under the standard normal curve

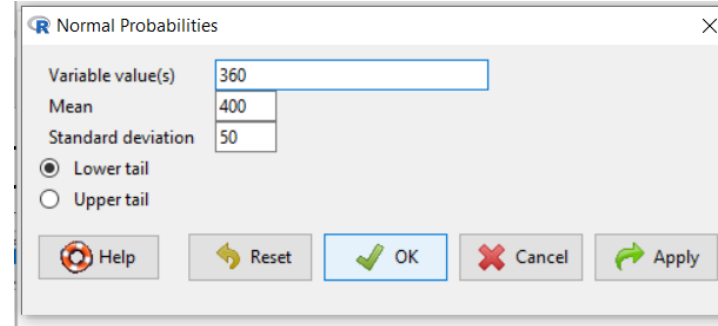
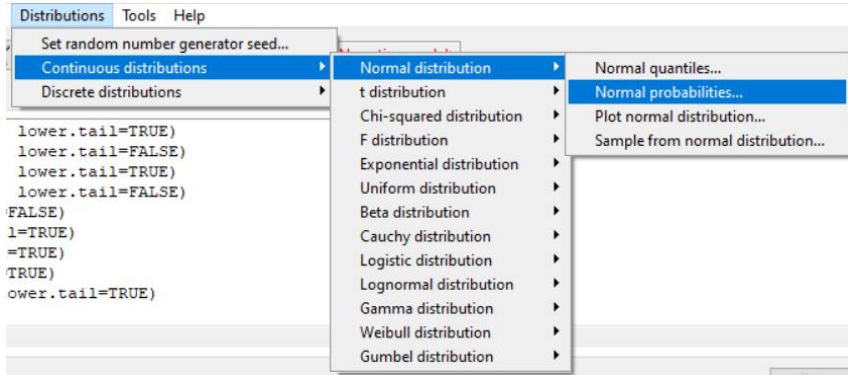
b) $P(0 < Z < 1.62) = ?$ The result is 0.4474

What about $P(Z = 1.62)$? **The result is zero.**

c) What about the $P(1.0 < Z < 2.0)$? $P(1.0 < Z < 2.0) = 0.1586 - 0.0227 = 0.1359$

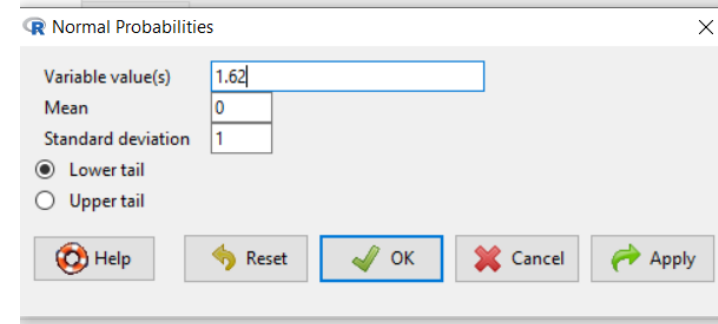
NORMAL DISTRIBUTION EXAMPLE

a)



```
> pnorm(c(360), mean=400, sd=50, lower.tail=TRUE)
[1] 0.2118554
```

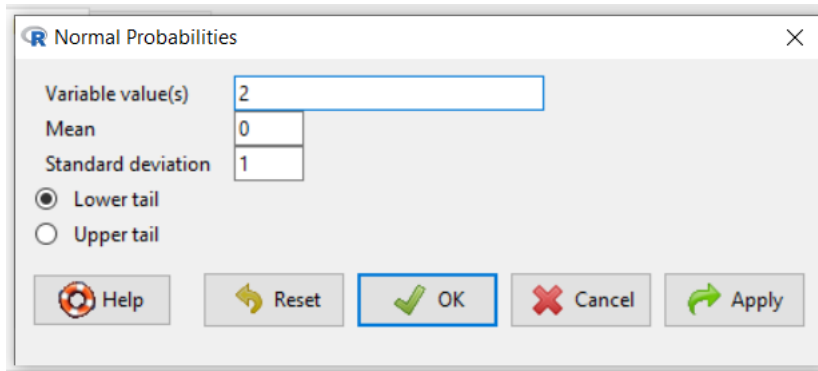
b)



```
> pnorm(c(1.62), mean=0, sd=1, lower.tail=TRUE)
[1] 0.9473839
```

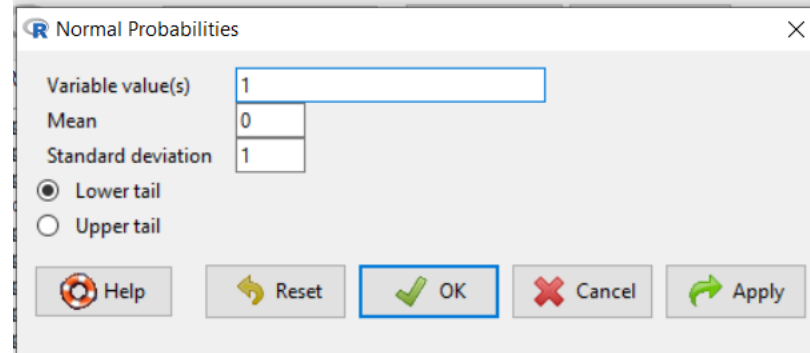
From 0.9473 subtract 0.5 to find the area between 0 and 1.62 $\Rightarrow 0.9473 - 0.5 = 0.4473$

c1)



```
> pnorm(c(2), mean=0, sd=1, lower.tail=TRUE)
[1] 0.9772499
```

c2)



```
> pnorm(c(1), mean=0, sd=1, lower.tail=TRUE)
[1] 0.8413447
```

From c1 subtract c2 to find the area.
 $= 0.9772 - 0.8413 = 0.1359$

NORMAL DISTRIBUTION ANOTHER EXAMPLE

An HEB store sells donuts they bake every day. All the donuts not sold are discarded. The demand for donuts follows a normal distribution with $\mu=35$ donuts and $\sigma=8$ donuts. How many donuts should HEB make every day to meet the demand 90% of the time?

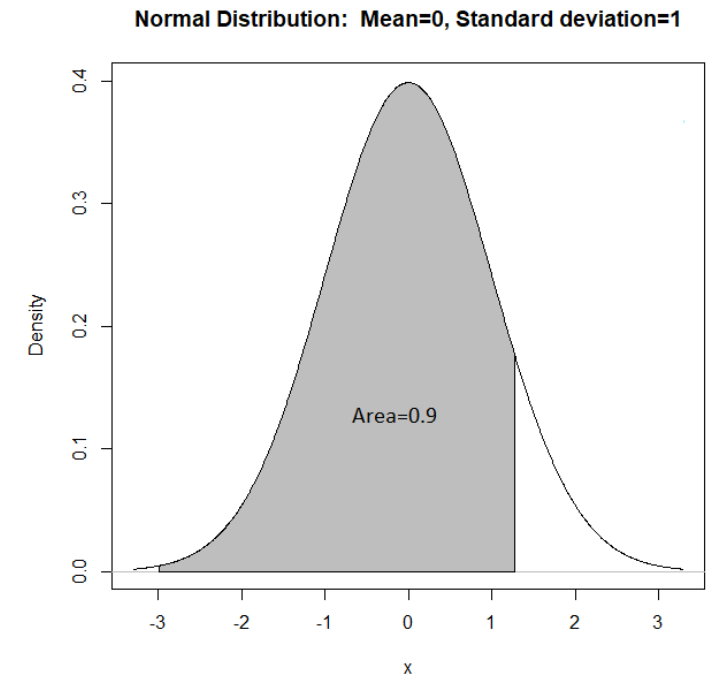
Solution:

x_0 = the number of donuts to bake

$$P(X < x_0) = 0.90 = 0.5 + 0.4 = 0.5 + P(0 \leq Z \leq 1.28)$$

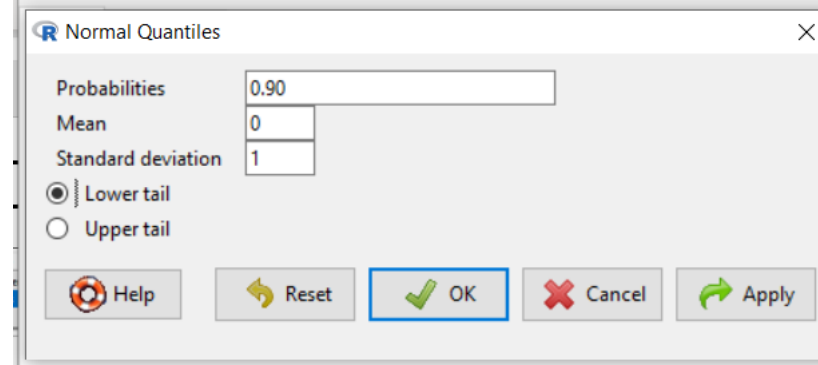
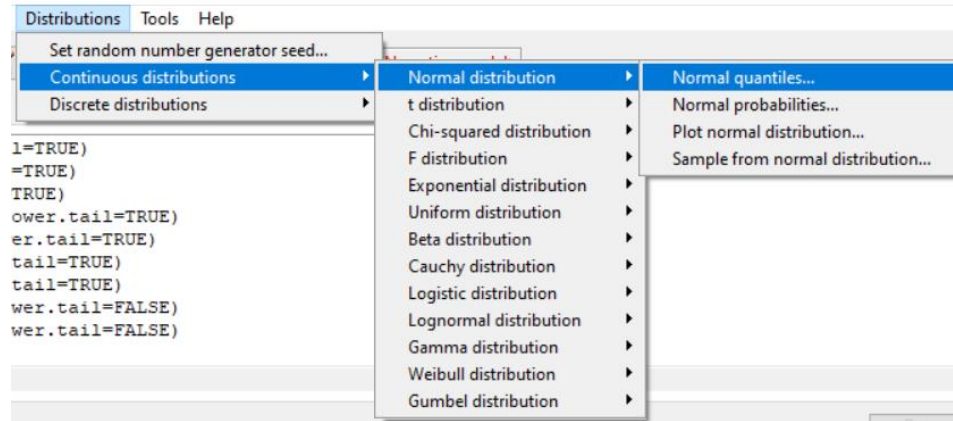
$$\Rightarrow P(Z \leq 1.28) = 0.4 + 0.5 = 0.9 \Rightarrow (x_0 - 35)/8 = 1.28$$

$$\Rightarrow x_0 = 35 + 1.28 * 8 = 45.24 \approx 46 \text{ donuts.}$$



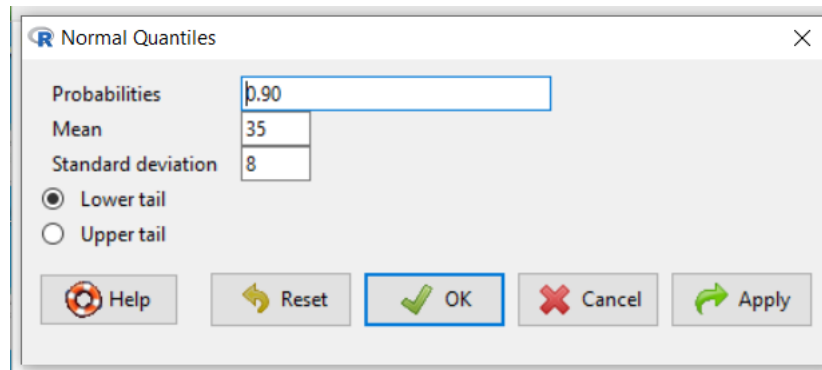
NORMAL DISTRIBUTION ANOTHER EXAMPLE

Find Z score

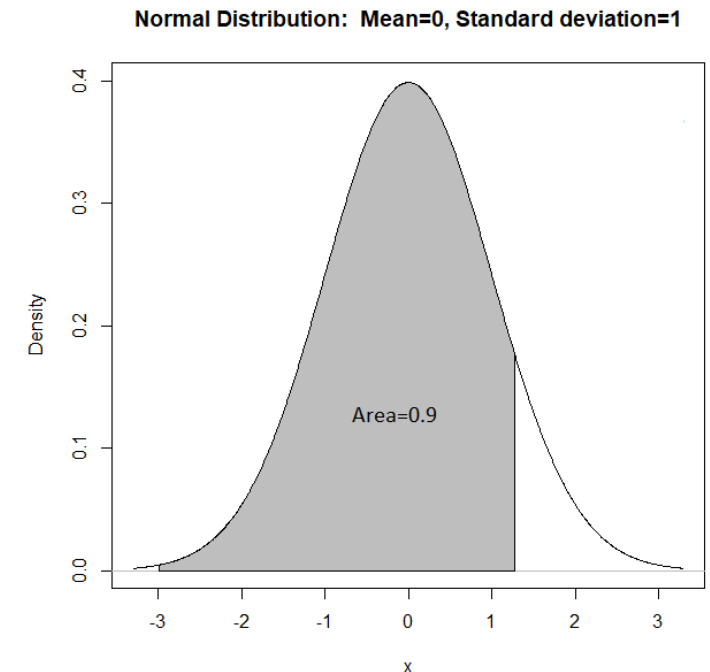


```
> qnorm(c(0.90), mean=0, sd=1, lower.tail=TRUE)
[1] 1.281552
```

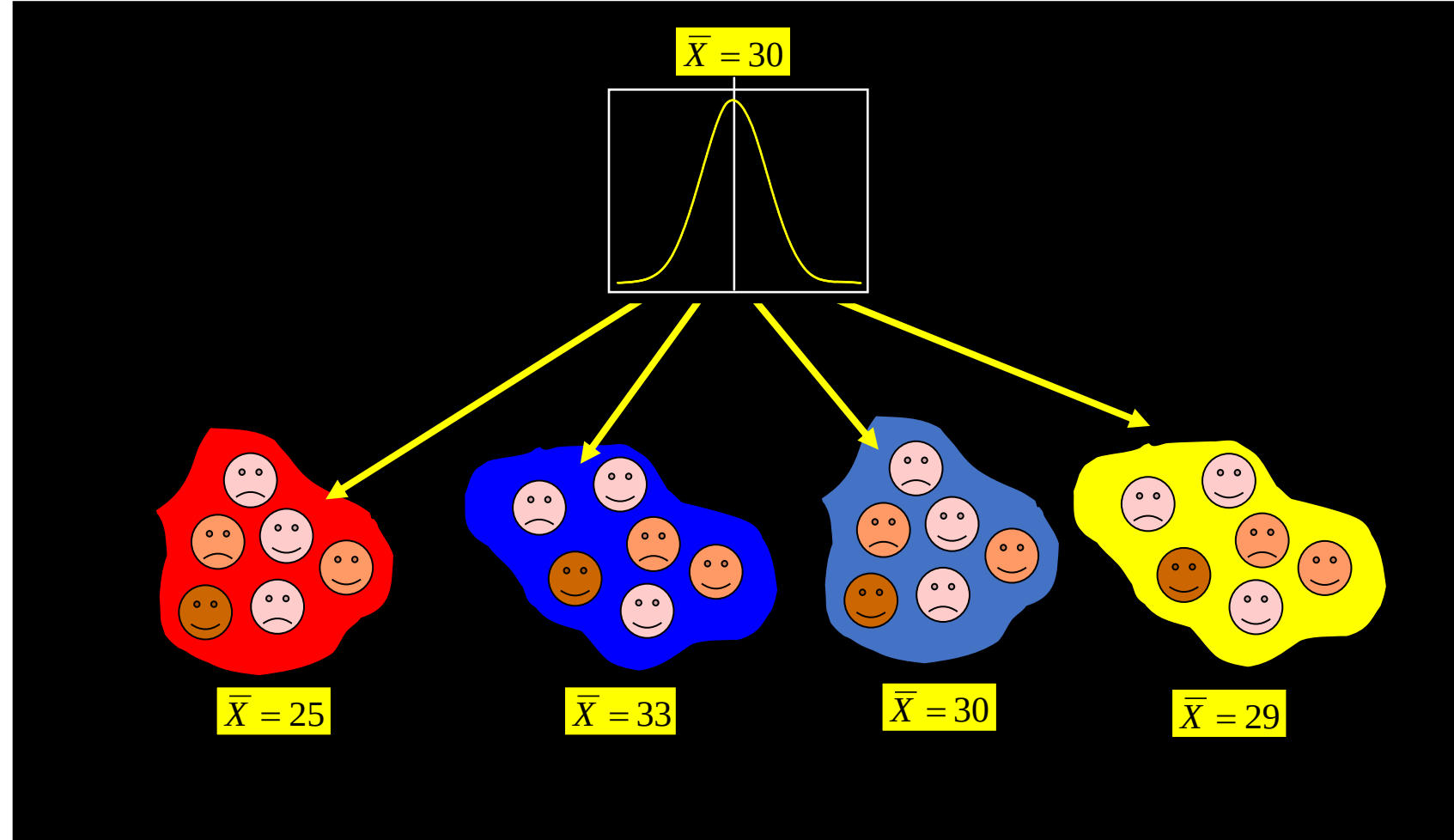
Find direct solution



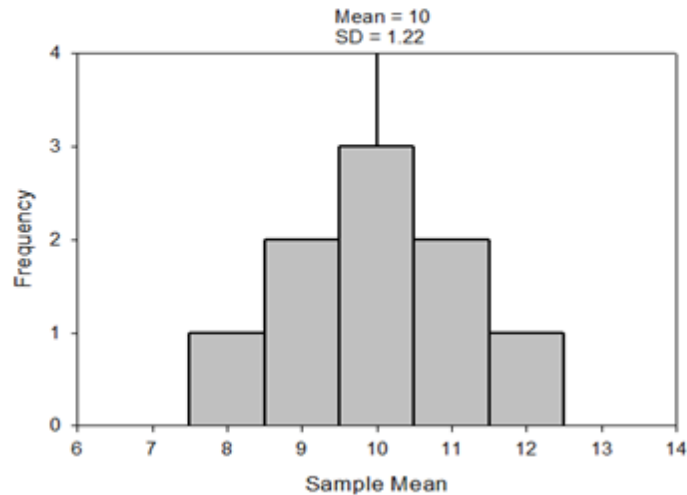
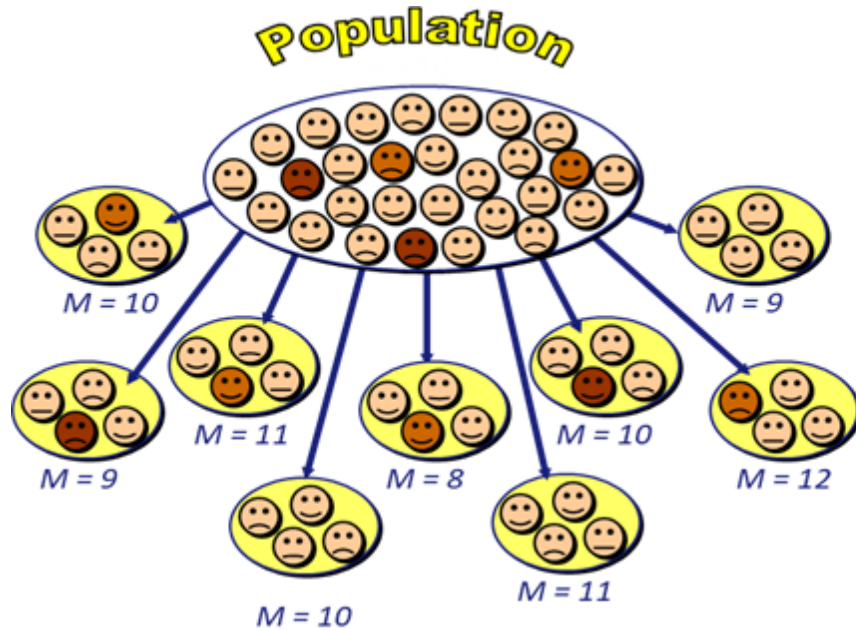
```
> qnorm(c(0.90), mean=35, sd=8, lower.tail=TRUE)
[1] 45.25241
```



STATISTICAL INFERENCE AND SMPLING



STATISTICAL INFERENCE AND SMPLING



$$\mu_{\bar{X}} = \mu$$

$$\sigma_{\bar{X}} = \frac{s}{\sqrt{n}}$$

CENTRAL LIMIT THEOREM

When obtaining large samples($n > 30$) from any population, the sample mean will follow an approximate normal distribution.

The standard error of the sample mean ($\bar{x}$) = standard deviation of the probability distribution for $\bar{x}$.

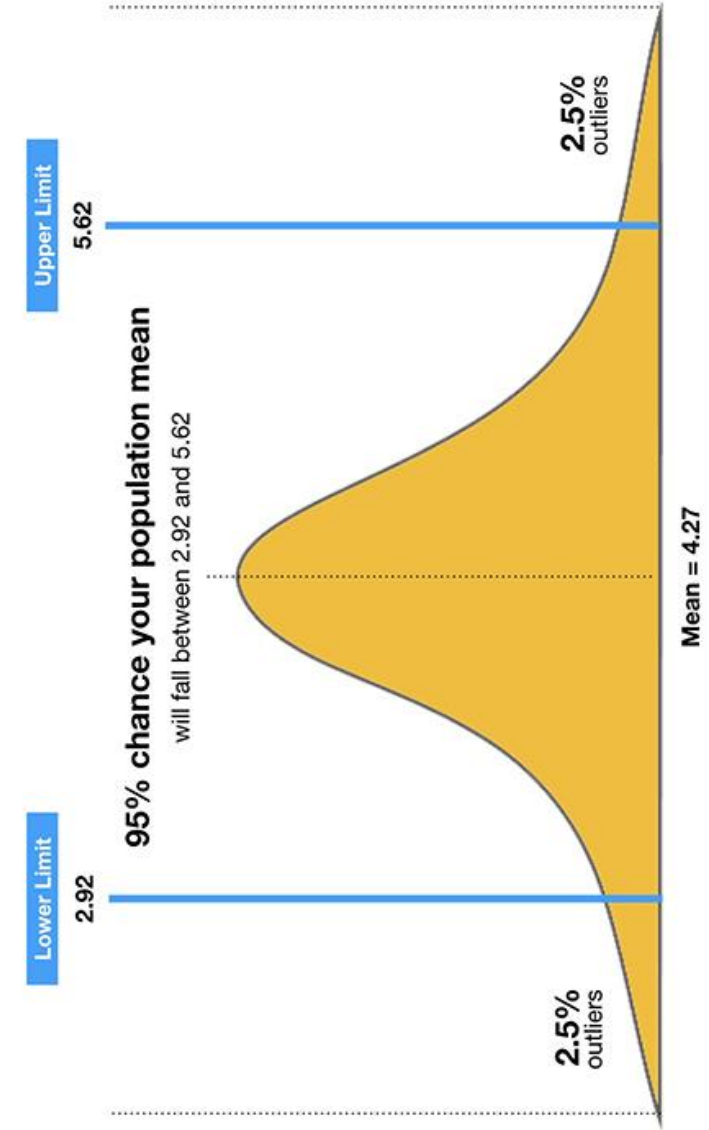
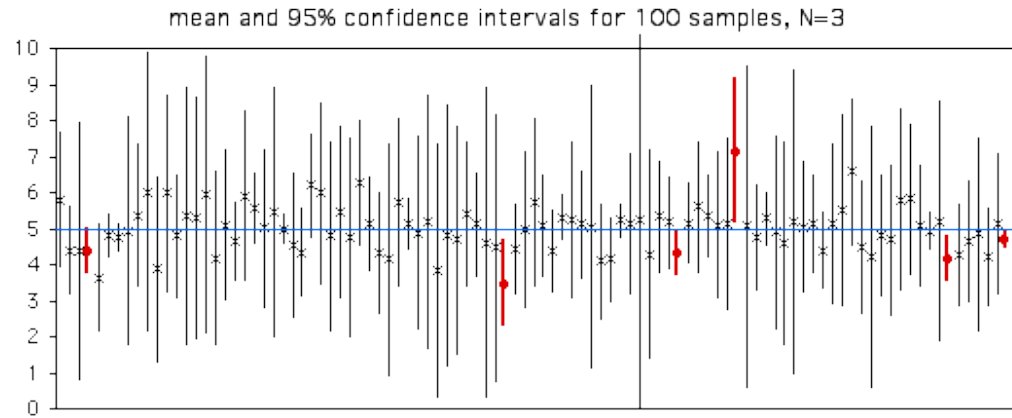
CONFIDENCE INTERVALS

After you've calculated the mean of a set of observations, you should give some indication of how close your estimate is likely to be to the parametric ("true") mean. One way to do this is with confidence limits. Confidence limits are the numbers at the upper and lower end of a confidence interval; for example, if your mean is 7.4 with confidence limits of 5.4 and 9.4, your confidence interval is 5.4 to 9.4. Most people use 95% confidence limits, although you could use other values. Setting 95% confidence limits means that if you took repeated random samples from a population and calculated the mean and confidence limits for each sample, the confidence interval for 95% of your samples would include the parametric mean.

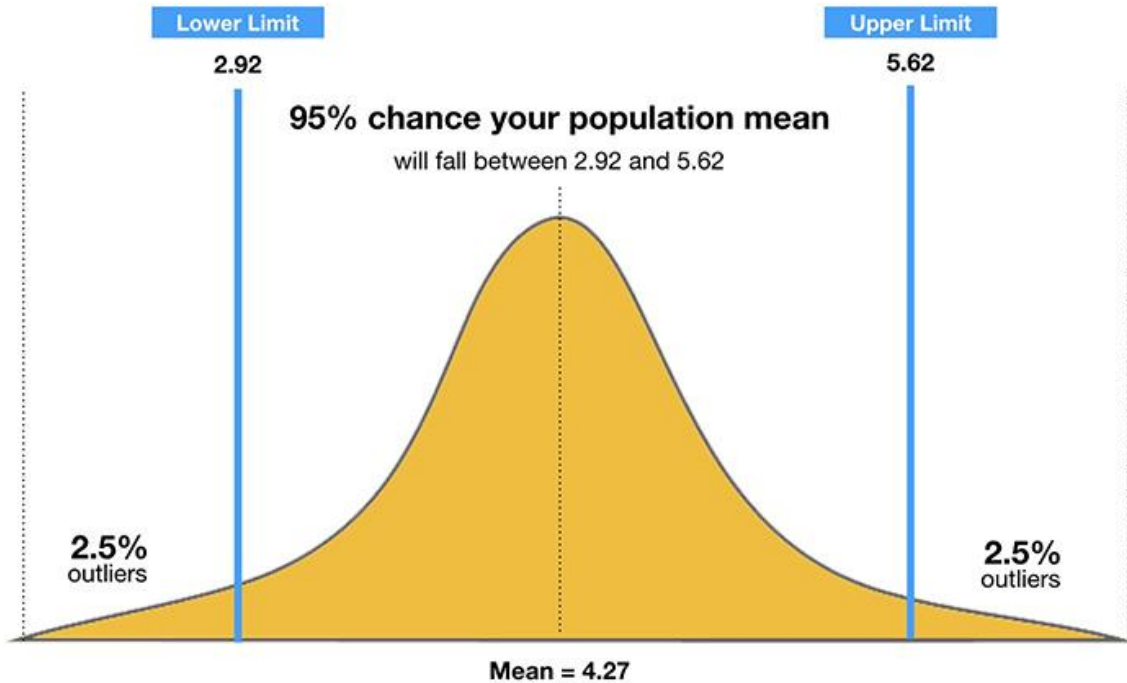
To illustrate this, here are the means and confidence intervals for 100 samples of 3 observations from a population with a parametric mean of 5. Of the 100 samples, 94 (shown with X for the mean and a thin line for the confidence interval) have the parametric mean within their 95% confidence interval, and 6 (shown with circles and thick lines) have the parametric mean outside the confidence interval.

CONFIDENCE INTERVALS (σ known)

We know how to estimate μ (the average of the entire population) because we can calculate the mean of the sample and we know the result is unbiased. The sample mean is also called a point estimate because it consist of a single number.



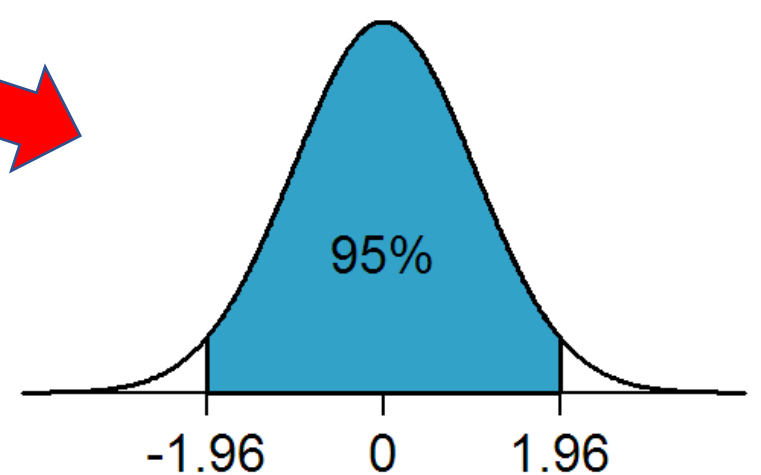
CONFIDENCE INTERVALS



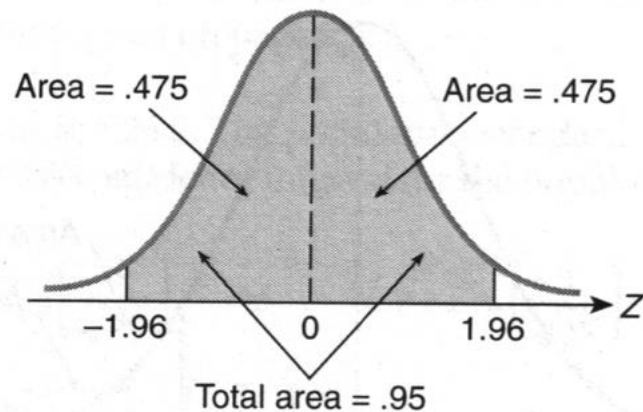
The 95% confidence interval is a range of values that you can be 95% confident contains the true mean of the population. Due to natural sampling variability, the sample mean (center of the CI) will vary from sample to sample.

For example, the probability of the population mean value being between -1.96 and +1.96 standard deviations (z-scores) from the sample mean is 95%.

Accordingly, there is a 5% chance that the population mean lies outside of the upper and lower confidence interval (as illustrated by the 2.5% of outliers on either side of the 1.96 z-scores).



CONFIDENCE INTERVALS



To calculate the confidence interval, start by computing the mean and standard error of the sample.

$$\mu_{\bar{X}} = \mu$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

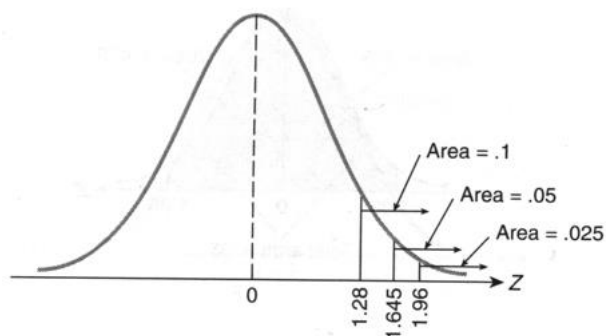
$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$P(-1.96 \leq Z \leq 1.96) = .95$$

$$P\left(-1.96 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq 1.96\right) = .95$$

$$P\left(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}\right) = .95$$

$$\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}}\right)$$



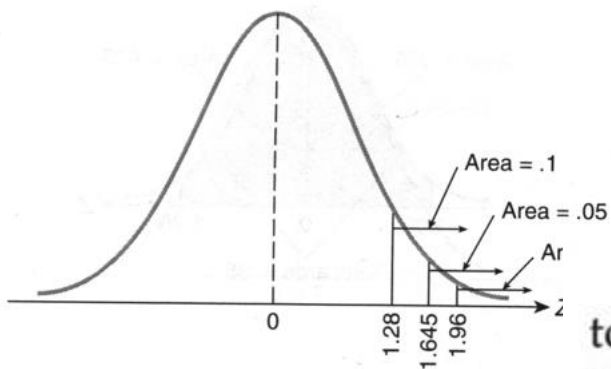
Using the values from our assembly-time example, we have $\bar{x} = 26.9$, $\sigma = 3$, and $n = 25$. The resulting 95% confidence interval is

$$\left(26.9 - 1.96 \cdot \frac{3}{\sqrt{25}}, 26.9 + 1.96 \cdot \frac{3}{\sqrt{25}}\right)$$

or

$$(25.72, 28.08)$$

CONFIDENCE INTERVALS



Let Z_a denote the value of Z such that the area *to the right* of this value is equal to a . How can we determine $Z_{.025}$, $Z_{.05}$, and $Z_{.1}$ (Figure 8.16)? Using Table A.4, $Z_{.025} = 1.96$, $Z_{.05} = 1.645$, and $Z_{.1} = 1.28$.

When defining a confidence interval for μ , we can define a 99% confidence interval, a 95% confidence interval, a 90% confidence interval, or whatever. The specific percentage represents the **confidence level**. The *higher* the confidence level, the *wider* the confidence interval. The confidence level is written as $(1 - \alpha) \cdot 100\%$, where $\alpha = .01$ for a 99% confidence interval, $\alpha = .05$ for a 95% confidence interval, and so on. Thus, a $(1 - \alpha) \cdot 100\%$ confidence interval for the mean of a normal population, μ , is

$$\left[\bar{x} - Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right), \bar{x} + Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \right]$$

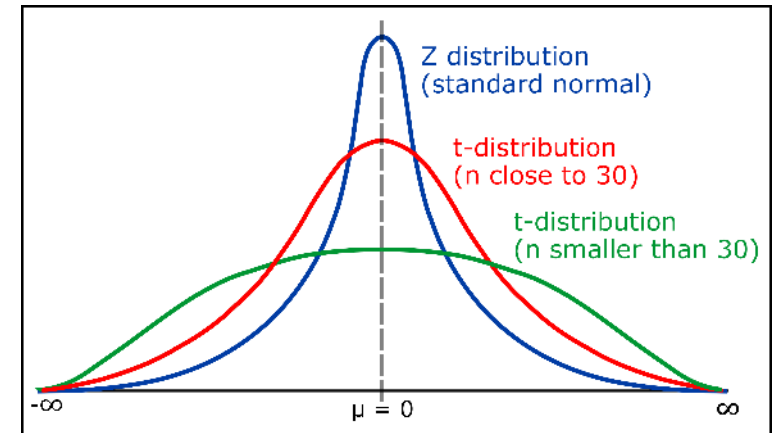
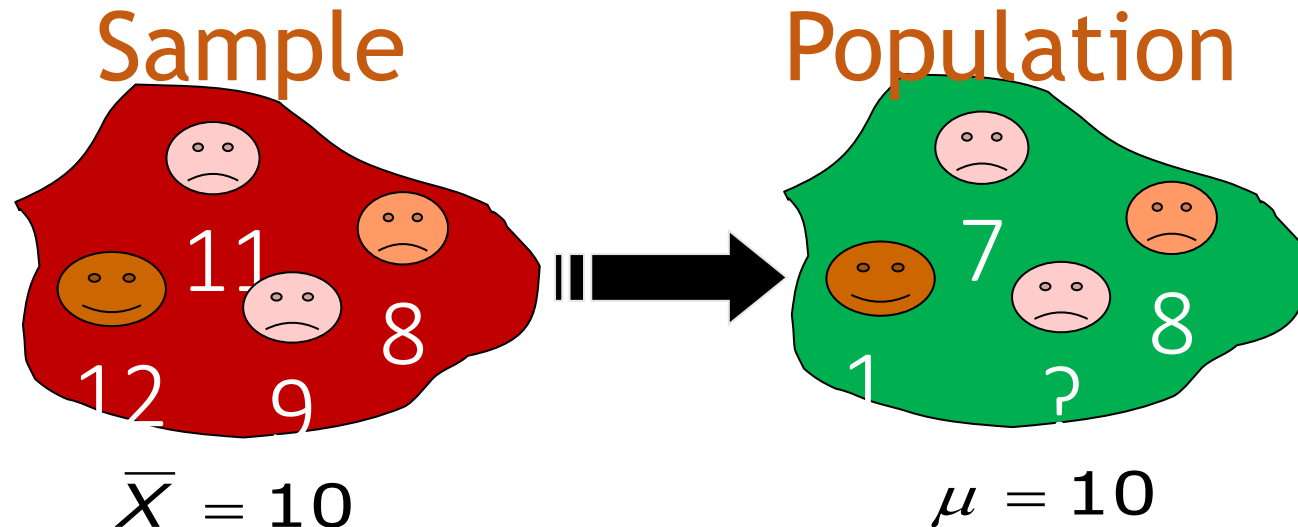
Margin of error

According to the Central Limit Theorem, formula provides an approximate confidence interval for the mean of any population, provided the sample size, n , is large

CONFIDENCE INTERVALS (σ unknown)

When σ unknown we cannot calculate the confidence interval using the previous formula and $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ becomes $\frac{\bar{X}-\mu}{s/\sqrt{n}}$ and this no longer follow a standard normal random variable Z. It follows a **t-distribution** where the shape depends on the sample size. To take into consideration the sample size we need to use the degrees of freedom(df). **Degrees of freedom = $df=n-1$** .

Degrees of freedom are the number of independent variables that can be estimated in a statistical analysis.



CONFIDENCE INTERVALS (σ unknown)

When **σ is estimated** from the sample standard deviation s , the

sampling distribution for $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$ will depend on the sample size.

The sample distribution of $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

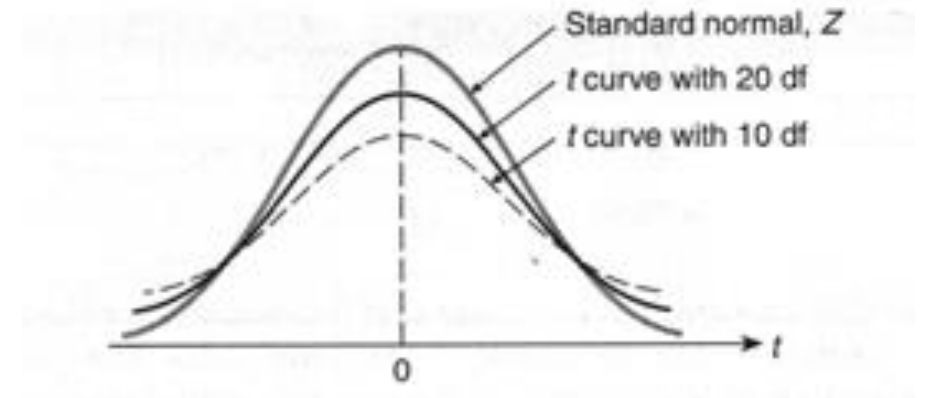
is a t distribution with $n - 1$ degrees of freedom.

The degrees of freedom (df) is a measure of how well s estimates σ . *The larger the degrees of freedom, the better σ is estimated.*

We use the t -tables to obtain these “critical” values.

The confidence interval is between

$$\bar{X} - t_{\alpha/2, n-1} \left(\frac{s}{\sqrt{n}} \right) \text{ and } \bar{X} + t_{\alpha/2, n-1} \left(\frac{s}{\sqrt{n}} \right)$$



CONFIDENCE INTERVALS (σ unknown)

The output voltage of power supplies manufactured by a company is believed to follow a normal distribution. The company believed that the average output voltage of a particular power supply unit is 10 volts. The company collects eighteen random observations from this power supply.

10.85, 11.40, 10.81, 10.24, 10.23, 9.49, 9.89, 10.11, 10.57, 11.21, 10.10, 11.22, 10.31, 11.24, 9.51, 10.52, 9.92, 8.33

What is the 95% confidence interval for the average output voltage for this power supply unit?

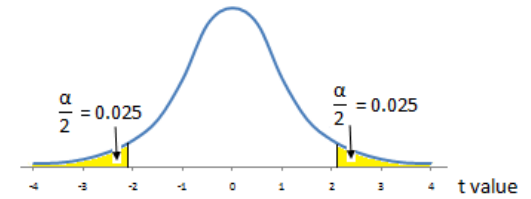
The point estimate of σ is $s = .7667$

The point estimate of μ is $\bar{x} = 10.3306$

$$\bar{x} - t_{.025,17} \left(\frac{.7667}{\sqrt{18}} \right) = 9.949 \quad \text{and} \quad \bar{x} + t_{.025,17} \left(\frac{.7667}{\sqrt{18}} \right) = 10.712$$

Student's t Distribution Table

For example, the t value for
18 degrees of freedom
is 2.101 for 95% confidence
interval (2-Tail $\alpha = 0.05$).



	90%	95%	97.5%	99%	99.5%	99.95%	1-Tail Confidence Level
	80%	90%	95%	98%	99%	99.9%	2-Tail Confidence Level
	0.100	0.050	0.025	0.010	0.005	0.0005	1-Tail Alpha
df	0.20	0.10	0.05	0.02	0.01	0.001	2-Tail Alpha
1	3.0777	6.3138	12.7062	31.8205	63.6567	636.6192	
2	1.8856	2.9200	4.3027	6.9646	9.9248	31.5991	
3	1.6377	2.3534	3.1824	4.5407	5.8409	12.9240	
4	1.5332	2.1318	2.7764	3.7469	4.6041	8.6103	
5	1.4759	2.0150	2.5706	3.3649	4.0321	6.8688	
6	1.4398	1.9432	2.4469	3.1427	3.7074	5.9588	
7	1.4149	1.8946	2.3646	2.9980	3.4995	5.4079	
8	1.3968	1.8595	2.3060	2.8965	3.3554	5.0413	
9	1.3830	1.8331	2.2622	2.8214	3.2498	4.7809	
10	1.3722	1.8125	2.2281	2.7638	3.1693	4.5869	
11	1.3634	1.7959	2.2010	2.7181	3.1058	4.4370	
12	1.3562	1.7823	2.1788	2.6810	3.0545	4.3178	
13	1.3502	1.7709	2.1604	2.6503	3.0123	4.2208	
14	1.3450	1.7613	2.1448	2.6245	2.9768	4.1405	
15	1.3406	1.7531	2.1314	2.6025	2.9467	4.0728	
16	1.3368	1.7459	2.1199	2.5835	2.9208	4.0150	
17	1.3334	1.7396	2.1098	2.5669	2.8982	3.9651	
18	1.3304	1.7341	2.1009	2.5524	2.8784	3.9216	
19	1.3277	1.7291	2.0930	2.5395	2.8609	3.8834	
20	1.3253	1.7247	2.0860	2.5280	2.8453	3.8495	
21	1.3232	1.7207	2.0796	2.5176	2.8314	3.8193	
22	1.3212	1.7171	2.0739	2.5083	2.8188	3.7921	
23	1.3195	1.7139	2.0687	2.4999	2.8073	3.7676	
24	1.3178	1.7109	2.0639	2.4922	2.7969	3.7454	
25	1.3163	1.7081	2.0595	2.4851	2.7874	3.7251	
26	1.3150	1.7056	2.0555	2.4786	2.7787	3.7066	
27	1.3137	1.7033	2.0518	2.4727	2.7707	3.6896	
28	1.3125	1.7011	2.0484	2.4671	2.7633	3.6739	
29	1.3114	1.6991	2.0452	2.4620	2.7564	3.6594	
30	1.3104	1.6973	2.0423	2.4573	2.7500	3.6460	

CONFIDENCE INTERVALS (σ unknown) using Rcmdr

Import your data from excel into the Rcmdr environment

Run numerical summaries: Statistics → Summaries → Numerical summaries

mean	sd	IQR	skewness	kurtosis	0%	25%	50%	75%	100%	n
10.33056	0.7667155	0.875	-0.8372635	1.332968	8.33	9.965	10.275	10.84	11.4	18

Run one sample t-test: Statistics → Means → Single sample t-test

```
One Sample t-test

data:  Data
t = 57.164, df = 17, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
  9.949277 10.711834
sample estimates:
mean of x
 10.33056
```

	Data
1	10.85
2	11.40
3	10.81
4	10.24
5	10.23
6	9.49
7	9.89
8	10.11
9	10.57
10	11.21
11	10.10
12	11.22
13	10.31
14	11.24
15	9.51
16	10.52
17	9.92
18	8.33

$$\left[\bar{x} - Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right), \bar{x} + Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \right]$$

Margin error (E) : How large our sample should be if we want our point estimate for $\bar{x}$ to be within 15 hours of the actual values of μ with 95% confidence interval ?

FIND NECESSARY SAMPLE SIZE

General formula $n = \left[\frac{Z_{\alpha/2} \cdot \sigma}{E} \right]^2$

For σ known

Question: How large a sample is needed if we want our point estimator $\bar{x}$ to be within 15 units of the actual value of μ with 95% confidence?

Answer:

$$E = 15 = Z_{\alpha/2} \left(\frac{50}{\sqrt{n}} \right) \text{ where } Z_{\alpha/2} = Z_{0.25} = 1.96 \Rightarrow n = 42.68 \approx 43$$

For σ unknown

Question: How large a sample is needed if we want our point estimator $\bar{x}$ to be within 15 units of the actual value of μ with 95% confidence?

Answer: a) preliminary sampling. Get s from the preliminary sample and replace σ with s . If $n > 30$ we can still use $Z_{\alpha/2}$

b) get a rough approximation of σ

We know 95.4% of the population is between $\mu - 2\sigma = L$ and $\mu + 2\sigma = H$.

$H - L = 4\sigma \Rightarrow \sigma = \frac{H - L}{4}$ We replace the calculated value in the E formula.

CONFIDENCE INTERVALS PROPORTIONS

Suppose the proportion of American adults who support the expansion of solar energy is $p = 0.88$, which is our parameter of interest. If we were to take a poll of 1000 American adults on this topic, the estimate would not be perfect, but how close might we expect the sample proportion in the poll would be to 88%?

CENTRAL LIMIT THEOREM AND THE SUCCESS-FAILURE CONDITION

When observations are independent and the sample size is sufficiently large, the sample proportion $\hat{p}$ will tend to follow a normal distribution with the following mean and standard error:

$$\mu_{\hat{p}} = p \qquad SE_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

In order for the Central Limit Theorem to hold, the sample size is typically considered sufficiently large when $np \geq 10$ and $n(1-p) \geq 10$, which is called the **success-failure condition**.

There are $n = 1000$ observations for each sample proportion $\hat{p}$, and each of those observations are independent draws. The most common way for observations to be considered independent is if they are from a simple random sample.

Success-failure condition. We can confirm the sample size is sufficiently large by checking the success-failure condition and confirming the two calculated values are greater than 10:

$$np = 1000 \times 0.88 = 880 \geq 10 \quad n(1-p) = 1000 \times (1 - 0.88) = 120 \geq 10$$

The independence and success-failure conditions are both satisfied, so the Central Limit Theorem applies, and it's reasonable to model $\hat{p}$ using a normal distribution

CONFIDENCE INTERVALS PROPORTIONS

Suppose the proportion of American adults who support the expansion of solar energy is $p = 0.88$, which is our parameter of interest. If we were to take a poll of 1000 American adults on this topic, the estimate would not be perfect, but how close might we expect the sample proportion in the poll would be to 88%?

Compute and interpret a 95% confidence interval for the population proportion.

$$\hat{p} \pm 1.96 \times SE_{\hat{p}} \rightarrow 0.887 \pm 1.96 \times 0.010 \rightarrow (0.8674, 0.9066)$$

CONFIDENCE INTERVAL FOR A SINGLE PROPORTION

Once you've determined a one-proportion confidence interval would be helpful for an application, there are four steps to constructing the interval:

Prepare. Identify $\hat{p}$ and n , and determine what confidence level you wish to use.

Check. Verify the conditions to ensure $\hat{p}$ is nearly normal. For one-proportion confidence intervals, use $\hat{p}$ in place of p to check the success-failure condition.

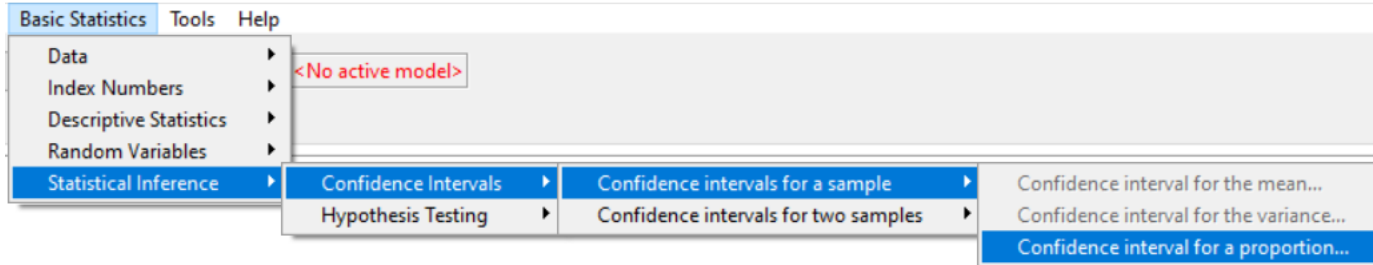
Calculate. If the conditions hold, compute SE using $\hat{p}$, find z^* , and construct the interval.

Conclude. Interpret the confidence interval in the context of the problem.

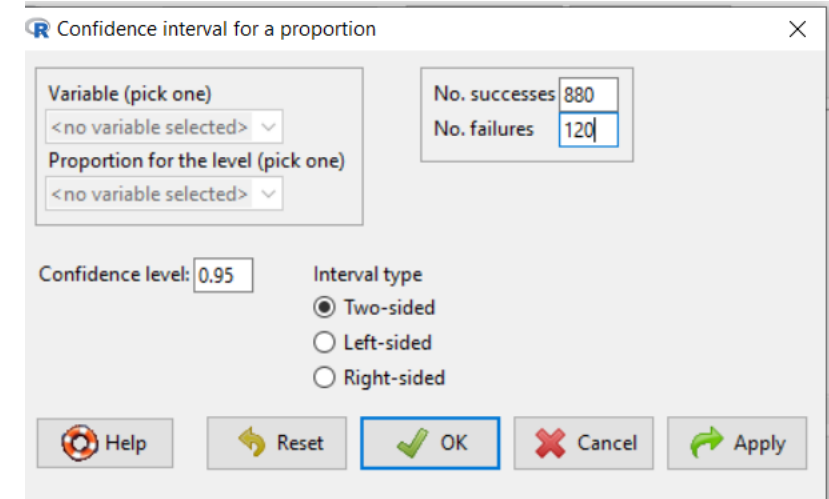
We are 95% confident that 86.74% to 90.66% of American adults support the expansion of solar power in 2018.

CONFIDENCE INTERVALS PROPORTIONS

Load [RcmdrPlugin.TeachStat](#) plugin and follow the menu Basic Statistics



```
Confidence Interval for a proportion
-----
Interval type: Two-sided
Confidence level: 95%
Sample: No. successes = 880 -- No. attempts = 1000
Sample estimate: proportion 0.88
Interval: ( 0.859859 , 0.900141 )
```



HYPOTHESES TESTING

A newspaper article claim the average height of adult males in the U.S. is not the same as it was 50 years ago. It was 5.9 feet. Supposing you are the owner of a clothing firm, so this info is important to you. To investigate the article claim you selected 75 males, measured their heights, and found $\bar{X} = 5.79$ feet. From previous experience you know that $\sigma = 0.4$ feet.

Null Hypothesis

$$H_0: \mu = 5.9$$

Alternative Hypothesis

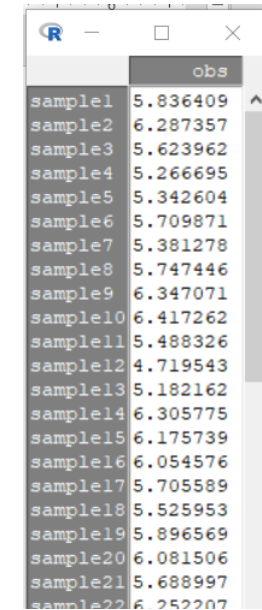
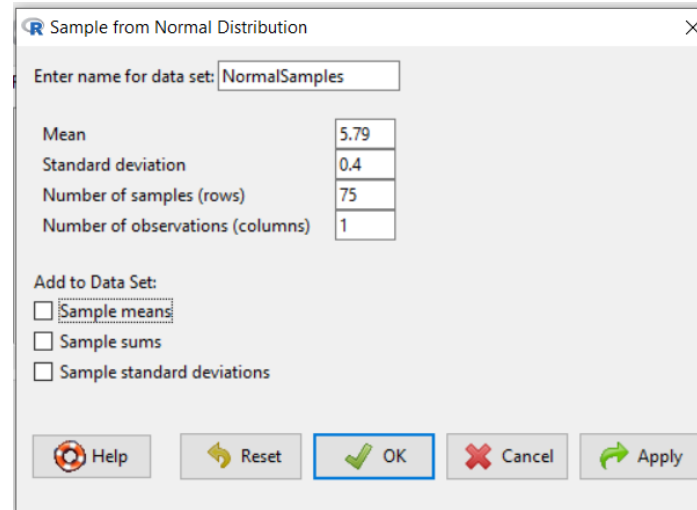
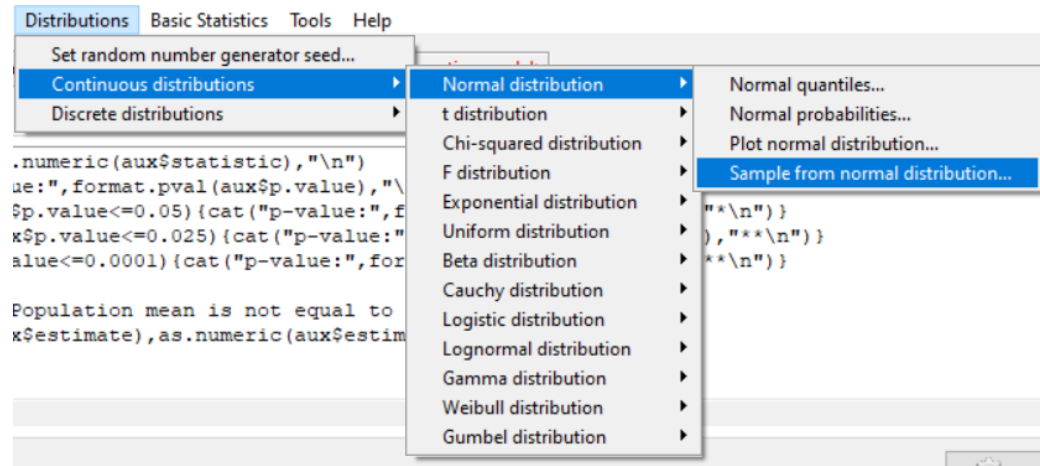
$$H_a: \mu \neq 5.9$$

Null hypothesis: A statement (equality or inequality) concerning a population parameter; the researcher want to discredit

Alternative hypothesis: A statement in contradiction to the null hypothesis; the researcher wishes to support

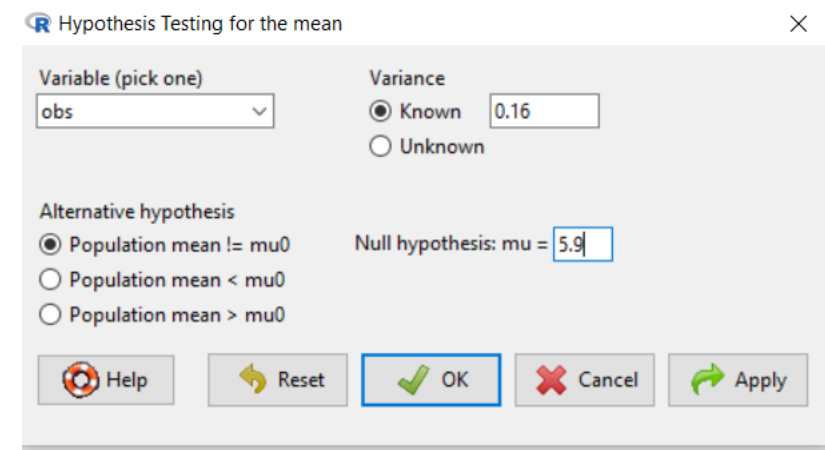
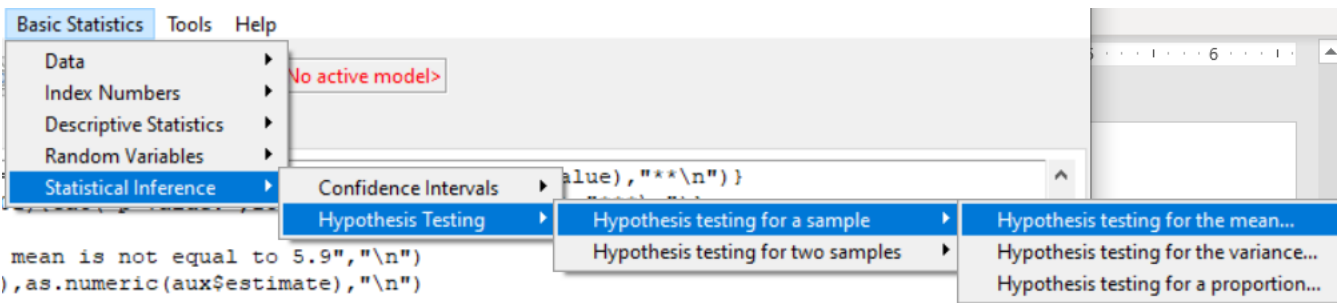
HYPOTHESES TESTING (HOW TO WITH Rcmdr)

1. Generate a sample with the corresponding statistics.

A screenshot of the R console window showing a list of 22 samples. Each sample has an 'obs' column with a numerical value.

	obs
sample1	5.836409
sample2	6.287357
sample3	5.623962
sample4	5.266695
sample5	5.342604
sample6	5.709871
sample7	5.381278
sample8	5.747446
sample9	6.347071
sample10	6.417262
sample11	5.488326
sample12	4.719543
sample13	5.182162
sample14	6.305775
sample15	6.175739
sample16	6.054576
sample17	5.705589
sample18	5.525953
sample19	5.896569
sample20	6.081506
sample21	5.688997
sample22	6.252207

2. Run hypothesis testing



```
mean is not equal to 5.9", "\n")
), as.numeric(aux$estimate), "\n")

Hypothesis Testing for the mean with known variance = 0.16
-----
Variable: obs
Distribution: N(0,1)
Test statistics value: -3.828735
p-value: 0.0001288 **
Alternative hypothesis: Population mean is not equal to 5.9
Sample estimate: mean of NormalSamples$obs 5.723158
```

HYPOTHESES TESTING Type I and Type II Errors

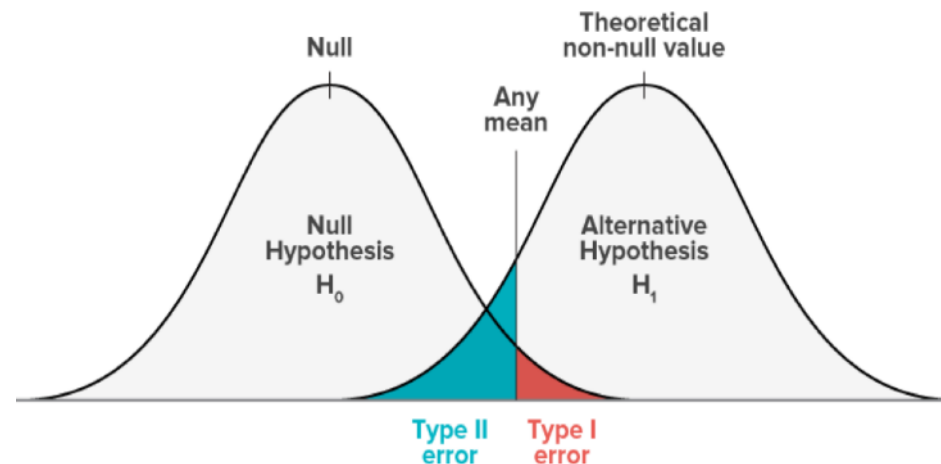
Type I Error: Rejecting the null hypothesis when in fact it is true.

Type II Error: Fail to reject the null hypothesis when in fact the null hypothesis is not true.

α = probability of rejecting H_0 when H_0 is true = P(Type I Error)

β = probability of failing to reject H_0 when H_0 is false = P(Type II Error)

Power of the test = $1 - \beta$



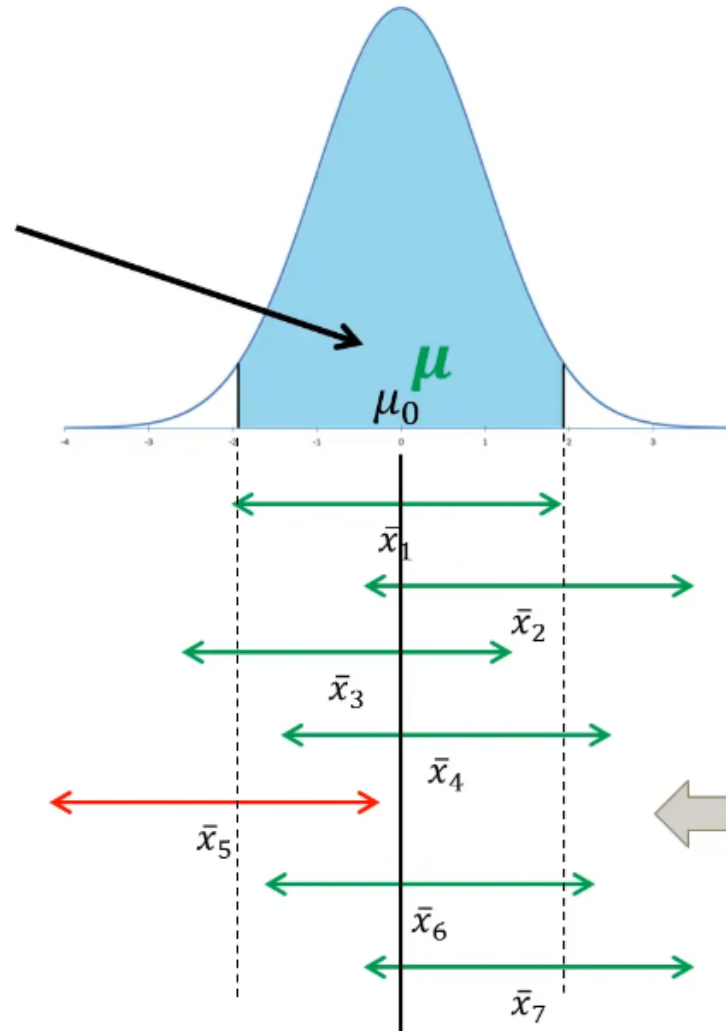
	THE TRUTH	
	The null hypothesis (H_0) is true (H_a is false)	The null hypothesis (H_0) is not true (H_a is true)
Reject H_0 (support H_a)	TYPE I (α) error/ Alpha Risk/ p – value Overreacting ($1 - \alpha$) = the Confidence level of the test	Correct Decision ($1 - \beta$) Power of the test
Fail to Reject H_0 (do not support H_a)	Correct Decision	TYPE II (β) error/ Beta Risk Underreacting

HYPOTHESES TESTING Type I Errors

$$\alpha = .05$$

95% of all sample means ($\bar{x}$) are *hypothesized to be* in this region.

Fail to reject null hypothesis
Fail to reject null hypothesis
Fail to reject null hypothesis
Fail to reject null hypothesis
Reject null hypothesis
Fail to reject null hypothesis
Fail to reject null hypothesis



$$H_0: \mu = \mu_0$$

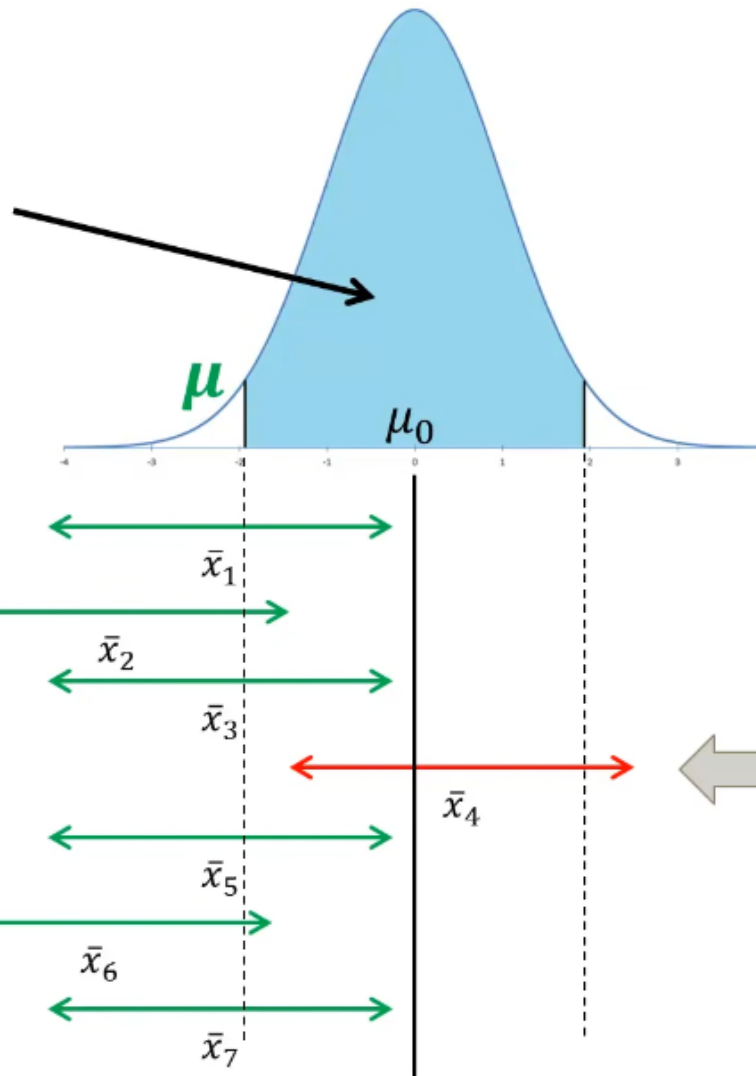
$$H_a: \mu \neq \mu_0$$

If we took a sample and it was by chance like $\bar{x}_5$, we would **incorrectly** reject the null hypothesis.

Type I Error

HYPOTHESES TESTING Type II Errors

95% of all sample means ($\bar{x}$) are hypothesized to be in this region.
 $\alpha = .05$



Reject null hypothesis

Reject null hypothesis

Reject null hypothesis

Fail to reject null hypothesis

Reject null hypothesis

Reject null hypothesis

Reject null hypothesis

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

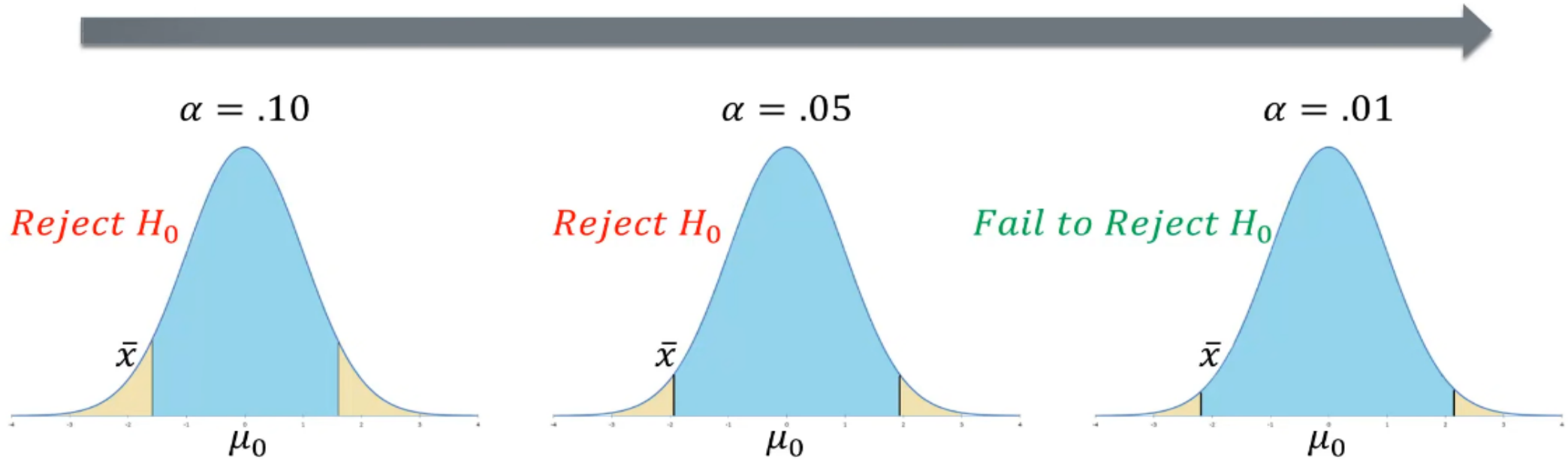
If we took a sample and it was by chance like $\bar{x}_4$, we would incorrectly “accept” the null hypothesis.

Type II Error

Beta (β) is the probability of committing a Type II error. The value of β varies with certain experimental factors.

HYPOTHESES TESTING Type I and II Errors

As α decreases so does the chance of Type I error. The critical value to reject the null hypothesis moves outward thus “capturing” more sample means. Kind of like moving the goal posts on a football pitch; the wider they are, more kicks go in.



However the move outward of the critical values may also “capture” a mean from a different population off to the side. We would fail to reject the null when indeed we should. Thus the chance of Type II error increases as α decreases.

HYPOTHESES TESTING Type I and Type II Errors - Example

We sample 16 values from a normally distributed population with $\sigma=8$ and μ unknown. We test: $H_0: \mu = 75$

$H_a: \mu \neq 75$ at $\alpha = 0.05$

We know $Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$ and ask, “for what values of Z we reject H_0 ?”

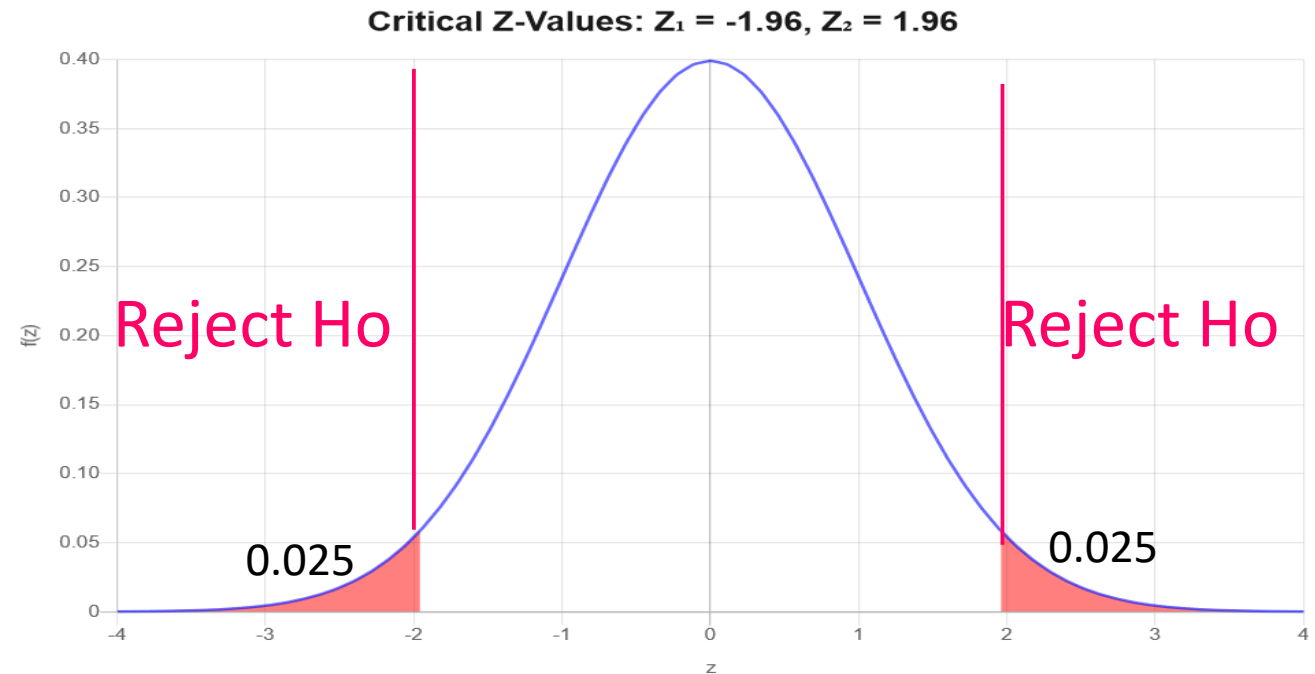
We reject the null if:

$$Z \leq -1.96 \text{ or } Z \geq 1.96$$

Written different:

$$\frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \leq -1.96 \text{ or } \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \geq 1.96$$

$$\bar{X} \leq 71.08 \text{ and } \bar{X} \geq 78.92$$



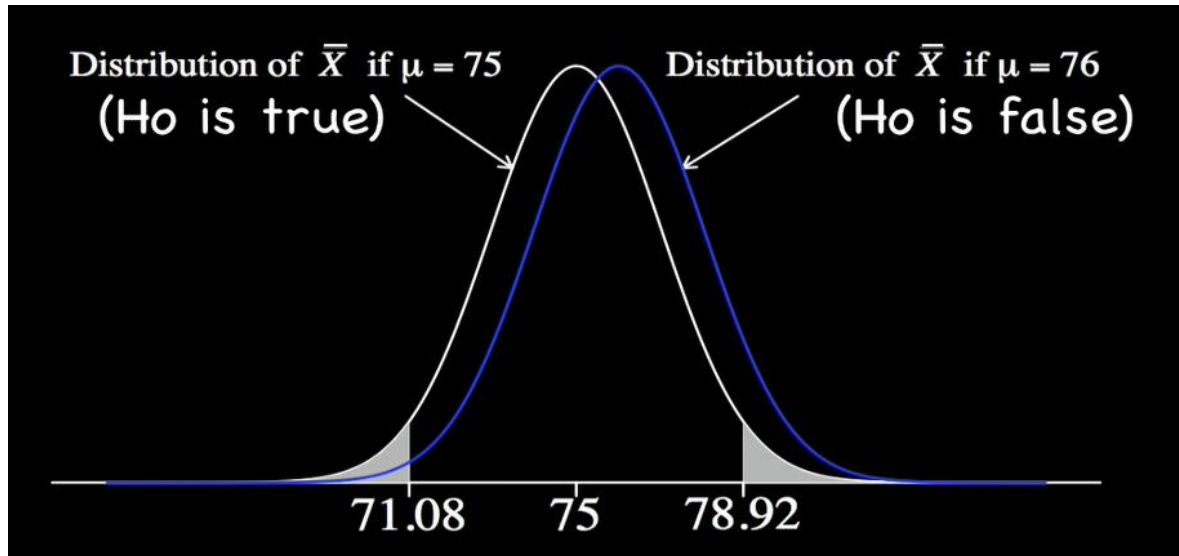
HYPOTHESES TESTING Type I and Type II Errors - Example

The picture shows the distribution of $\bar{X}$ for $\mu=75$ (H_0 is true)

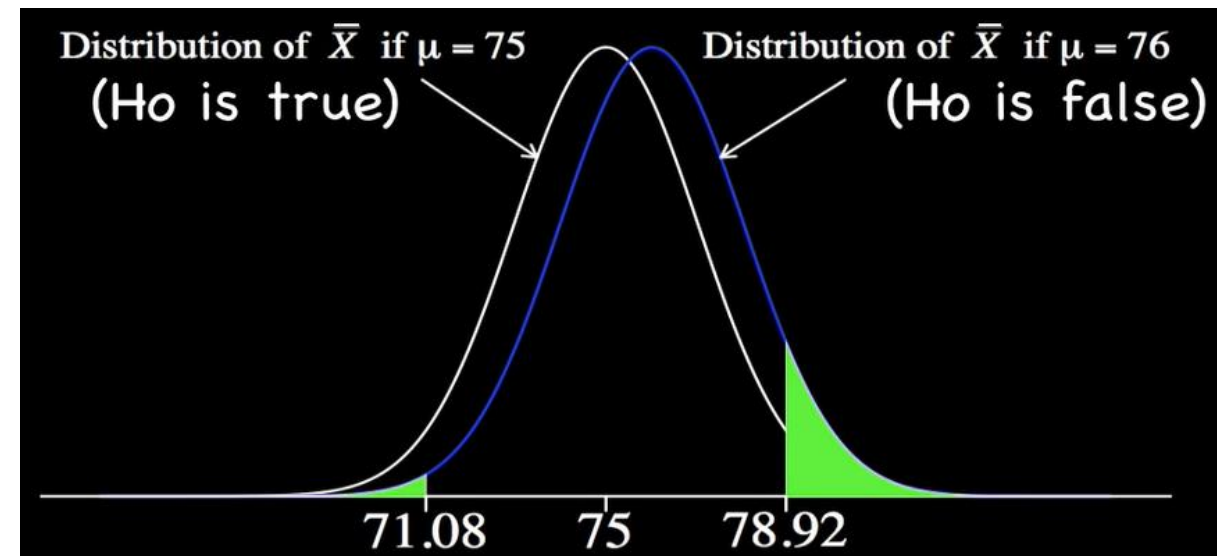
We can ask "If $\mu=76$, what is the power of our test knowing that:

$H_0: \mu = 75$ and $H_a: \mu \neq 75$

Power = $P(\text{Reject } H_0 | \mu = 76)$



Grey areas = Type I Error



Green area = Power of test

$$P(\bar{X} \leq 71.8 | \mu = 76) \text{ and } P(\bar{X} \geq 78.92 | \mu = 76)$$

Using the same Z formula, we get $P(Z \leq -2.46) = 0.0069$ and $P(Z \geq 1.46) = 0.0721$ The total sum of the two green areas is 0.079

What would be the probability of a Type 2 Error. Well $P(\text{Type 2 Error}) = \beta = 1 - \text{Power of test} = 1 - 0.079 = 0.921$

What if we are asked to calculate the $P(\text{Type 2 Error})$ directly.

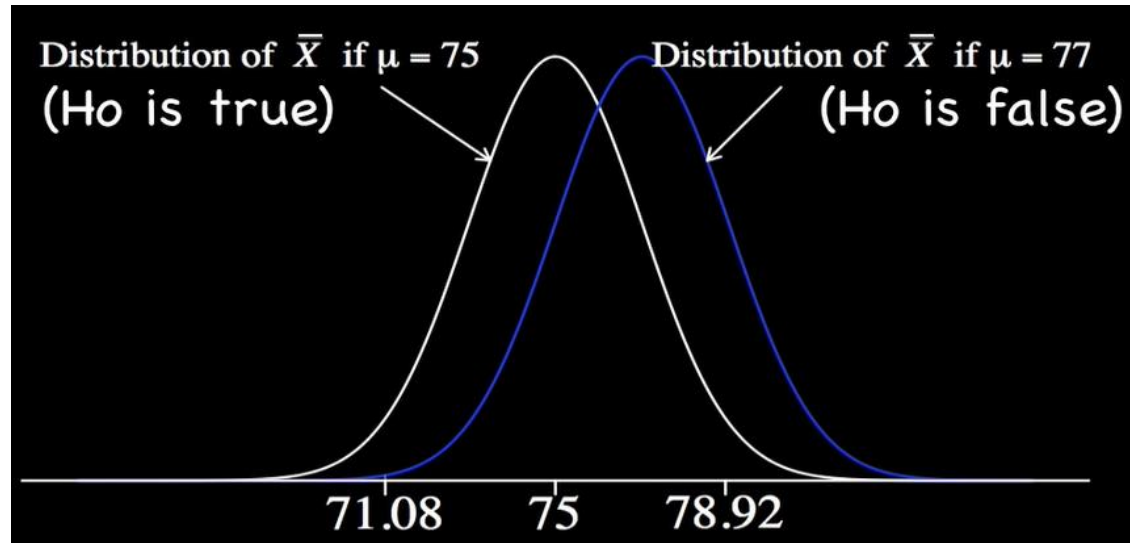
HYPOTHESES TESTING Type I and Type II Errors - Example

The picture shows the distribution of $\bar{X}$ for $\mu=75$ (H_0 is true)

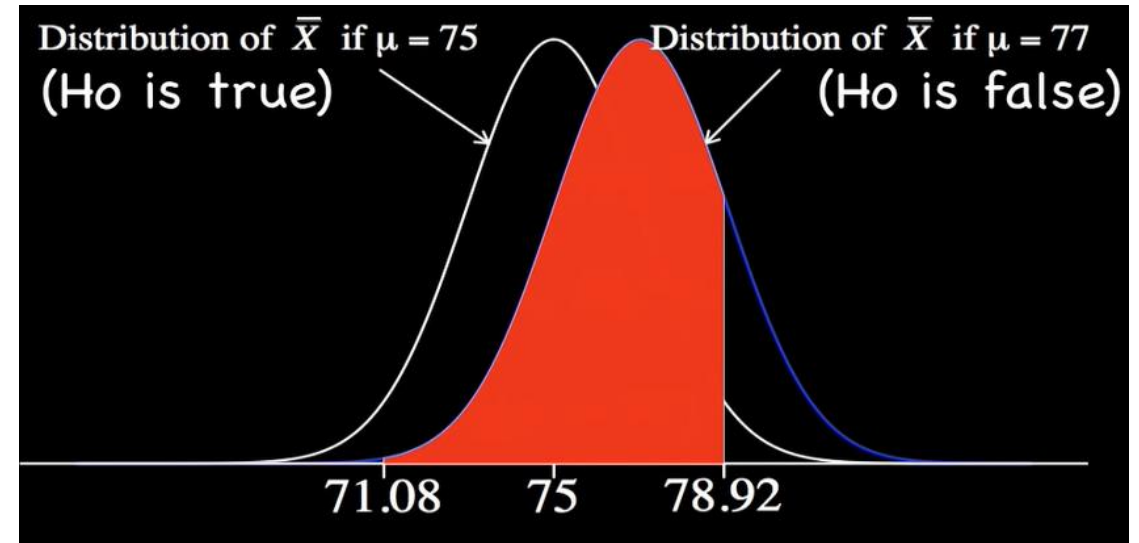
We can ask "If $\mu=77$, what is the power of our test knowing that:

$H_0: \mu = 75$ and $H_a: \mu \neq 75$

$\beta = P(\text{Not reject } H_0 \mid \mu=77)$



Grey areas = Type I Error



Red area = Type 2 Error

$$P(\bar{X} \leq 71.8 \mid \mu = 77) \text{ and } P(\bar{X} \geq 78.92 \mid \mu = 77)$$

Using the same Z formula, we get $P(-2.96 \leq Z \leq 0.96) = 0.830$ Power of the test = $1 - \beta = 1 - 0.830 = 0.170$

The Power of test increase as the true value of μ gets farther from the hypothesis.

STEPS FOR HYPOTHESIS TESTING

1. Set the null hypothesis. If the purpose of the hypothesis test is to see whether the population mean is equal o some value μ_0 , the Ho hypothesis always has the equal sign, and the alternative Ha, the unequal sign.
2. Define the test statistic. If σ is known for the population, we use the Z formula: $Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$
3. Define a rejection region using α =the significance level. If the test value is in this region, we reject Ho.
4. Calculate the value of the test statistic and perform the test. Conclude either to “reject Ho” or “fail to reject Ho”
5. Give a conclusion in terms of the original problem. We do not use here statistic terms.

HYPOTHESES TESTING PROPORTIONS

The following question comes from a book written by Hans Rosling, Anna Rosling Rönnlund, and Ola Rosling called *Factfulness*:

How many of the world's 1 year old children today have been vaccinated against some disease:

- a. 20%*
- b. 50%*
- c. 80%*

When considering Roslings' question about infant vaccination, the null hypothesis represents the notion that the people we will be considering – college-educated adults – are as accurate as random guessing. That is, the proportion p of respondents who pick the correct answer, that 80% of 1 year olds have been vaccinated against some disease, is about 33.3% (or 1-in-3 if wanting to be perfectly precise). The alternative hypothesis is that this proportion is something other than 33.3%. While it's helpful to write these hypotheses in words, it can be useful to write them using mathematical notation:

$$H_0: p = 0.333$$

$$H_A: p \neq 0.333$$

In this hypothesis setup, we want to make a conclusion about the population parameter p .

HYPOTHESES TESTING PROPORTIONS

We will use the `rosling.responses` data set to evaluate the hypothesis test evaluating whether college-educated adults who get the question about infant vaccination correct is different from 33.3%. This data set summarizes the answers of 50 college-educated adults. Of these 50 adults, 24% of respondents got the question correct that 80% of 1 year olds have been vaccinated against some disease.

The conditions are met for $\hat{p}$ to be approximately normal: the data come from a simple random sample (satisfies independence), and $n\hat{p} = 12$ and $n(1 - \hat{p}) = 38$ are both at least 10 (success-failure condition).

To construct the confidence interval, we will need to identify the point estimate ($\hat{p} = 0.24$), the critical value for the 95% confidence level ($z^* = 1.96$), and the standard error of $\hat{p}$ ($SE_{\hat{p}} = \sqrt{\hat{p}(1 - \hat{p})/n} = 0.060$). With those pieces, the confidence interval for p can be constructed:

$$\begin{aligned}\hat{p} \pm z^* \times SE_{\hat{p}} \\ 0.24 \pm 1.96 \times 0.060 \\ (0.122, 0.358)\end{aligned}$$

We are 95% confident that the proportion of all college-educated adults to correctly answer this particular question about infant vaccination is between 12.2% and 35.8%.

Because the null value in the hypothesis test is $p_0 = 0.333$, which falls within the range of plausible values from the confidence interval, we cannot say the null value is implausible.¹⁶ That is, the data do not provide sufficient evidence to reject the notion that the performance of college-educated adults was different than random guessing, and we do not reject the null hypothesis, H_0 .

CONFIDENCE INTERVALS & HYPOTHESES TESTING PROPORTIONS

Basic Statistics Tools Help

Data
Index Numbers
Descriptive Statistics
Random Variables
Statistical Inference

<No active model>

Confidence Intervals
Hypothesis Testing

Confidence intervals for a sample
Confidence intervals for two samples
Confidence interval for a proportion...

```
n(n")
- \n")

Confidence Interval for a proportion
-----
Interval type: Two-sided
Confidence level: 95%
Sample: No. successes = 12 -- No. attempts = 50
Sample estimate: proportion 0.24
Interval: ( 0.1216208 , 0.3583792 )
```

Confidence interval for a proportion

Variable (pick one)
<no variable selected>

No. successes 12
No. failures 38

Proportion for the level (pick one)
<no variable selected>

Confidence level: 0.95

Interval type
☒ Two-sided
☐ Left-sided
☐ Right-sided

Help Reset OK Cancel Apply

Basic Statistics Tools Help

Data
Index Numbers
Descriptive Statistics
Random Variables
Statistical Inference

<No active model>

Confidence Intervals
Hypothesis Testing

Hypothesis testing for a sample
Hypothesis testing for two samples
Hypothesis testing for a proportion...

```
n(n")
- \n")

Hypothesis Testing for a proportion
-----
Sample: No. successes = 12 -- No. attempts = 50
Distribution: Normal(0,1)
Test statistics value: -1.395349
p-value: 0.16291
Alternative hypothesis: Population proportion is not equal to 0.333
Sample estimate: proportion 0.24
```

Hypothesis Testing for a proportion

Variable (pick one)
<no variable selected>

No. successes 12
No. failures 38

Proportion for the level (pick one)
<no variable selected>

Alternative hypothesis
☒ Population proportion != p0
☐ Population proportion < p0
☐ Population proportion > p0

Null hypothesis: p = 0.333

Help Reset OK Cancel Apply

HYPOTHESES TESTING PROPORTIONS

Let's move onto a second question posed by the Roslings:

There are 2 billion children in the world today aged 0-15 years old, how many children will there be in year 2100 according to the United Nations?

- a. 4 billion.
- b. 3 billion.
- c. 2 billion.

Set up appropriate hypotheses to evaluate whether college-educated adults are better than random guessing on this question. Also, see if you can guess the correct answer before checking the answer in the footnote!¹⁷

Compute a 95% confidence interval for the fraction of college-educated adults who answered the children-in-2100 question correctly, and evaluate the hypotheses in Guided Practice 5.22.

To compute the standard error, we'll again use $\hat{p}$ in place of p for the calculation:

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.149(1 - 0.149)}{228}} = 0.024$$

In Guided Practice 5.23, we found that $\hat{p}$ can be modeled using a normal distribution, which ensures a 95% confidence interval may be accurately constructed as

$$\hat{p} \pm z^* \times SE \rightarrow 0.149 \pm 1.96 \times 0.024 \rightarrow (0.103, 0.195)$$

Because the null value, $p_0 = 0.333$, is not in the confidence interval, a population proportion of 0.333 is implausible and we reject the null hypothesis. That is, the data provide statistically significant evidence that the actual proportion of college adults who get the children-in-2100 question correct is different from random guessing. Because the entire 95% confidence interval is below 0.333, we can conclude college-educated adults do *worse* than random guessing on this question.

¹⁷The appropriate hypotheses are:

H_0 : the proportion who get the answer correct is the same as random guessing: 1-in-3, or $p = 0.333$.

H_A : the proportion who get the answer correct is different than random guessing, $p \neq 0.333$.

The correct answer to the question is 2 billion. While the world population is projected to increase, the average age is also expected to rise. That is, the majority of the population growth will happen in older age groups, meaning people are projected to live longer in the future across much of the world.

¹⁸We check both conditions, which are satisfied, so it is reasonable to use a normal distribution for $\hat{p}$:

Independence. Since the data are from a simple random sample, the observations are independent.

Success-failure. We'll use $\hat{p}$ in place of p to check: $n\hat{p} = 34$ and $n(1 - \hat{p}) = 194$. Both are greater than 10, so the success-failure condition is satisfied.

REPORT TESTING RESULTS USING A p-value

P-value is the observed α or the observed significant level. This value tells us if a test result is barely significant or overwhelmingly significant.

The p-value is the value of α at which the hypothesis test procedure changes conclusions based on a given set of data. It is the largest value of α for which you fail to reject H_0 .

P-value is determined by replacing the area corresponding to α with the area corresponding to computed Z test.

INTERPRETING THE P-VALUE

- Reject H_0 if p-value $< \alpha$
- Fail to reject H_0 if p-value $\geq \alpha$

RULE OF THUMB

- Reject H_0 if the p-value is small ($p < 0.01$)
- Fail to reject H_0 if the p-value is large ($p > 0.1$)
- The test is inconclusive if $0.01 \leq p\text{-value} \leq 0.1$

TWO OR MORE POPULATIONS

COMPARING TWO OR MORE POPULATIONS

Independent sample: nothing relates observations in one sample with observations in the other sample

Dependent sample: observations in both samples are somehow related (comparison of before vs after, observations with matching characteristics, observations matched by locations, etc.)

For the next couples of slides, we must install and use the NMBU plugin for R-commander.

COMPARING TWO MEANS USING TWO INDEPENDENT SAMPLES (σ 's KNOWN)

Constructing confidence interval for $(\mu_1 - \mu_2)$ for σ 's known:

μ_1 and μ_2 are both unknown, but σ_1 and σ_2 are known the Z-test is:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

The confidence interval is:

$$(\bar{X}_1 - \bar{X}_2) - Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \text{ to } (\bar{X}_1 - \bar{X}_2) + Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Remember:

$$P(-1.96 < Z < 1.96) = .95$$

HYPOTESIS TESTING FOR μ_1 and μ_2 (σ 's KNOWN)

The assumptions for the Independent Samples Z-Test include:

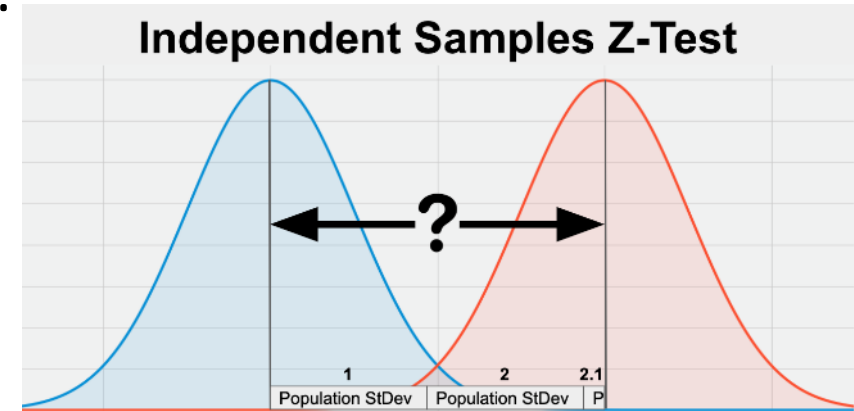
1. Continuous
2. Normally Distributed
3. Random Sample
4. Enough Data
5. Similar Spread Between Groups

Example: $H_0: \mu_1 \geq \mu_2$ and $H_a: \mu_1 < \mu_2$

The Z-test is: $Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ Same steps as in previous hypothesis testing. Reject H_0 if $Z > Z_\alpha$

Example: $H_0: \mu_1 \leq \mu_2$ and $H_a: \mu_1 > \mu_2$

The Z-test is: $Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ Same steps as in previous hypothesis testing. Reject H_0 if $Z < -Z_\alpha$



Two sample procedure for any specified value: $H_0: \mu_1 - \mu_2 = 0$; $H_a: \mu_1 - \mu_2 \neq 0$.

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - D_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

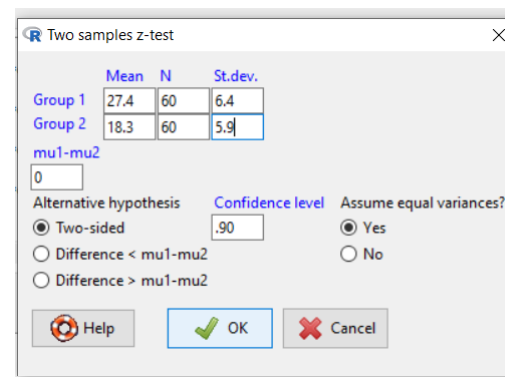
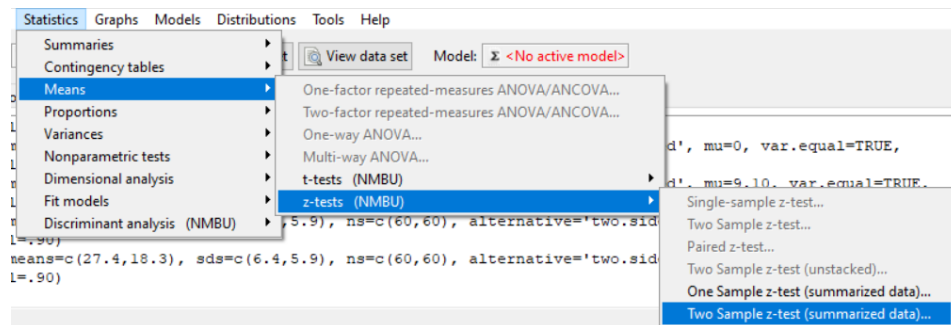
It is similar for the case of a t-test (σ 's unknown)

HYPOTESIS TESTING FOR μ_1 and μ_2 (σ 's KNOWN)

A production manager wants to determine if there is a difference in the number of units produced by the night shift and the day shift. A random sample of night and day shifts is selected and the number of production unit in each shift is recorded.

	$\bar{X}$	σ	n
Day shift	27.4	6.4	60
Night shift	18.3	5.9	60

First: do the calculation by hand using the formulas in the previous slides. After that, you can use Rcmdr to verify your results.



```
Two Sample z-test
data: two samples
z = 8.0978, df = Inf, p-value = 5.595e-16
alternative hypothesis: true difference in means is not equal to 0
90 percent confidence interval:
 7.251582 10.948418
sample estimates:
mean of x mean of y
 27.4      18.3
```

COMPARING TWO MEANS USING TWO INDEPENDENT SAMPLES (σ 's UNKNOWN)

Constructing confidence interval for $(\mu_1 - \mu_2)$ for σ 's unknown:

μ_1 and μ_2 are both unknown, but σ_1 and σ_2 are unknown the t-test is:

$$t' = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad \text{with } df = \frac{\left[\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right]^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}}$$

If df is not an integer we rounded it to the nearest integer.

The confidence interval is:

$$(\bar{X}_1 - \bar{X}_2) - t_{\alpha/2, df} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \quad \text{to} \quad (\bar{X}_1 - \bar{X}_2) + t_{\alpha/2, df} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

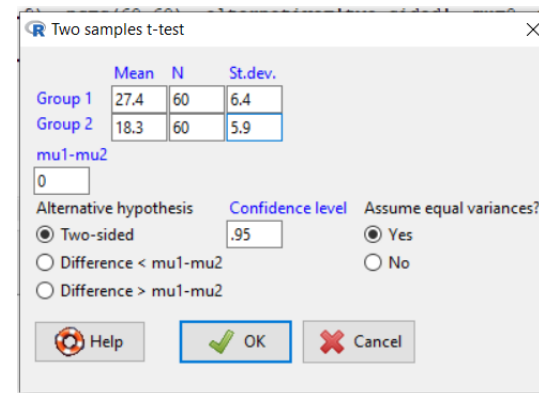
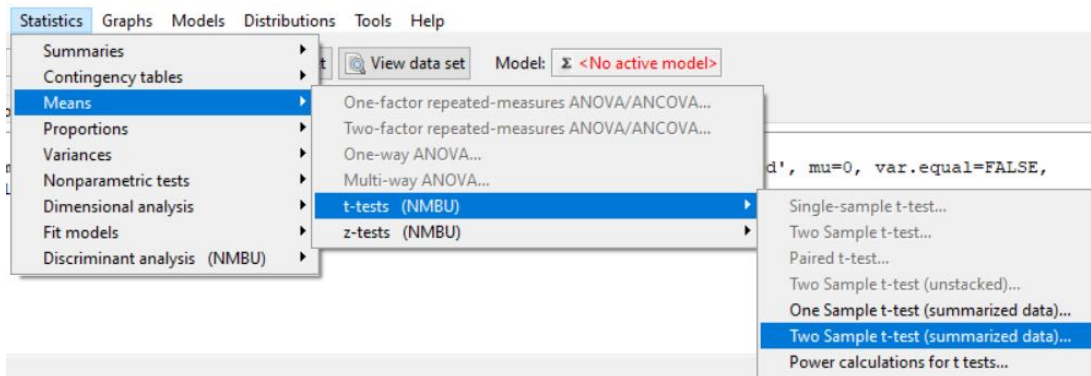
When hypothesis testing, we reject the H_0 if $|t'| > t_{\alpha/2, df}$

HYPOTESIS TESTING FOR μ_1 and μ_2 (σ 's UNKNOWN)

A production manager wants to determine if there is a difference in the number of units produced by the night shift and the day shift. A random sample of night and day shifts is selected and the number of production unit in each shift is recorded. (T-test assume equal variances) For UNEQUAL VARIANCES see the Welch t-test next slide.

	$\bar{X}$	s	n
Day shift	27.4	6.4	60
Night shift	18.3	5.9	60

First: do the calculation by hand using the formulas in the previous slides. After that, you can use Rcmdr to verify your results.



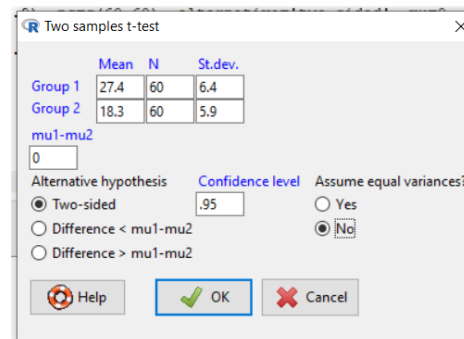
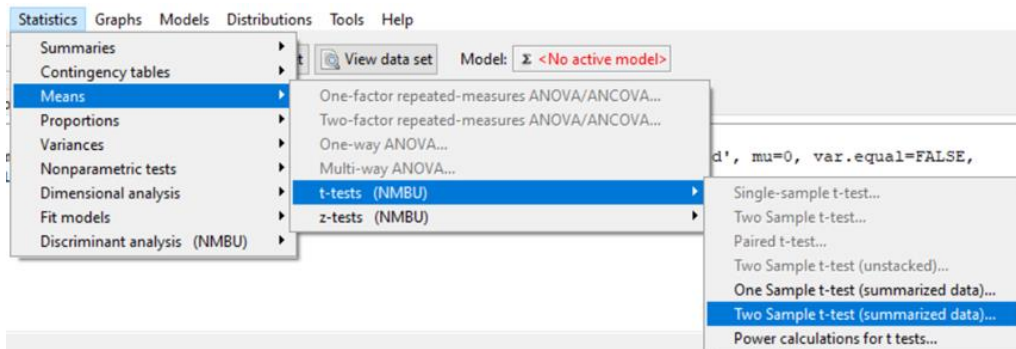
```
Two Sample t-test
data: two samples
t = 8.0978, df = 118, p-value = 5.733e-13
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 6.874652 11.325348
sample estimates:
mean of x mean of y
 27.4      18.3
```

COMPARING TWO MEANS USING TWO INDEPENDENT SAMPLES (σ 's UNKNOWN) – UNEQUAL VARIANCE CASE

A production manager wants to determine if there is a difference in the number of units produced by the night shift and the day shift. A random sample of night and day shifts is selected and the number of production unit in each shift is recorded.

Welch t-test.

	$\bar{X}$	s	n
Day shift	27.4	6.4	60
Night shift	18.3	5.9	60



Welch Two Sample t-test

```
data: two samples
t = 8.0978, df = 117.23, p-value = 5.932e-13
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 6.8745 11.3255
sample estimates:
mean of x mean of y
 27.4      18.3
```

COMPARING TWO MEANS USING TWO INDEPENDENT SAMPLES (σ 's UNKNOWN) – EQUAL VARIANCE CASE

If we believe σ_1 and σ_2 are equal from previous experience, or because we used another sample to test the equality of the variances, we can use a better test, more powerful. Therefore, if $\sigma_1 = \sigma_2 = \sigma$ we combine the estimate s_1 and s_2 into one sample variance (pooled sample variance):

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Confidence intervals for $\mu_1 - \mu_2$ is tested using the test:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}} \quad \text{with } df = n_1 + n_2 - 2$$

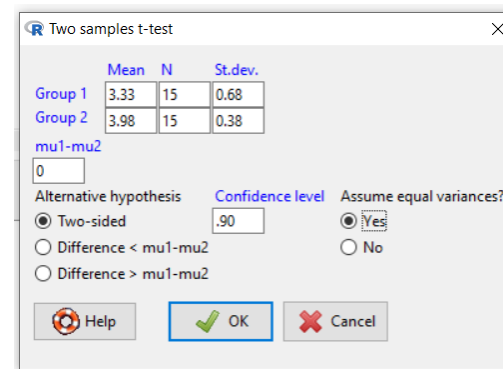
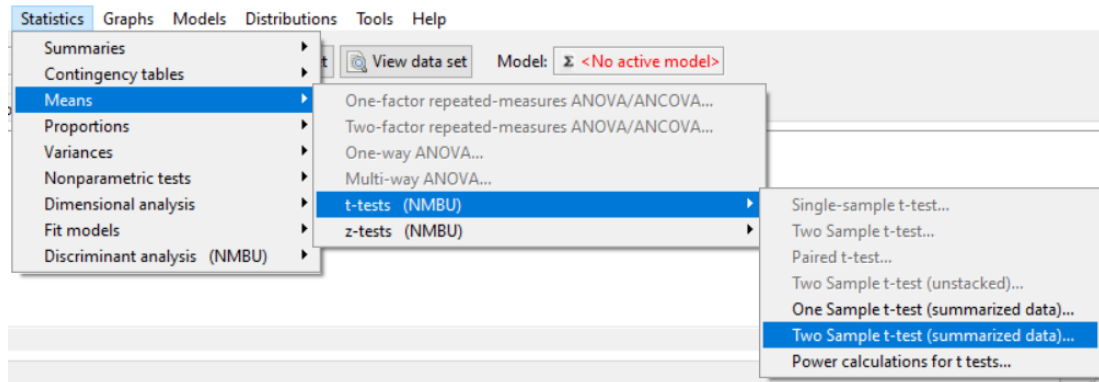
$$\text{C.I. } (\bar{X}_1 - \bar{X}_2) - t_{\alpha/2, df} \sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}} \text{ to } (\bar{X}_1 - \bar{X}_2) + t_{\alpha/2, df} \sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}$$

HYPOTESIS TESTING FOR μ_1 and μ_2 (σ 's UNKNOWN)

A production manager wants to determine if there is a difference in the number of units produced by the night shift and the day shift. A random sample of night and day shifts is selected and the number of production unit in each shift is recorded.

	$\bar{X}$	s	n
Day shift	3.33	0.68	15
Night shift	3.98	0.38	15

First: do the calculation by hand using the formulas in the previous slides. After that, you can use Rcmdr to verify your results.

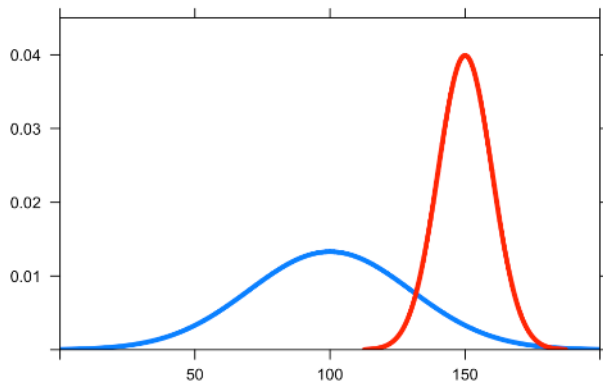


```
Two Sample t-test
data: two samples
t = -3.2317, df = 28, p-value = 0.003142
alternative hypothesis: true difference in means is not equal to 0
90 percent confidence interval:
 -0.9921487 -0.3078513
sample estimates:
mean of x mean of y
 3.33      3.98
```

COMPARING VARIANCES OF TWO NORMAL POPULATIONS USING INDEPENDENT SAMPLES

Of interest now is the variance of the two populations. Such variances can be obtained from a separate sample. It is not safe to test for both the mean and the variance equality using the same dataset.

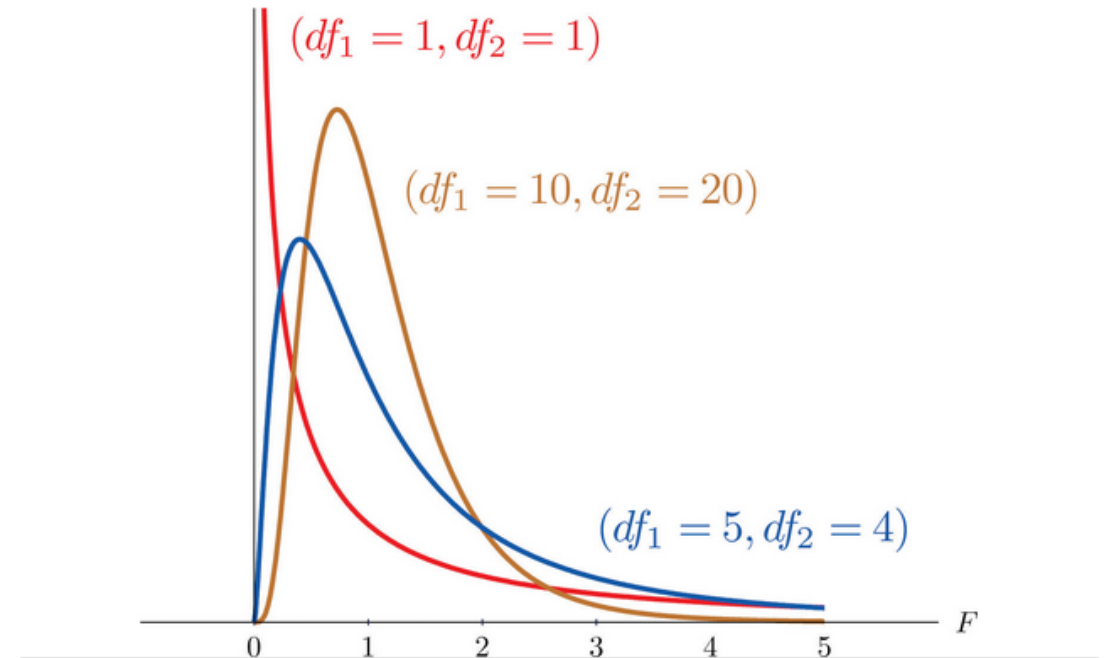
Normal distributions with different variances



F-Distributions

Another important and useful family of distributions in statistics is the family of F -distributions. Each member of the F -distribution family is specified by a pair of parameters called *degrees of freedom* and denoted df_1 and df_2 . [Figure 11.7 "Many"](#) shows several F -distributions for different pairs of degrees of freedom. An **F random variable** is a random variable that assumes only positive values and follows an F -distribution.

Figure 11.7 Many F-Distributions



The parameter df_1 is often referred to as the *numerator* degrees of freedom and the parameter df_2 as the *denominator* degrees of freedom. It is important to keep in mind that they are not interchangeable. For example, the F -distribution with degrees of freedom $df_1 = 3$ and $df_2 = 8$ is a different distribution from the F -distribution with degrees of freedom $df_1 = 8$ and $df_2 = 3$.

COMPARING VARIANCES OF TWO NORMAL POPULATIONS USING INDEPENDENT SAMPLES

Test Statistic for Hypothesis Tests Concerning the Difference Between Two Population Variances

$$F = \frac{s_1^2}{s_2^2}$$

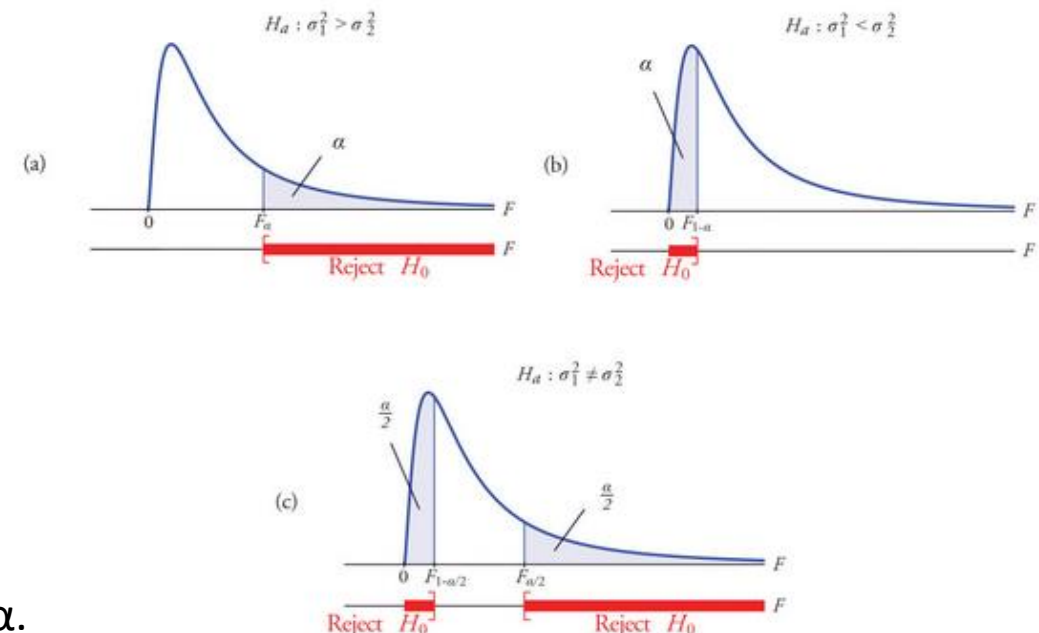
If the two populations are normally distributed and if $H_0 : \sigma_1^2 = \sigma_2^2$ is true then under independent sampling F approximately follows an F -distribution with degrees of freedom $df_1 = n_1 - 1$ and $df_2 = n_2 - 1$.

If $F_u(r, s)$ denotes the value of the F -distribution with degrees of freedom $df_1 = r$ and $df_2 = s$ that cuts off a right tail of area u , then

$$F_c(k, \ell) = \frac{1}{F_{1-c}(\ell, k)}$$

Terminology	Alternative Hypothesis	Rejection Region
Right-tailed	$H_a : \sigma_1^2 > \sigma_2^2$	$F \geq F_\alpha$
Left-tailed	$H_a : \sigma_1^2 < \sigma_2^2$	$F \leq F_{1-\alpha}$
Two-tailed	$H_a : \sigma_1^2 \neq \sigma_2^2$	$F \leq F_{1-\alpha/2}$ or $F \geq F_{\alpha/2}$

Figure 11.9 Rejection Regions: (a) Right-Tailed; (b) Left-Tailed; (c) Two-Tailed



There are different F Tables for different levels of significance α .

COMPARING VARIANCES OF TWO NORMAL POPULATIONS USING INDEPENDENT SAMPLES

Example: variations of volumes, variations of time in line, variations of waiting time.

Note: s1 and s2 are actually s1² and s2²

Set 1	Set 2		Ho: $\sigma_1 \leq \sigma_2$
n1=25	n2=20		Ha: $\sigma_1 > \sigma_2$
x1=31.8	x2=32.1		
s1=1.21	s2=0.72		F=s1/s2=larger variance/smaller variance
df1=25-1	df2=20-1		F _{0.05, 24, 19} =2.19
α=0.05			Reject Ho if F _{calc} > F _{table} = F _{0.05, 24, 19}
			F _{calc} = (1.21) ² / (0.72) ² = 2.82
			Because 2.82 > 2.11, we reject Ho

Keep in mind that:

F-value(df1, df2) having a left –tail area of α = 1/(F-value(df2, df1) having a right tail area of α

Confidence interval for $\frac{s_1^2}{s_2^2}$ when $F_R = F_{\alpha/2, df_1, df_2}$ and $F_L = \frac{1}{F_{\alpha/2, df_2, df_1}}$ are the right and left values is $\frac{s_1^2/s_2^2}{F_R}$ to $\frac{s_1^2/s_2^2}{F_L}$

COMPARING THE MEANS OF TWO NORMAL POPULATIONS USING PAIRED SAMPLE

Suppose chemical engineers wish to compare the fuel economy obtained by two different formulations of gasoline. Since fuel economy varies widely from car to car, if the mean fuel economy of two independent samples of vehicles run on the two types of fuel were compared, even if one formulation were better than the other the large variability from vehicle to vehicle might make any difference arising from difference in fuel difficult to detect. Just imagine one random sample having many more large vehicles than the other. Instead of independent random samples, it would make more sense to select pairs of cars of the same make and model and driven under similar circumstances and compare the fuel economy of the two cars in each pair. Thus, the data would look something like the table below, where the first car in each pair is operated on one formulation of the fuel (call it Type 1 gasoline) and the second car is operated on the second (call it Type 2 gasoline).

Make and Model	Car 1	Car 2
Buick LaCrosse	17.0	17.0
Dodge Viper	13.2	12.9
Honda CR-Z	35.3	35.4
Hummer H 3	13.6	13.2
Lexus RX	32.7	32.5
Mazda CX-9	18.4	18.1
Saab 9-3	22.5	22.5
Toyota Corolla	26.8	26.7
Volvo XC 90	15.1	15.0

The first column of numbers form a sample from Population 1, the population of all cars operated on Type 1 gasoline; the second column of numbers form a sample from Population 2, the population of all cars operated on Type 2 gasoline.

COMPARING THE MEANS OF TWO NORMAL POPULATIONS USING PAIRED SAMPLE

It would be incorrect to analyze the data using the formulas from the previous section, however, since the samples were not drawn independently. What is correct is to compute the difference in the numbers in each pair (subtracting in the same order each time) to obtain the third column of numbers as shown in the table below and treat the differences as the data. At this point, the new sample of differences $d_1=0.0, \dots, d_9=0.1$ in the third column of the table below may be considered as a random sample of size $n=9$ selected from a population with mean $\mu_d=\mu_1-\mu_2$. This approach essentially transforms the paired two-sample problem into a one-sample problem as discussed in the previous slides.

Make and Model	Car 1	Car 2	Difference
Buick LaCrosse	17.0	17.0	0.0
Dodge Viper	13.2	12.9	0.3
Honda CR-Z	35.3	35.4	-0.1
Hummer H 3	13.6	13.2	0.4
Lexus RX	32.7	32.5	0.2
Mazda CX-9	18.4	18.1	0.3
Saab 9-3	22.5	22.5	0.0
Toyota Corolla	26.8	26.7	0.1
Volvo XC 90	15.1	15.0	0.1

When the population of differences is normally distributed the following formula for a confidence interval for $\mu_d=\mu_1-\mu_2$ is valid. 100(1- α)% Confidence Interval for the Difference Between Two Population Means: Paired Difference Samples

$$\bar{d} \pm t_{\alpha/2} \frac{s_d}{\sqrt{n}}$$

where there are n pairs, $\bar{d}$ is the mean and s_d is the standard deviation of their differences. $df = n - 1$.

The population of differences must be *normally distributed*.

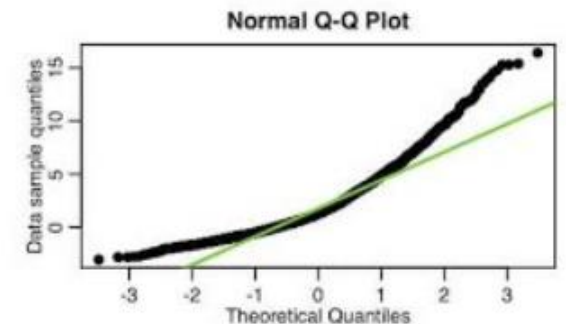
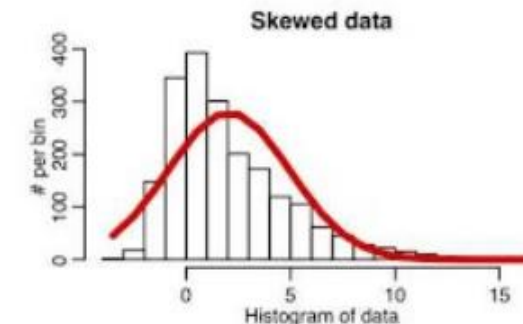
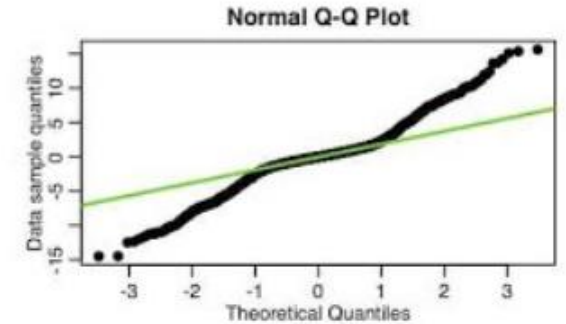
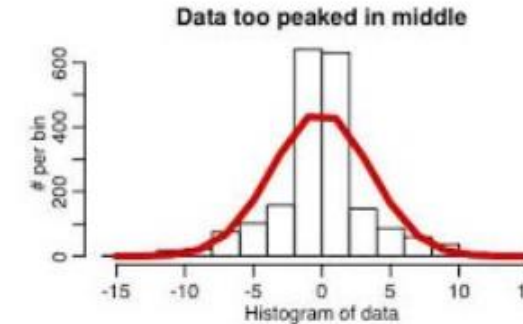
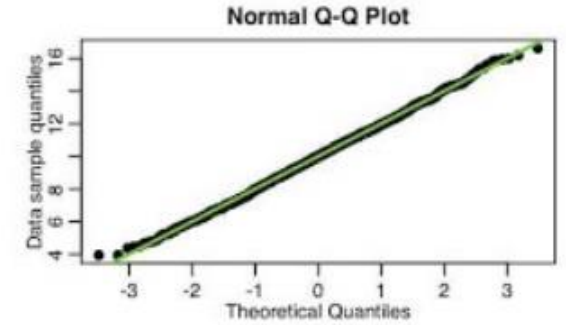
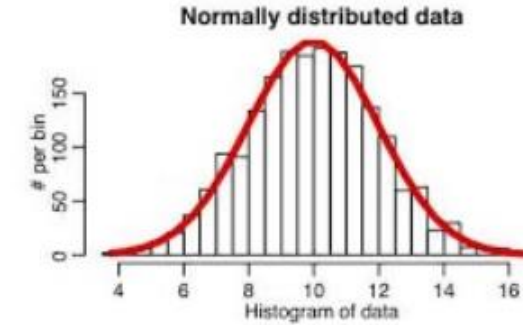
TESTs FOR NORMALLY DISTRIBUTED POPULATIONS

Visual test: QQ-plot

We plot the theoretical quantiles or basically known as the standard normal variate (a normal distribution with mean=0 and standard deviation=1) on the x-axis and the ordered values for the random variable which we want to find whether it is Gaussian distributed or not, on the y-axis. Which gives a very beautiful and a smooth straight line like structure from each point plotted on the graph.

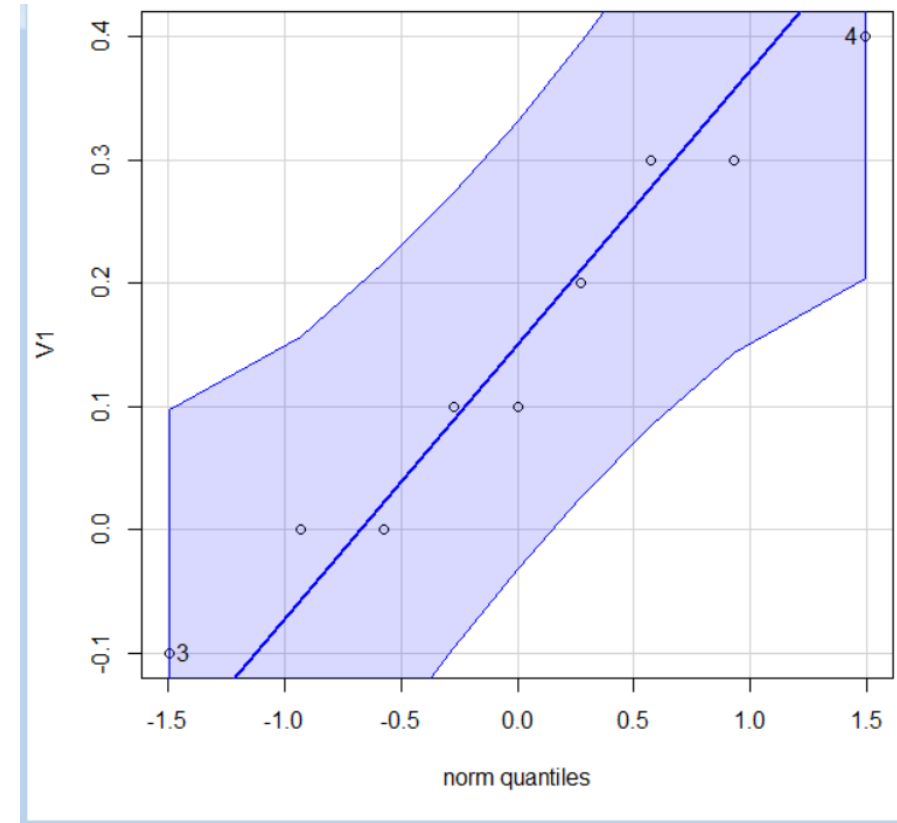
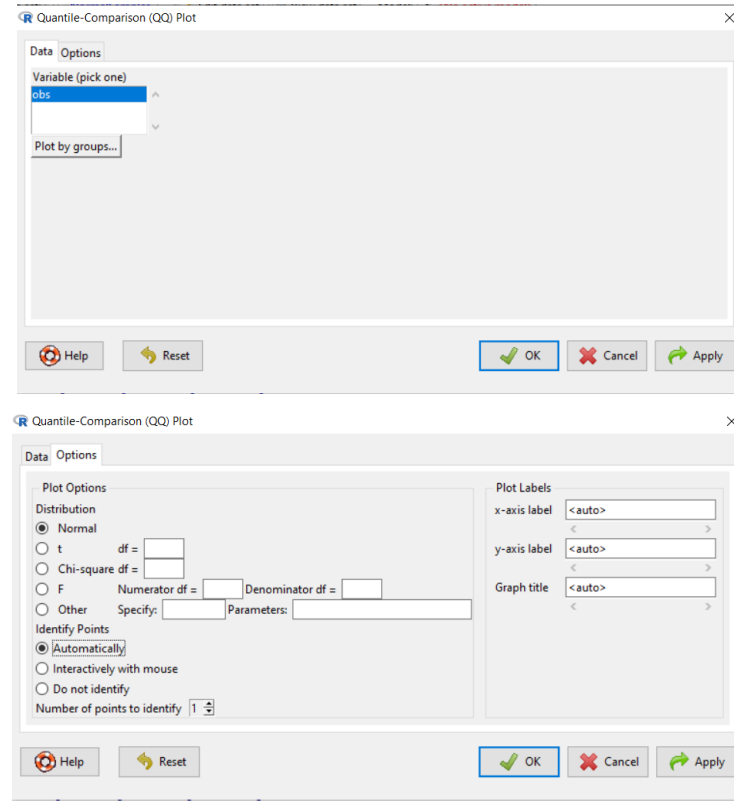
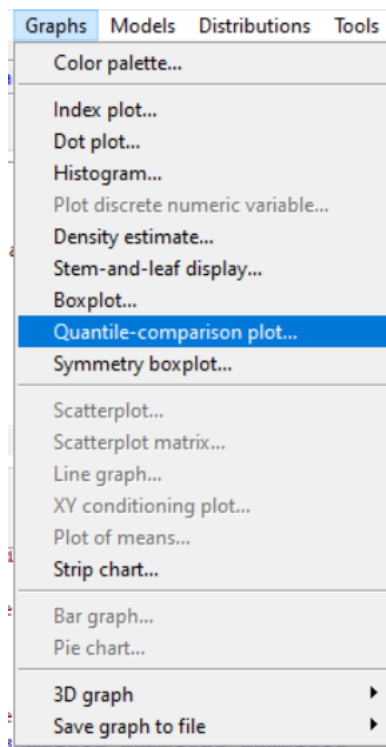
Now we have to focus on the ends of the straight line. If the points at the ends of the curve formed from the points are not falling on a straight line but indeed are scattered significantly from the positions then we cannot conclude a relationship between the x and y axes which clearly signifies that our ordered values which we wanted to calculate are not Normally distributed.

If all the points plotted on the graph perfectly lies on a straight line then we can clearly say that this distribution is Normally distribution because it is evenly aligned with the standard normal variate which is the simple concept of Q-Q plot.



TESTs FOR NORMAL DISTRIBUTED POPULATIONS

	V1
1	0.0
2	0.3
3	-0.1
4	0.4
5	0.2
6	0.3
7	0.0
8	0.1
9	0.1



The data looks normal. However, we can use a better test (Shapiro-Wilks)

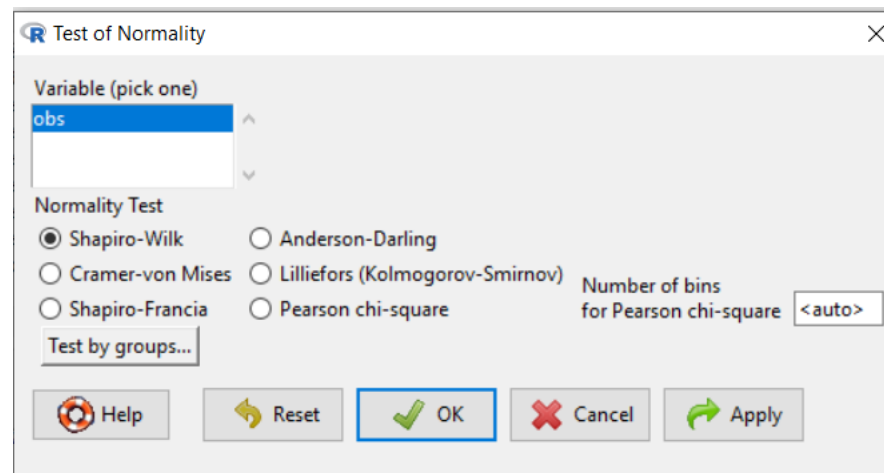
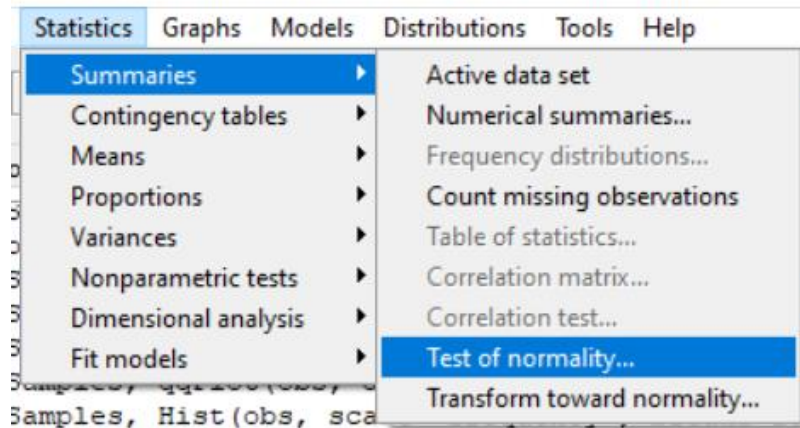
TESTs FOR NORMALLY DISTRIBUTED POPULATIONS

The **Shapiro-Wilks** test for normality is one of three general normality tests designed to detect all departures from normality.

Shapiro-Wilk test: H_0 : Data is normally distributed

H_a : Data is not normally distributed

If the test is non-significant ($p > .05$) it tells us that the distribution of the sample is not significantly different from a normal distribution. If, however, the test is significant ($p < .05$) then the distribution in question is significantly different from a normal distribution.



```
Shapiro-Wilk normality test
data:  V1
W = 0.95459, p-value = 0.7403
```

We fail to reject the null hypothesis, and we can say with 95% confidence that our data set is normally distributed.

GOING BACK TO ... COMPARING THE MEANS OF TWO NORMAL POPULATIONS USING PAIRED SAMPLE

Create a confidence interval of 95%.

The mean and standard deviation of the differences are

$$\bar{d} = \frac{\sum d}{n} = \frac{1.3}{9} = 0.1\bar{4}$$
$$s_d = \sqrt{\frac{\sum d^2 - \frac{1}{n}(\sum d)^2}{n-1}} = \sqrt{\frac{0.41 - \frac{1}{9}(1.3)^2}{8}} = 0.1\bar{6}$$

The point estimate of $\mu_1 - \mu_2 = \mu_d$ is

$$\bar{d} = 0.14$$

To apply the formula for the confidence interval, we must find $t_{\alpha/2}$. The 95% confidence level means that $\alpha=1-0.95=0.05$ so that $t_{\alpha/2}=t_{0.025}$. The degrees of freedom are $df=9-1=8$ we read from the t-table that $t_{0.025}=2.306$

$$\bar{d} \pm t_{\alpha/2} \frac{s_d}{\sqrt{n}} = 0.14 \pm 2.306 \left(\frac{0.1\bar{6}}{\sqrt{9}} \right) \approx 0.14 \pm 0.13$$

We are 95% confident that the difference in the population means lies in the interval $[0.01, 0.27]$, in the sense that in repeated sampling 95% of all intervals constructed from the sample data in this manner will contain $\mu_d = \mu_1 - \mu_2$. Stated differently, we are 95% confident that mean fuel economy is between 0.01 and 0.27 mpg greater with Type 1 gasoline than with Type 2 gasoline.

GOING BACK TO ... COMPARING THE MEANS OF TWO NORMAL POPULATIONS USING PAIRED SAMPLE

Hypothesis testing: Testing hypotheses concerning the difference of two population means using paired difference samples is done precisely as it is done for independent samples, although now the null and alternative hypotheses are expressed in terms of μ_d instead of $\mu_1 - \mu_2$. Thus, the null hypothesis will always be written $H_0: \mu_d = D_0$

Form of H_a	Terminology
$H_a : \mu_d < D_0$	Left-tailed
$H_a : \mu_d > D_0$	Right-tailed
$H_a : \mu_d \neq D_0$	Two-tailed

Standardized Test Statistic for Hypothesis Tests Concerning the Difference Between Two Population Means: Paired Difference Samples

$$T = \frac{\bar{d} - D_0}{s_d / \sqrt{n}}$$

where there are n pairs, $\bar{d}$ is the mean and s_d is the standard deviation of their differences.

The test statistic has Student's t -distribution with $df = n - 1$ degrees of freedom.

The population of differences must be normally distributed.

GOING BACK TO ... COMPARING THE MEANS OF TWO NORMAL POPULATIONS USING PAIRED SAMPLE

Solution:

The only part of the table that we use is the third column, the differences.

- **Step 1.** Since the differences were computed in the order Type 1 mpg – Type 2 mpg, better fuel economy with Type 1 fuel corresponds to $\mu_d = \mu_1 - \mu_2 > 0$. Thus the test is

$$\begin{array}{c} H_0 : \mu_d = 0 \\ \text{vs.} \\ H_a : \mu_d > 0 @ \alpha = 0.05 \end{array}$$

(If the differences had been computed in the opposite order then the alternative hypotheses would have been $H_a : \mu_d < 0$.)

- **Step 2.** Since the sampling is in pairs the test statistic is

$$T = \frac{\bar{d} - D_0}{s_d / \sqrt{n}}$$

- **Step 3.** We have already computed $\bar{d}$ and s_d in the previous example. Inserting their values and $D_0 = 0$ into the formula for the test statistic gives

$$T = \frac{\bar{d} - D_0}{s_d / \sqrt{n}} = \frac{0.14}{0.16 / \sqrt{3}} = 2.600$$

- **Step 4.** Since the symbol in H_a is ">" this is a right-tailed test, so there is a single critical value, $t_\alpha = t_{0.05}$ with 8 degrees of freedom, which from the row labeled $df = 8$ in Figure 7.1.6 we read off as 1.860. The rejection region is $[1.860, \infty)$.
- **Step 5.** As shown in Figure 9.3.1 the test statistic falls in the rejection region. The decision is to reject H_0 . In the context of the problem our conclusion is:

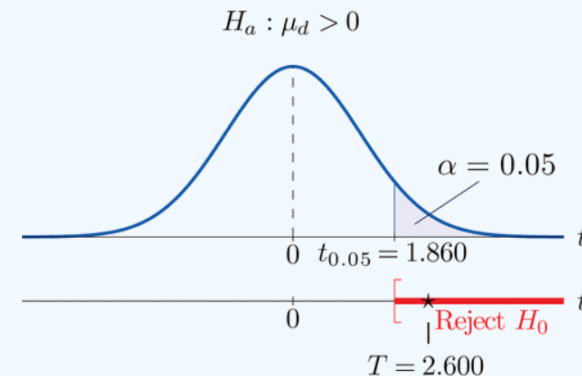
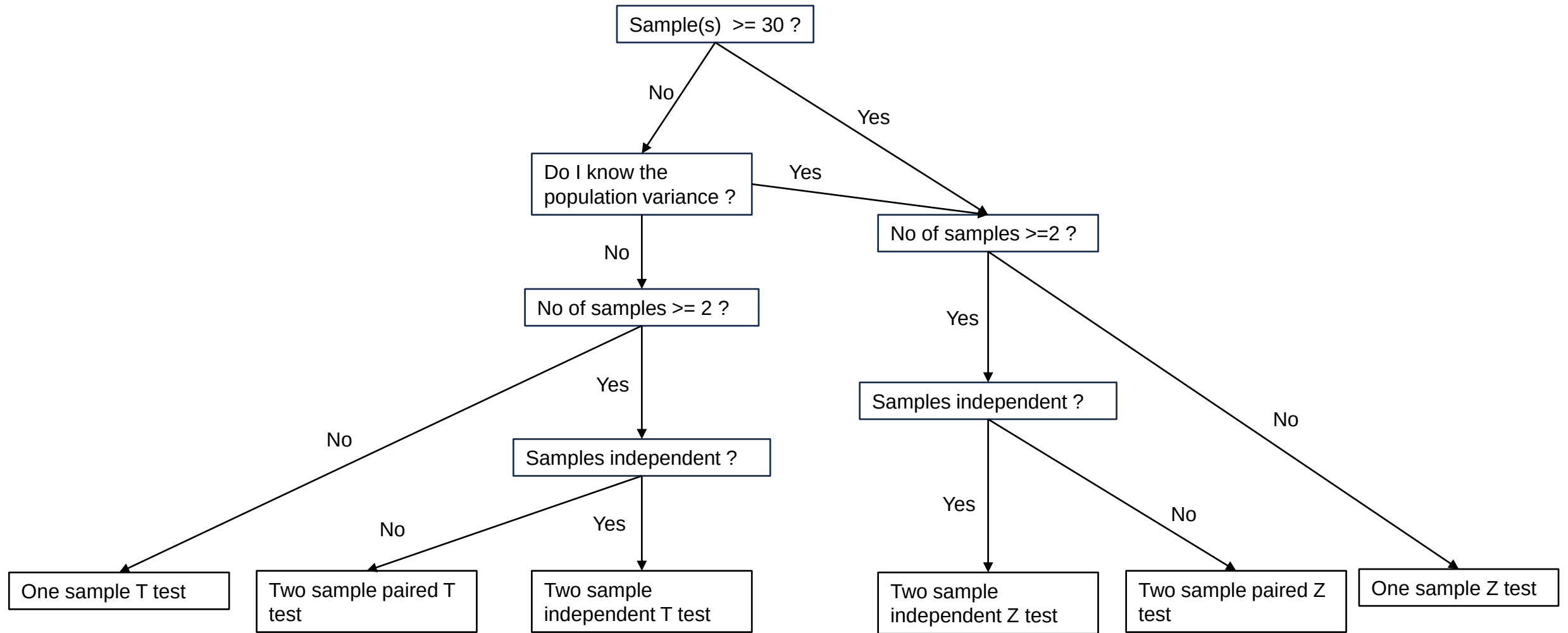
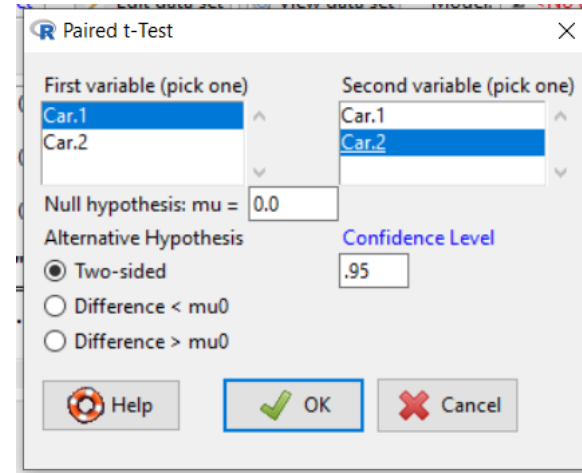
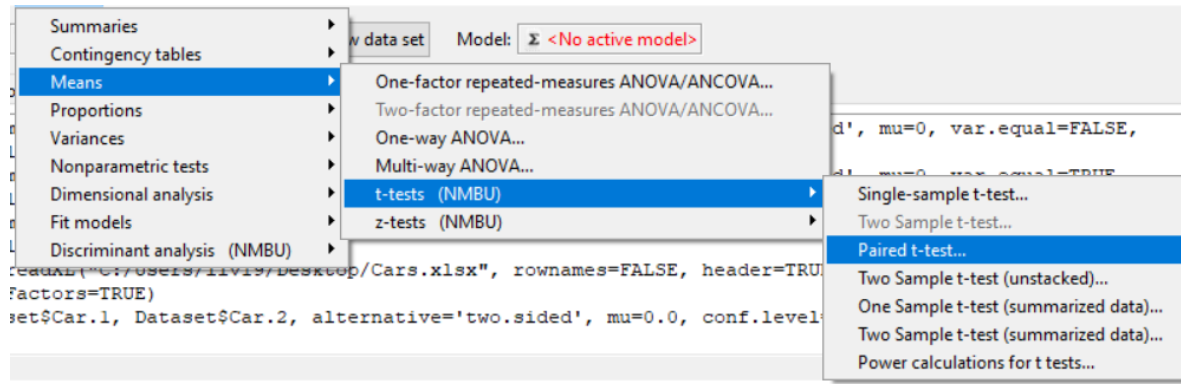


Figure 9.3.1: Rejection Region and Test Statistic for "Example 9.3.2"

The data provide sufficient evidence, at the 5% level of significance, to conclude that the mean fuel economy provided by Type 1 gasoline is greater than that for Type 2 gasoline.



GOING BACK TO ... COMPARING THE MEANS OF TWO NORMAL POPULATIONS USING PAIRED SAMPLE



```
Paired t-test

data: Dataset$Car.1 and Dataset$Car.2
t = 2.6, df = 8, p-value = 0.03162
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.0163331 0.2725558
sample estimates:
 mean of the differences  std.dev. of the differences
               0.1444444                0.1666667
```

COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

A one-way ANOVA is a statistical test used to determine whether or not there is a significant difference between the means of three or more independent groups.

Assumptions:

1. Normality – Each sample is drawn from a normally distributed population.

Check the assumption visually using histograms or Q-Q plots.

Check the assumption using formal statistical tests like Shapiro-Wilk, Kolmogorov-Smirnov, Jarque-Barre, or D'Agostino-Pearson.

In general, a one-way ANOVA is considered to be fairly robust against violations of the normality assumption as long as the sample sizes are sufficiently large.

If the normality assumption is *severely* violated or if you just want to be extra conservative, you have two choices:

(1) Transform the response values of your data so that the distributions are more normally distributed.

(2) Perform an equivalent non-parametric test such as a [Kruskal-Wallis Test](#) that doesn't require the assumption of normality.

COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

A one-way ANOVA is a statistical test used to determine whether or not there is a significant difference between the means of three or more independent groups.

Assumptions:

2. Equal Variances – The variances of the populations that the samples come from are equal.

Check the assumption visually using boxplots.

Check the assumption using a formal statistical tests like Bartlett's Test.

In general, a one-way ANOVA is considered to be fairly robust against violations of the equal variances assumption as long as each group has the same sample size.

However, if the sample sizes are not the same and this assumption is severely violated, you could instead run a [Kruskal-Wallis Test](#), which is the non-parametric version of the one-way ANOVA.

COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

A one-way ANOVA is a statistical test used to determine whether or not there is a significant difference between the means of three or more independent groups.

Assumptions:

3. Independence – The observations in each group are independent of each other and the observations within groups were obtained by a random sample.

The observations in each group are independent of the observations in every other group.

The observations within each group were obtained by a random sample.

There is no formal test you can use to verify that the observations in each group are independent and that they were obtained by a random sample. The only way this assumption can be satisfied is if a randomized design was used.

If this assumption is violated, the best thing to do is to set up the experiment again in a way that uses a randomized design.

COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

A one-way ANOVA is a statistical test used to determine whether or not there is a significant difference between the means of three or more independent groups.

Ho: All μ 's are equal

Ha: Not all μ 's are equal

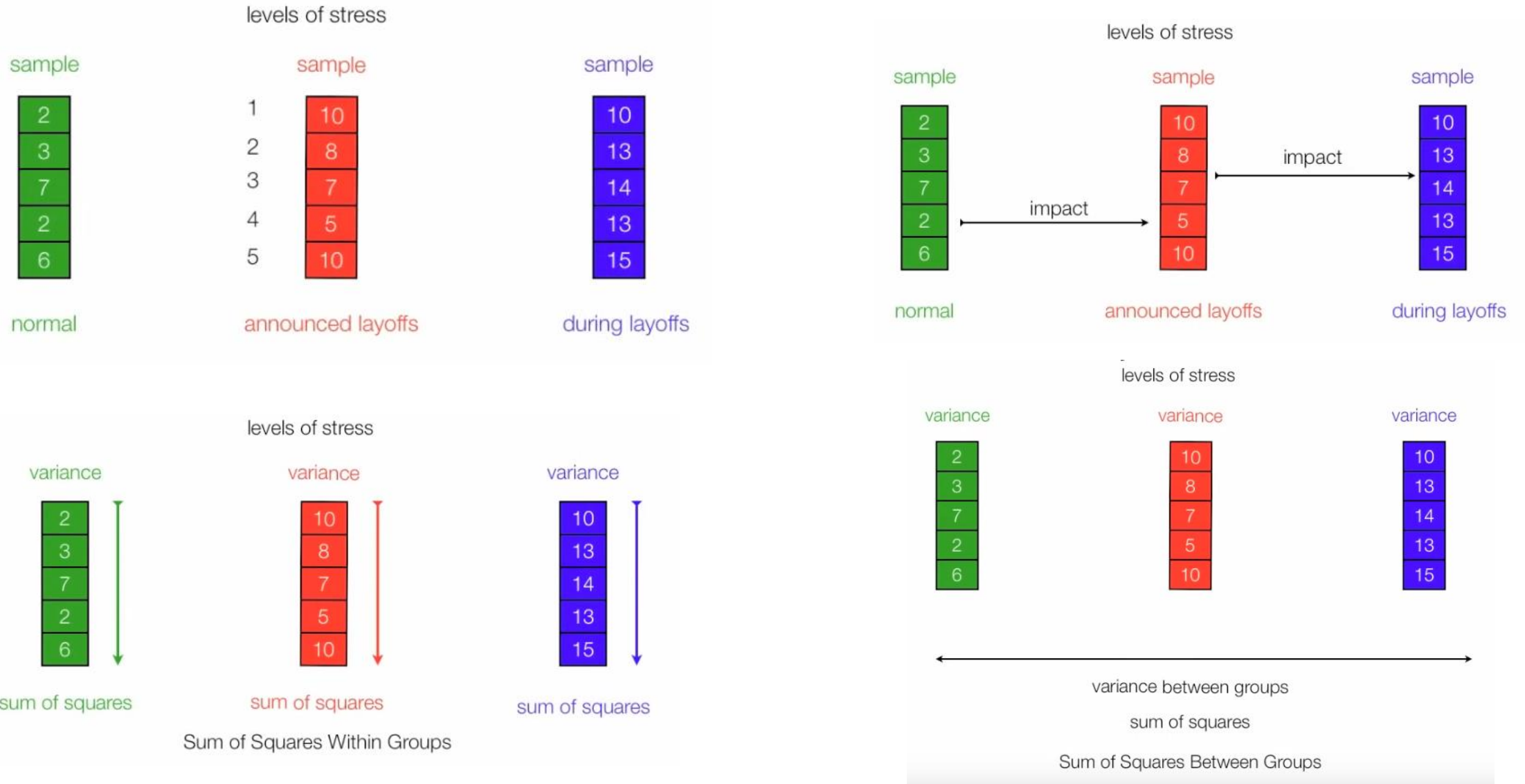
However, this doesn't tell us which groups are different from each other. It simply tells us that not all of the group means are equal.

In order to find out exactly which groups are different from each other, we must conduct a post hoc test (also known as a multiple comparison test), which will allow us to explore the difference between multiple group means while also controlling for the family-wise error rate.

In a hypothesis test, there is always a type I error rate, which is defined by our significance level (α) and tells us the probability of rejecting a null hypothesis that is actually true. In other words, it's the probability of getting a "false positive", i.e. when we claim there is a statistically significant difference among groups, but there actually isn't.

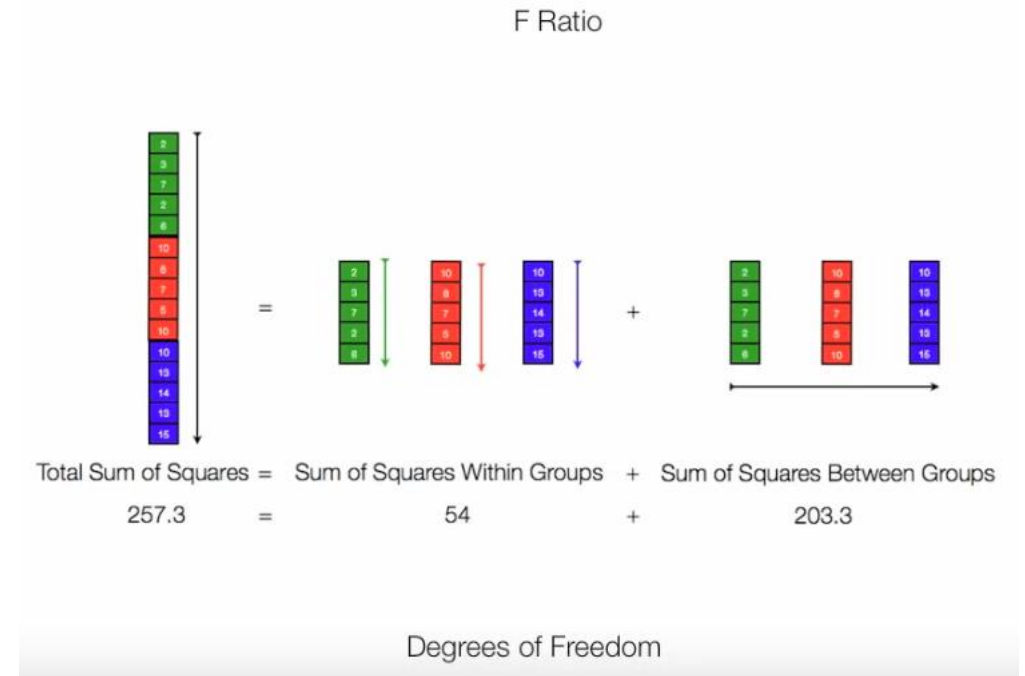
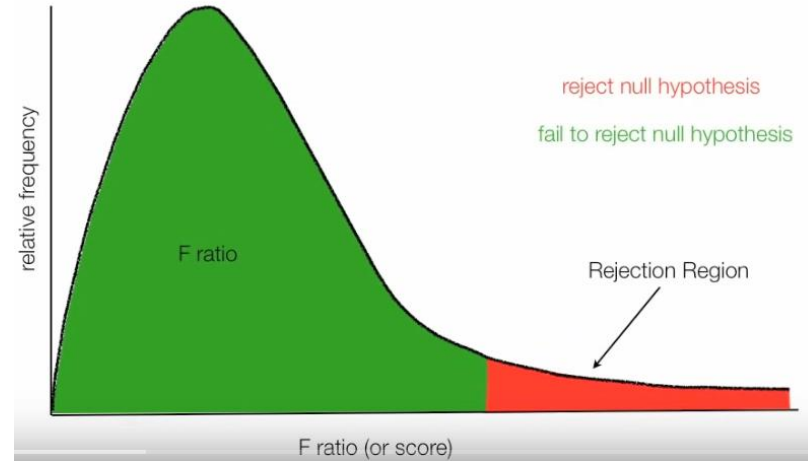
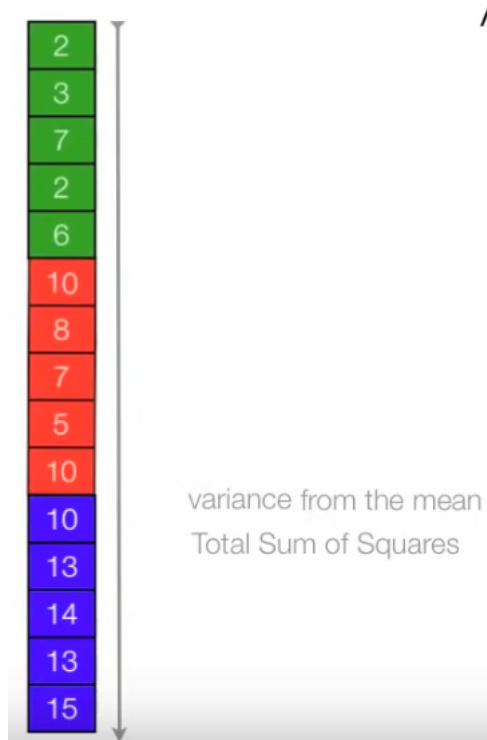
COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA



COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA



COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

Sum of Squares Within Groups

sample

2
3
7
2
6

sample

10
8
7
5
10

sample

10
13
14
13
15

Sum of Squares Within Groups

sample

2	- 4	= -2 ²
3	- 4	= -1 ²
7	- 4	= 3 ²
2	- 4	= -2 ²
6	- 4	= 2 ²

4

1

9

4

4

22

sample

10	- 8	= 2 ²
8	- 8	= 0 ²
7	- 8	= -1 ²
5	- 8	= -3 ²
10	- 8	= 2 ²

4

0

1

9

4

18

sample

10	- 13	= -3 ²
13	- 13	= 0 ²
14	- 13	= 1 ²
13	- 13	= 0 ²
15	- 13	= 2 ²

9

0

1

0

4

14

mean

4

8

13

Sum of **S**quares **W**ithin Groups = 22 + 18 + 14 = 54

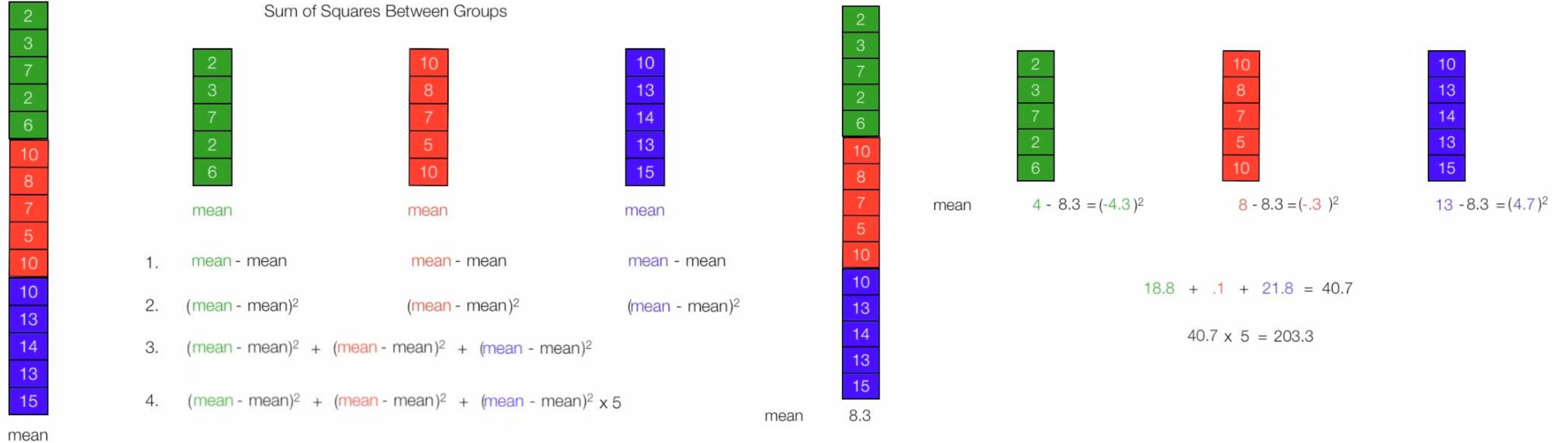
COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

observation		mean	observation - mean	(observation - mean) ²	Total Sum of Squares
2	-	8.3	= -6.3	40.1	SST = 257.3
3	-	8.3	= -5.3	28.4	
7	-	8.3	= -1.3	1.8	
2	-	8.3	= -6.3	40.1	
6	-	8.3	= -2.3	5.4	
10	-	8.3	= 1.7	2.7	
8	-	8.3	= -0.3	0.1	
7	-	8.3	= -1.3	1.8	
5	-	8.3	= -3.3	11.1	
10	-	8.3	= 1.7	2.8	
10	-	8.3	= 1.7	2.8	
13	-	8.3	= 4.7	21.8	
14	-	8.3	= 5.7	32.1	
13	-	8.3	= 4.7	21.8	
15	-	8.3	= 6.7	44.4	
mean		8.3			

COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA



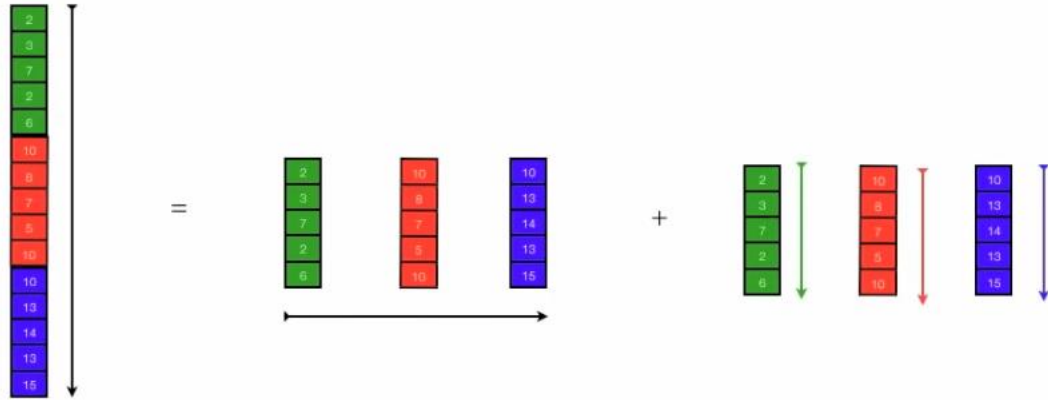
COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

$$SS_T = \sum \left[(X - \bar{X}_T)^2 \right] = SS_B = \sum_{EachGroup} \left[(\bar{X}_{group} - \bar{X}_T)^2 * (n_{group}) \right] + SS_W = \sum_{EachGroup} \left[\sum \left((X - \bar{X}_{group})^2 \right) \right]$$

COMPARING MORE THAN TWO MEANS

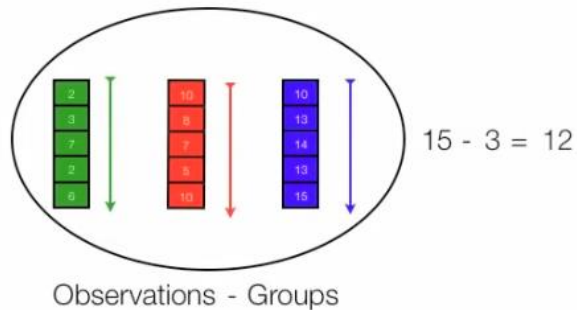
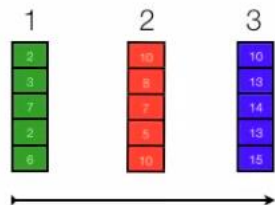
Analysis of Variance - ANOVA



$$\begin{array}{rcl} \text{Total Sum of Squares} & = & \text{Sum of Squares Between Groups} + \text{Sum of Squares Within Groups} \\ 257.3 & = & 203.3 + 54 \end{array}$$

$$\frac{\text{Sum of Squares Between Groups}}{\text{degrees of freedom}} = \frac{203.3}{2}$$

$$\begin{array}{l} \text{Groups} - 1 \\ 3 - 1 = \end{array}$$



$$\frac{\text{Sum of Squares Within Groups}}{\text{degrees of freedom}} = \frac{54}{12}$$

Final Calculations

$$\frac{\text{Sum of Squares Between Groups}}{\text{degrees of freedom}} = \frac{203.3}{2} = 101.667$$

$$\frac{\text{Sum of Squares Within Groups}}{\text{degrees of freedom}} = \frac{54}{12} = 4.5$$

$$F = \frac{101.667}{4.5} = 22.59$$

COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

$$\frac{\text{Sum of Squares Between Groups}}{\text{degrees of freedom}} = \frac{203.3}{2} = 101.667$$

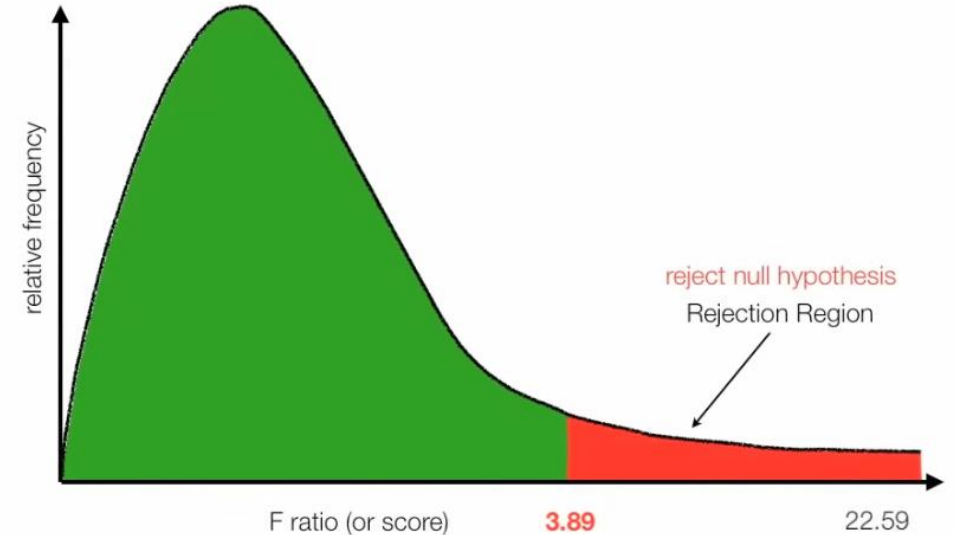
$$F = \frac{101.667}{4.5} = 22.59$$

$$\frac{\text{Sum of Squares Within Groups}}{\text{degrees of freedom}} = \frac{54}{12} = 4.5$$

F Distribution $F(2,12) = 22.59, p < .05$

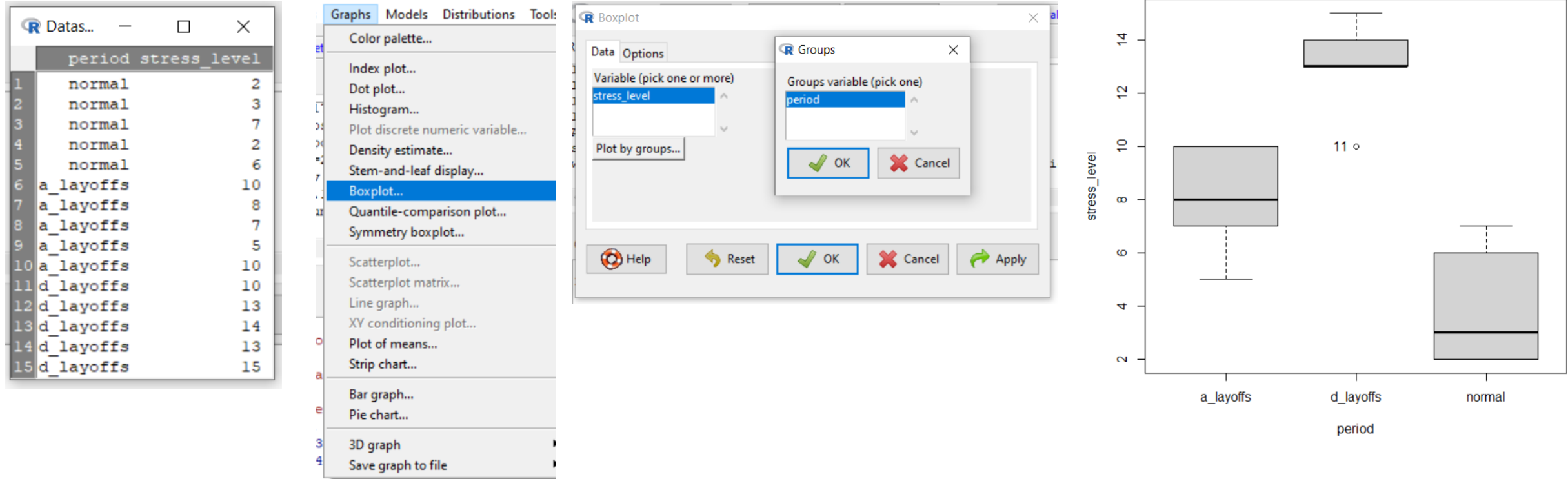
degrees of freedom numerator

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	161.5	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	246.0	248.0	249.1	250.1
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25



COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

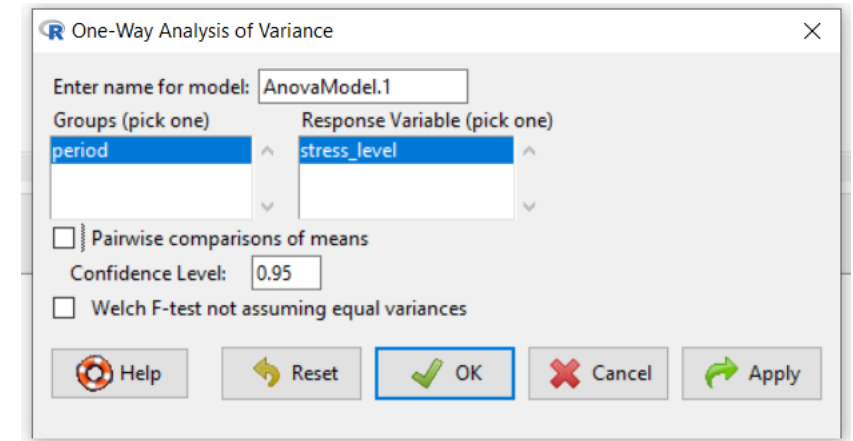
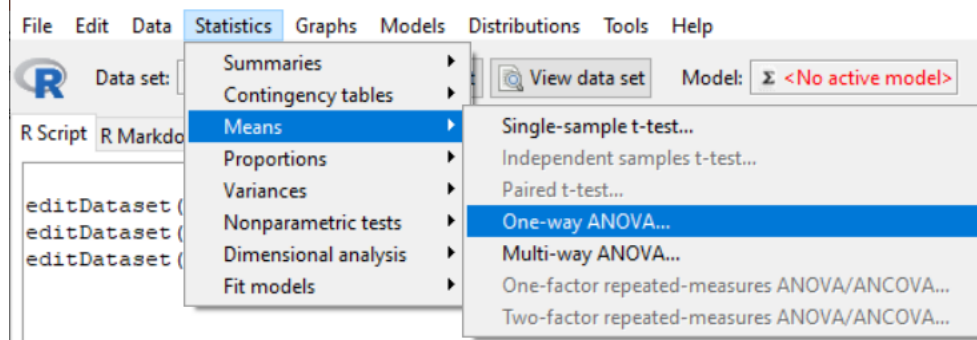


Since the variances of the three groups are not equal we should perform a Kruskal-Wallis test which is more conservative. I first carry on an ANOVA analysis using R-commander just for demonstration purposes. After that I carry out a Kruskal-Wallis test.

COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

	period	stress_level
1	normal	2
2	normal	3
3	normal	7
4	normal	2
5	normal	6
6	a_layoffs	10
7	a_layoffs	8
8	a_layoffs	7
9	a_layoffs	5
10	a_layoffs	10
11	d_layoffs	10
12	d_layoffs	13
13	d_layoffs	14
14	d_layoffs	13
15	d_layoffs	15



```
> summary(AnovaModel.1)
            Df Sum Sq Mean Sq F value    Pr(>F)
period      2  203.3   101.7    22.59 0.0000854 ***
Residuals  12   54.0     4.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

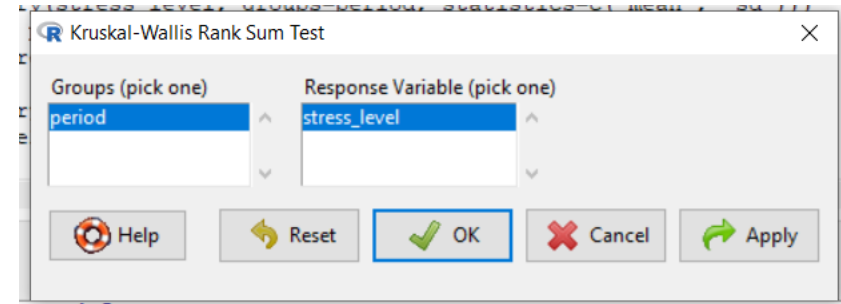
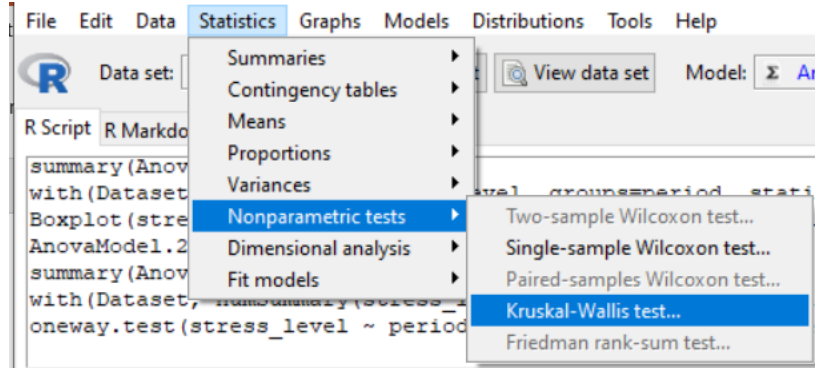
> with(Dataset, numSummary(stress_level, groups=period, statistics=c("mean", "sd")))
      mean      sd data:n
a_layoffs  8 2.121320     5
d_layoffs 13 1.870829     5
normal    4 2.345208     5
```

Reject the null hypothesis. That means
at least one mean is different than the other means.

COMPARING MORE THAN TWO MEANS

Analysis of Variance – Kruskal-Wallis

	period	stress_level
1	normal	2
2	normal	3
3	normal	7
4	normal	2
5	normal	6
6	a_layoffs	10
7	a_layoffs	8
8	a_layoffs	7
9	a_layoffs	5
10	a_layoffs	10
11	d_layoffs	10
12	d_layoffs	13
13	d_layoffs	14
14	d_layoffs	13
15	d_layoffs	15



Kruskal-Wallis rank sum test

```
data: stress_level by period
Kruskal-Wallis chi-squared = 10.982, df = 2, p-value = 0.004123
```

Reject the null hypothesis. That means
at least one MEDIAN is different than the other medians.

COMPARING MORE THAN TWO MEANS

Analysis of Variance - ANOVA

In a hypothesis test, there is always a type I error rate, which is defined by our significance level (alpha) and tells us the probability of rejecting a null hypothesis that is actually true. In other words, it's the probability of getting a "false positive", i.e. when we claim there is a statistically significant difference among groups, but there actually isn't.

For example, suppose we have four groups: A, B, C, and D. This means there are a total of six pairwise comparisons we want to look at with a post hoc test:

A – B (the difference between the group A mean and the group B mean)

A – C

A – D

B – C

B – D

C – D

Groups	Comparisons (Groups*(Groups-1))/2	Family-Wise Error Rate $1-(1-\alpha)^{\text{comparisons}}$
3	3	0.1426
4	6	0.2649
5	10	0.4013
6	15	0.5367
7	21	0.6594
8	28	0.7622
9	36	0.8422
10	45	0.9006

Notice that the family-wise error rate increases rapidly as the number of groups (and consequently the number of pairwise comparisons) increases. In fact, once we reach six groups, the probability of us getting a false positive is actually above 50%!

This means we would have serious doubts about our results if we were to make this many pairwise comparisons, knowing that our family-wise error rate was so high.

Fortunately, post hoc tests provide us with a way to make multiple comparisons between groups while controlling the family-wise error rate.

COMPARING MORE THAN TWO MEANS

Analysis of Variance – ANOVA – Tukey Test

Tukey's Test – useful when you want to make every possible pairwise comparison

1. Calculate $Q = \frac{\max(\bar{x}_i) - \min(\bar{x}_i)}{\sqrt{MS(error)/n_r}}$ where $MS(error)$ = pooled sample variance and n_r is the number of observations in each sample. Tukey considered n_r equal in all groups.

2. Find $Q_{\alpha,k,v}$ in the corresponding table.

3. Compute $D = Q_{\alpha,k,v} \cdot \sqrt{\frac{MS(error)}{n_r}}$

4. Place the sample mean in order from the smallest to largest

5. If two sample differ by more than D , the conclusion is that population means are unequal. In other words if $|\bar{x}_i - \bar{x}_j| > D \Rightarrow \mu_i \neq \mu_j$

● 0.05 level				
● 0.01 level				
Critical Values of the Studentized				
dfe	2	3	4	5
2	6.0849	8.3308	9.7980	10.8810
3	4.5007	5.9096	6.8245	7.5016
4	3.9265	5.0403	5.7571	6.2870
5	3.6354	4.6017	5.2185	5.6731
6	3.4605	4.3390	4.8956	5.3049
7	3.3439	4.1648	4.6812	5.0601
8	3.2612	4.0410	4.5288	4.8858
9	3.1991	3.9485	4.4149	4.7554
10	3.1511	3.8768	4.3266	4.6543
11	3.1127	3.8195	4.2561	4.5736
12	3.0813	3.7728	4.1985	4.5076
13	3.0553	3.7341	4.1509	4.4529
14	3.0332	3.7014	4.1105	4.4066
15	3.0143	3.6734	4.0760	4.3670
16	2.9980	3.6491	4.0461	4.3327
17	2.9837	3.6280	4.0200	4.3027
18	2.9712	3.6093	3.9970	4.2763
19	2.9600	3.5927	3.9766	4.2528
20	2.9500	3.5779	3.9583	4.2319
21	2.9410	3.5646	3.9419	4.2130
22	2.9329	3.5526	3.9270	4.1959
23	2.9255	3.5417	3.9136	4.1805
24	2.9188	3.5317	3.9013	4.1663
25	2.9126	3.5226	3.8900	4.1534

COMPARING MORE THAN TWO MEANS

Analysis of Variance – ANOVA – Tukey Test

In a hypothesis test, there is always a type I error rate, which is defined by our significance level (alpha) and tells us the probability of rejecting a null hypothesis that is actually true. In other words, it's the probability of getting a “false positive”, i.e. when we claim there is a statistically significant difference among groups, but there actually isn't.

Tukey's Test – useful when you want to make every possible pairwise comparison

The image shows a screenshot of the R Studio interface. On the left, a data frame is displayed with two columns: 'period' and 'stress_level'. The 'period' column has values 'normal', 'a_layoffs', and 'd_layoffs' repeated for different rows. The 'stress_level' column contains numerical values. In the center, the 'Statistics' menu is open, showing options like 'Means', 'Proportions', 'Variances', 'Nonparametric tests', 'Dimensional analysis', and 'Fit models'. The 'Means' submenu is also open, highlighting 'One-way ANOVA...'. On the right, the 'One-Way Analysis of Variance' dialog box is shown. It has fields for 'Enter name for model' (AnovaModel.2), 'Groups (pick one)' (period), and 'Response Variable (pick one)' (stress_level). The 'Pairwise comparisons of means' checkbox is checked, and the 'Confidence Level' is set to 0.95. The 'Welch F-test not assuming equal variances' checkbox is unchecked. At the bottom, there are buttons for 'Help', 'Reset', 'OK', 'Cancel', and 'Apply'.

period	stress_level
normal	2
normal	3
normal	7
normal	2
normal	6
a_layoffs	10
a_layoffs	8
a_layoffs	7
a_layoffs	5
a_layoffs	10
d_layoffs	10
d_layoffs	13
d_layoffs	14
d_layoffs	13
d_layoffs	15

COMPARING MORE THAN TWO MEANS

Analysis of Variance – ANOVA – Tukey Test

Simultaneous Tests for General Linear Hypotheses

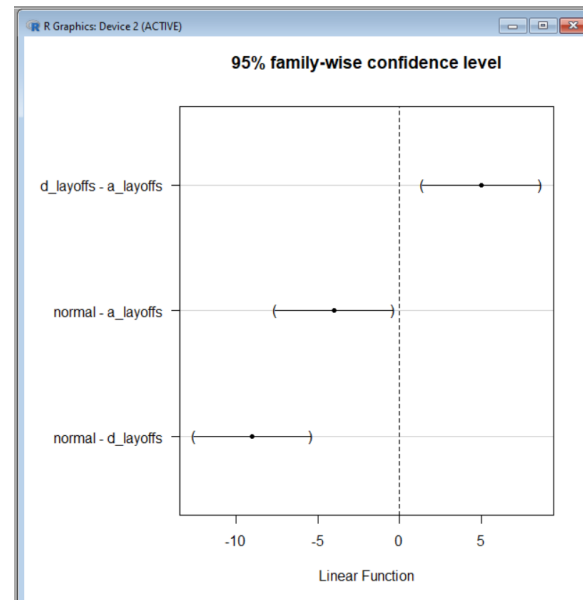
Multiple Comparisons of Means: Tukey Contrasts

Fit: aov(formula = stress_level ~ period, data = Dataset)

Linear Hypotheses:

	Estimate	Std. Error	t value	Pr(> t)	
d_layoffs - a_layoffs == 0	5.000	1.342	3.727	0.0078	**
normal - a_layoffs == 0	-4.000	1.342	-2.981	0.0290	*
normal - d_layoffs == 0	-9.000	1.342	-6.708	<0.001	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)



Simultaneous Confidence Intervals

Multiple Comparisons of Means: Tukey Contrasts

Fit: aov(formula = stress_level ~ period, data = Dataset)

Quantile = 2.666

95% family-wise confidence level

Linear Hypotheses:

	Estimate	lwr	upr
d_layoffs - a_layoffs == 0	5.0000	1.4232	8.5768
normal - a_layoffs == 0	-4.0000	-7.5768	-0.4232
normal - d_layoffs == 0	-9.0000	-12.5768	-5.4232

a_layoffs	d_layoffs	normal
"b"	"c"	"a"

VARIANCE, COVARIANCE

Variance measures how spread-out values are in a dataset.

$$s^2 = \sum (x_i - \bar{x})^2 / (n-1)$$

Example: Dataset: 6, 7, 10, 13, 14, 14, 18, 19, 22, 24 => $s^2=36678$

Covariance measures how changes in one variable are associated with changes in a second variable.

$$\text{COV}(X, Y) = \sum (x_i - \bar{x})(y_i - \bar{y}) / (n-1)$$

X	Y
3	6
5	7
6	7
6	13
7	16
8	15
12	17
14	20
15	24
19	27

Cov(X,Y)=35.33

X	Y
3	28
5	25
6	24
6	25
7	19
8	15
12	13
14	15
15	4
19	3

Cov(X,Y)=-42.83

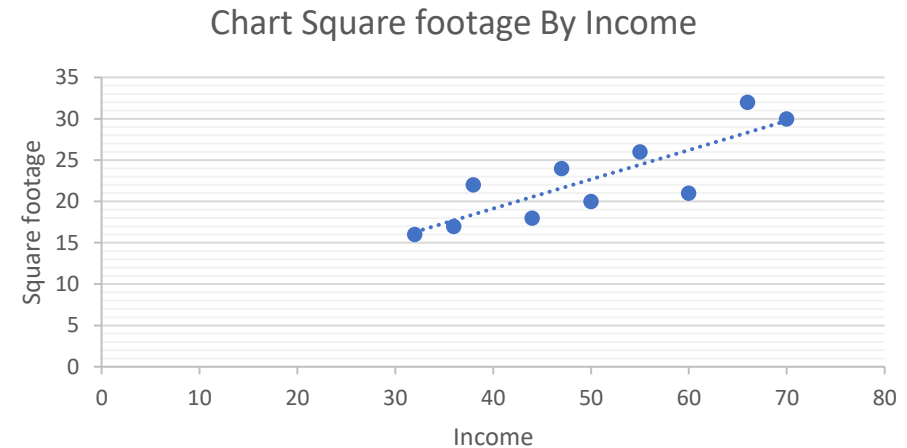
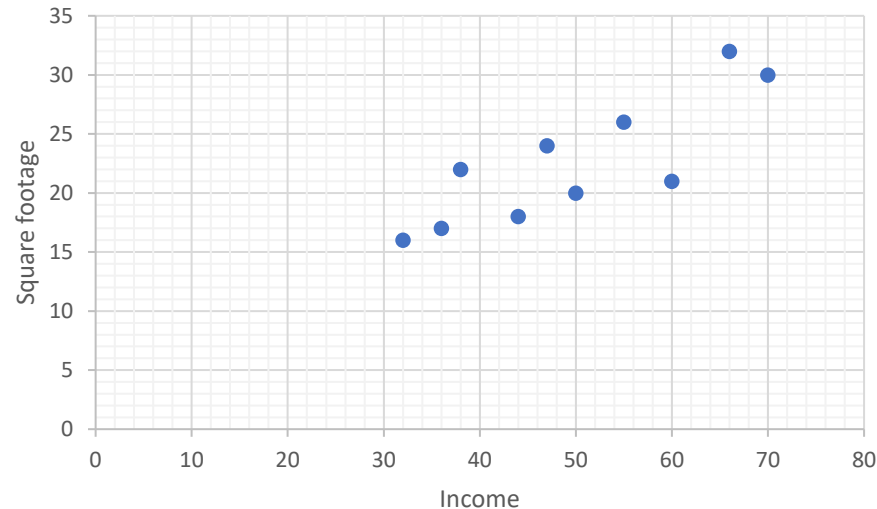
Problem with covariance is measure dependent.

E.g. the covariance of two variables measured in miles might be 4.25, but if the same scores are converted to kilometres, the covariance is 11. **Solution: Standardize the covariance and the result is coefficient of correlation**

COEFFICIENT OF CORRELATION

Suppose we have a dataset with family income (X, in thousands of dollars) and the square footage of their homes (Y, in hundreds of square feet). We want to see if there is any relationship between the two measures.

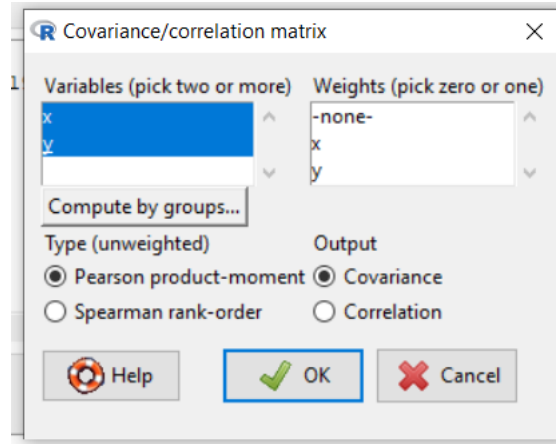
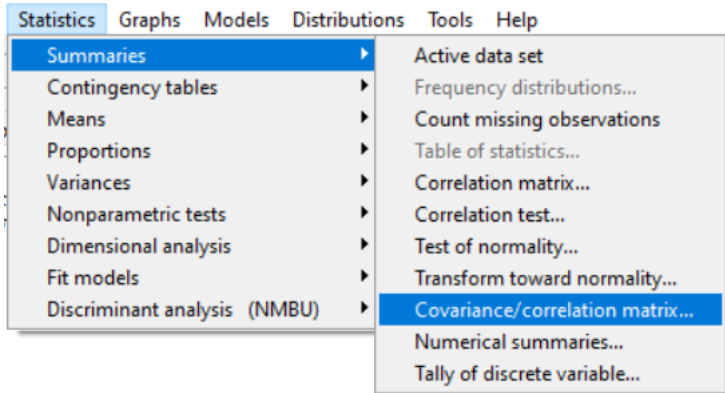
Income (X)	32	36	55	47	38	60	66	44	70	50
Square footage (Y)	16	17	26	24	22	21	32	18	30	20



The strength of the linear relationship between two variables is measured by the coefficient of correlation.

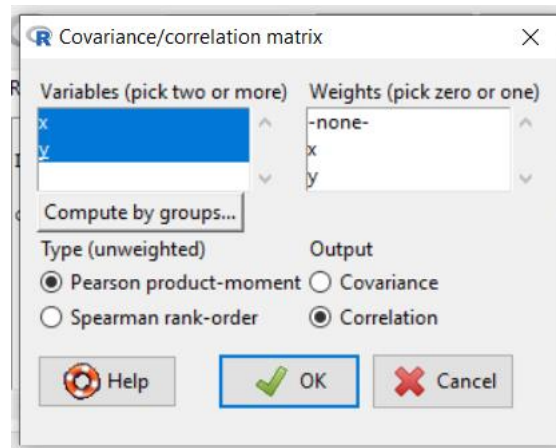
$$r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\sqrt{\Sigma(x - \bar{x})^2} \sqrt{\Sigma(y - \bar{y})^2}}$$

VARIANCE, COVARIANCE & CORRELATION



```
> cov(Dataset[,c("x","y")], use="complete.obs")
      x      y
x 26.94444 35.33333
y 35.33333 51.95556
```

	x	y
1	3	6
2	5	7
3	6	7
4	6	13
5	7	16
6	8	15
7	12	17
8	14	20
9	15	24
10	19	27

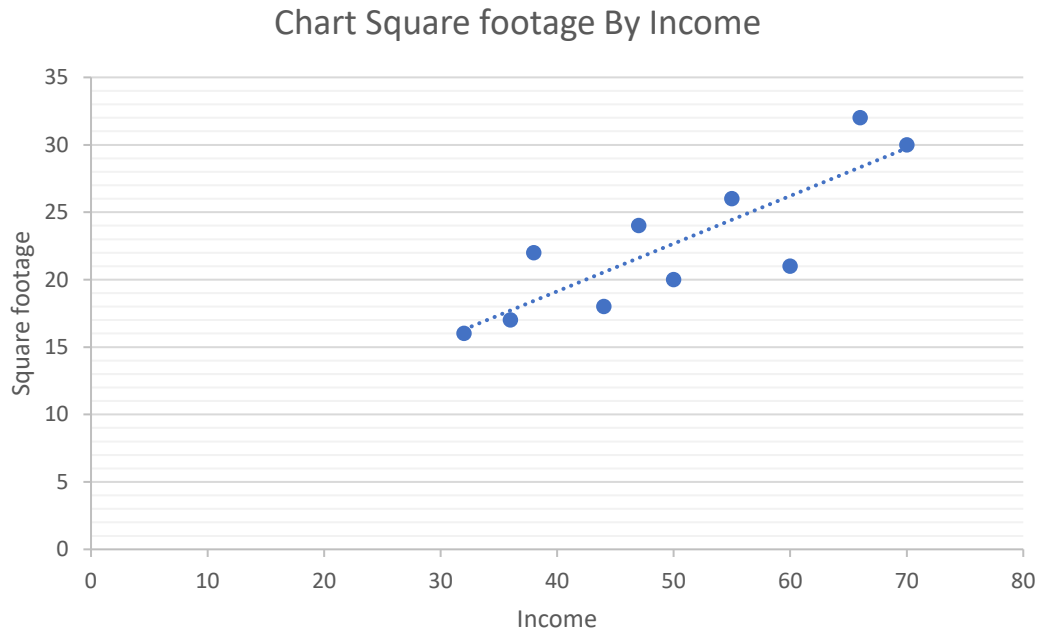


```
> cor(Dataset[,c("x","y")], use="complete.obs")
      x      y
x 1.0000000 0.9443522
y 0.9443522 1.0000000
```

COEFFICIENT OF CORRELATION

Suppose we have a dataset with family income (X, in thousands of dollars) and the square footage of their homes (Y, in hundreds of square feet). We want to see if there is any relationship between the two measures.

Income (X)	32	36	55	47	38	60	66	44	70	50
Square footage (Y)	16	17	26	24	22	21	32	18	30	20



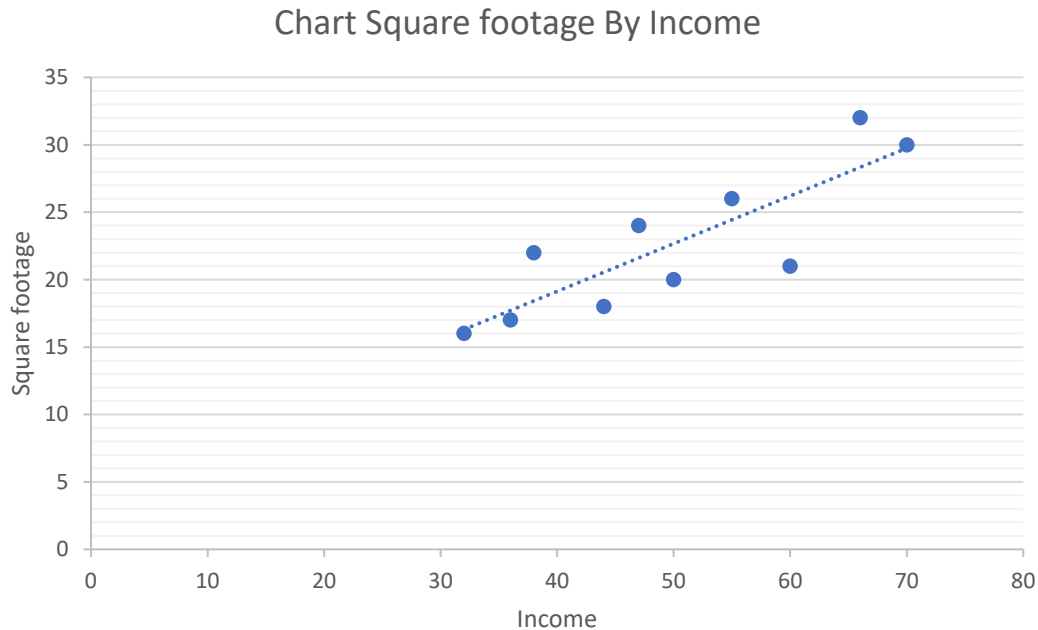
$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}} = \frac{\sum xy - (\sum x)(\sum y)/n}{\sqrt{\sum x^2 - (\sum x)^2/n} \sqrt{\sum y^2 - (\sum y)^2/n}}$$

- Comments:
- a) r is between -1 and 1 inclusive
 - b) the larger $|r|$, the stronger the linear relationship
 - c) the sign of r tells if the relationship between X and Y is positive (direct) or negative (inverse)
 - d) if $r=1$ or $r=-1$ then X and Y are perfectly correlated

COEFFICIENT OF CORRELATION

Suppose we have a dataset with family income (X, in thousands of dollars) and the square footage of their homes (Y, in hundreds of square feet). We want to see if there is any relationship between the two measures.

Income (X)	32	36	55	47	38	60	66	44	70	50
Square footage (Y)	16	17	26	24	22	21	32	18	30	20



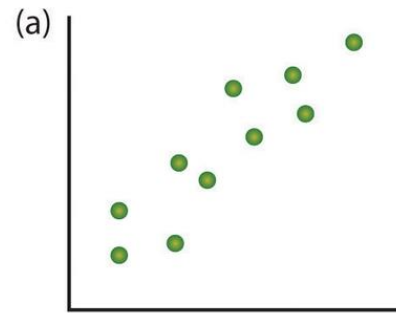
$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}} = \frac{\sum xy - (\sum x)(\sum y)/n}{\sqrt{\sum x^2 - (\sum x)^2/n} \sqrt{\sum y^2 - (\sum y)^2/n}}$$

- Comments:
- a) r is between -1 and 1 inclusive
 - b) the larger $|r|$, the stronger the linear relationship
 - c) the sign of r tells if the relationship between X and Y is positive (direct) or negative (inverse)
 - d) if $r=1$ or $r=-1$ then X and Y are perfectly correlated

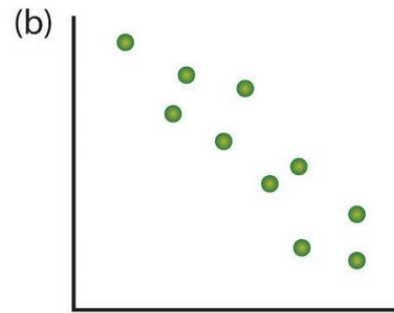
COEFFICIENT OF CORRELATION

The strength of the linear relationship between two variables is measured by the coefficient of correlation.

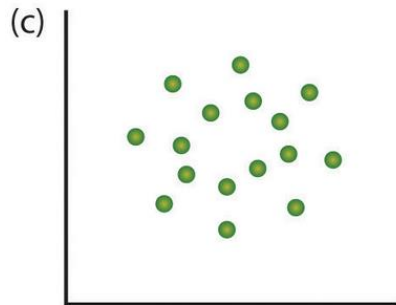
$$r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\sqrt{\Sigma(x - \bar{x})^2} \sqrt{\Sigma(y - \bar{y})^2}}$$



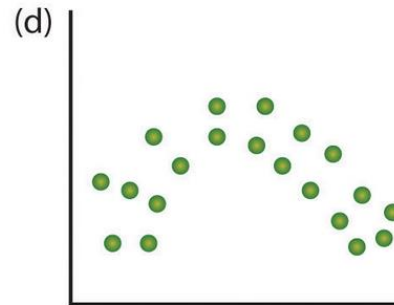
Positive linear
 $r = +.82$



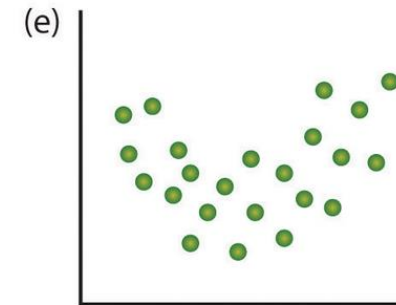
Negative linear
 $r = -.70$



Independent
 $r = 0.00$



Curvilinear
 $r = 0.00$



Curvilinear
 $r = 0.00$

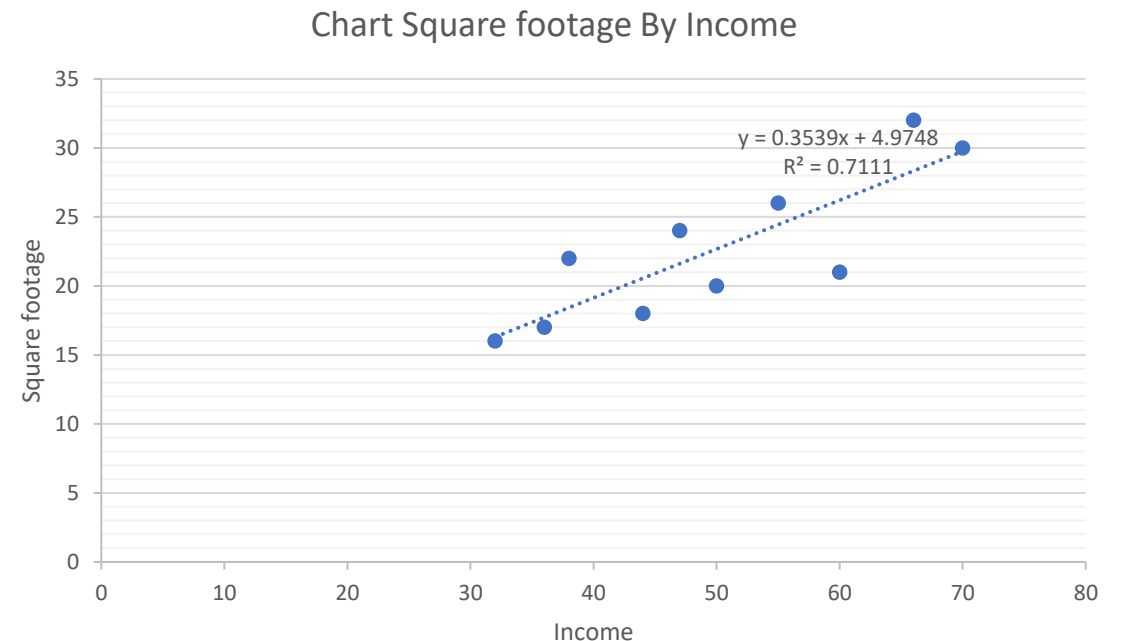
COEFFICIENT OF CORRELATION & LINEAR REGRESSION

Suppose we have a dataset with family income (X, in thousands of dollars) and the square footage of their homes (Y, in hundreds of square feet). We want now not only to see to what extent the two variables are correlated but also to see if we can use a better approach that would allow us to also do predictions.

Income (X)	32	36	55	47	38	60	66	44	70	50
Square footage (Y)	16	17	26	24	22	21	32	18	30	20

The graph on the right can be created using Excel. Notice the relationship of the two variables expressed by the equation line.

Let's understand how this line is found.

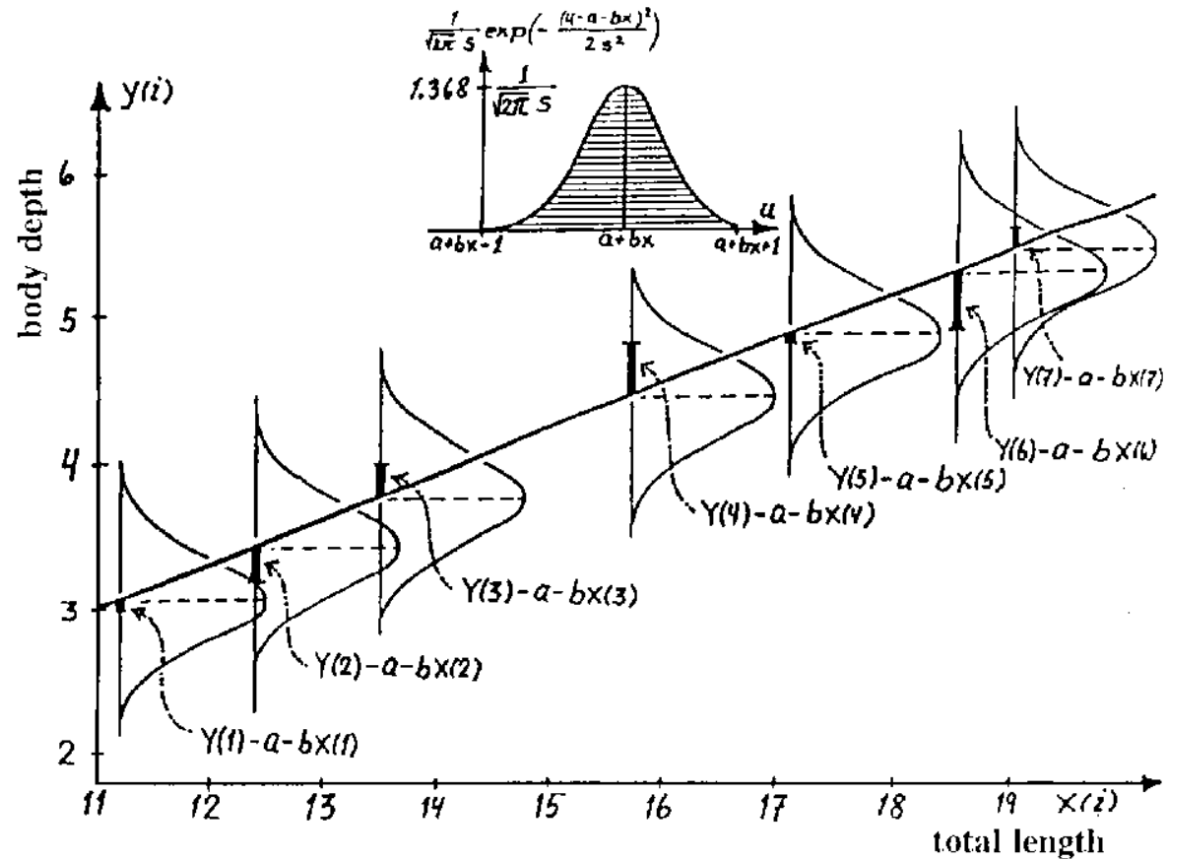


LINEAR REGRESSION INTRO

Simply defined, linear regression is a statistical method that describes the relationship between two or more variables. One of the variables is called the dependent variable while one or more other variables are called independent variables. The dependent variable values are considered as being influenced by the independent variable values. Usually, a dependent variable is noted with Y while an independent variable with X or $X_1, X_2 \dots X_n$.

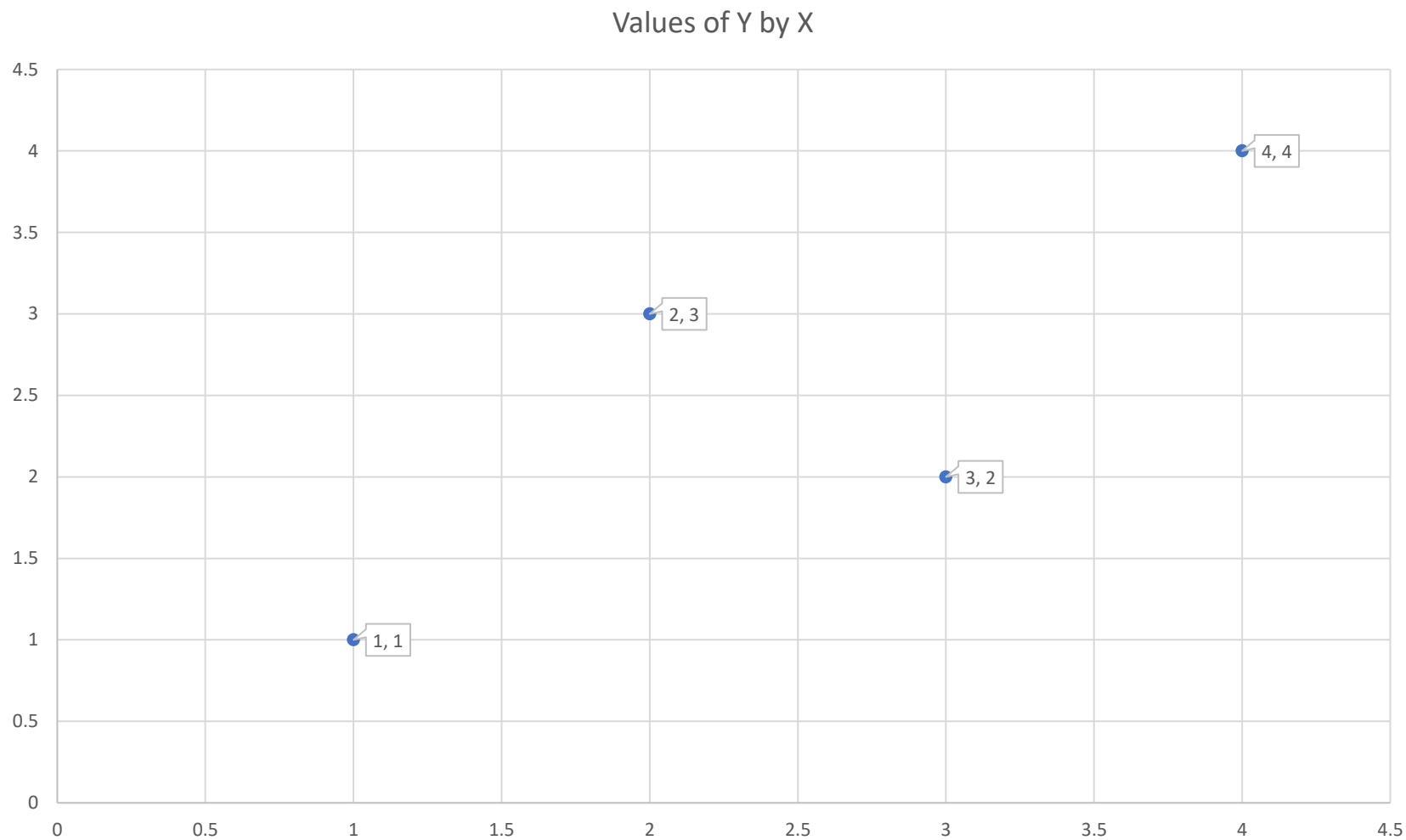
The term “linear” shows that between the dependent and independent variable values is a linear type of relationship. Simply said each value Y_i can be described by a value of X_i using the equation $Y_i(\text{dependent}) = b_0 + b_1 * X_i(\text{independent})$.

As we have seen in the previous slides not all the points fall exactly on the line. That means that the equation above should also include an error term. The equation becomes: $Y_i = b_0 + b_1 * X_i + \epsilon_i$



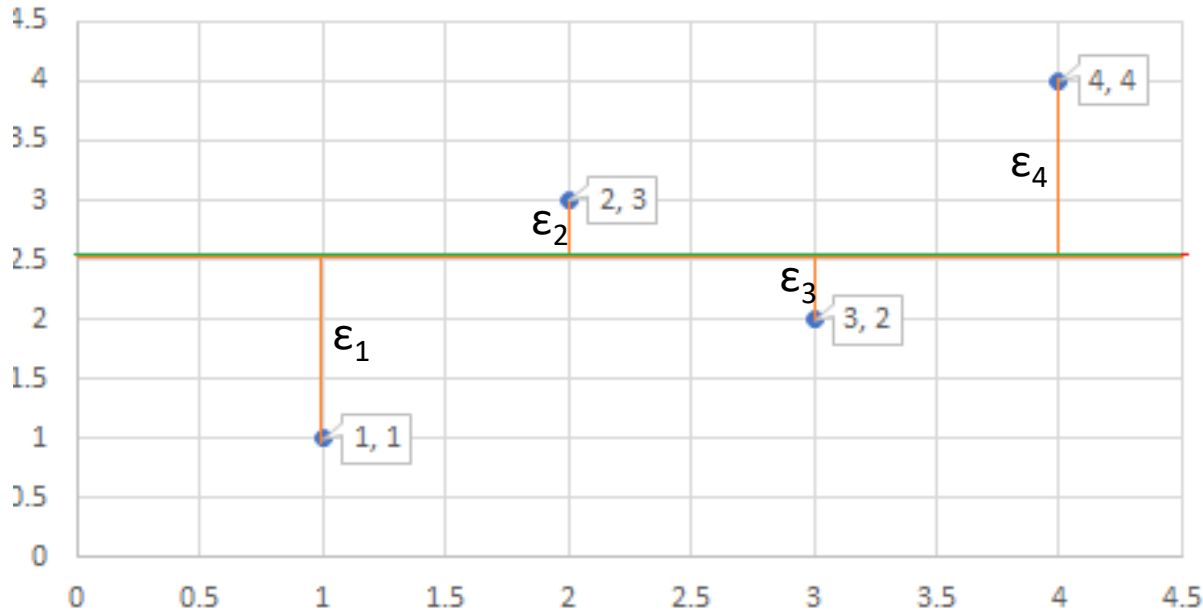
LINEAR REGRESSION INTRO

X	Y
1	1
2	3
3	2
4	4

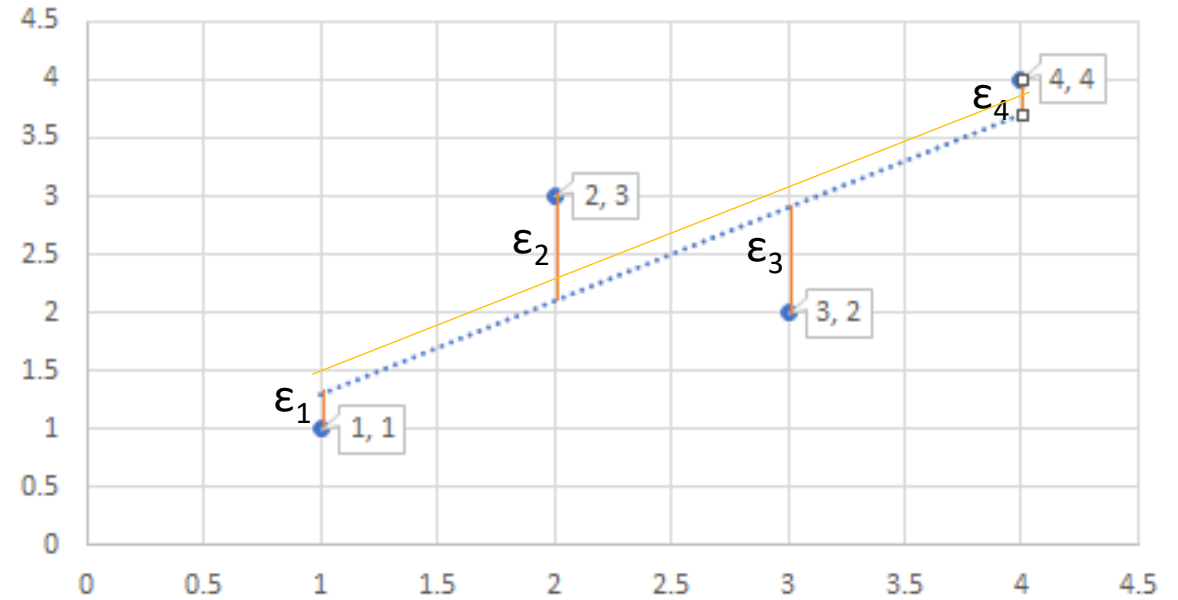


LINEAR REGRESSION INTRO

Values of Y by X



Values of Y by X



Why not the sum of errors ? Well let's compute the sum of errors for the two pictures using approximations.

For the left picture $\sum_{i=1}^4 \epsilon_i = -1.5 + 0.5 - 0.5 + 1.5 = 0$. For the right picture $\sum_{i=1}^4 \epsilon_i = -0.25 + 0.9 - 0.9 + 0.25 = 0$ for the dotted blue line and $\sum_{i=1}^4 \epsilon_i = -0.5 + 0.75 - 3 + 0.15 = -2.4$ for the yellow continuous line.

Let's try using the $\sum_{i=1}^4 |\epsilon_i|$. Then, for the left picture we have $\sum_{i=1}^4 |\epsilon_i| = 4$. For the right picture $\sum_{i=1}^4 |\epsilon_i| = 2.3$ for the blue dotted line and $\sum_{i=1}^4 |\epsilon_i| = 4.4$ for the yellow line. What to do? Solution: use the sum of square errors to find the best line that represent the relationship of the two variables Y and X.

LINEAR REGRESSION INTRO

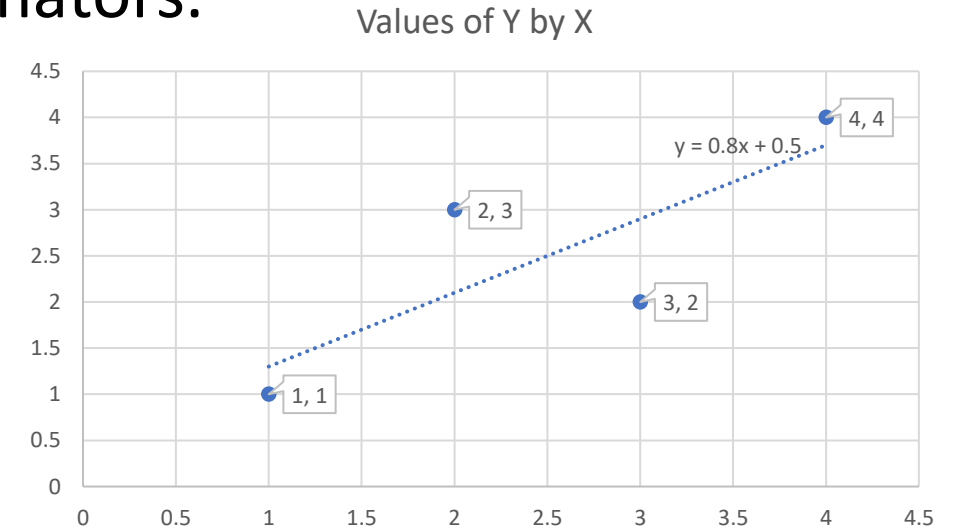
In order to find the best line that represent the relationship between X and Y we must be able to find the equation of the line $Y_i = b_0 + b_1 * X_i + \varepsilon_i$ where ε_i is the error term. We must therefore compute $\sum_{i=1}^n \varepsilon_i^2$ and find the parameters b_0 and b_1 so that $\sum_{i=1}^n \varepsilon_i^2 \approx 0$. There are many approaches to solve this problem. One approach would be to use the least square estimators.

The coefficients formula is:

$$b_0 = \bar{Y} - b_1 \bar{X}$$
$$b_1 = \frac{\sum (x_i - \bar{x})(y - \bar{y})}{\sum (x_i - \bar{x})^2}$$

which is $b_0=0.5$ and $b_1=0.8$ in our case.

What to do now that we have found the equation of the line. We can build a least square line through any dataset. However, to be able to do predictions we need to check if some assumptions related to our model holds or not.

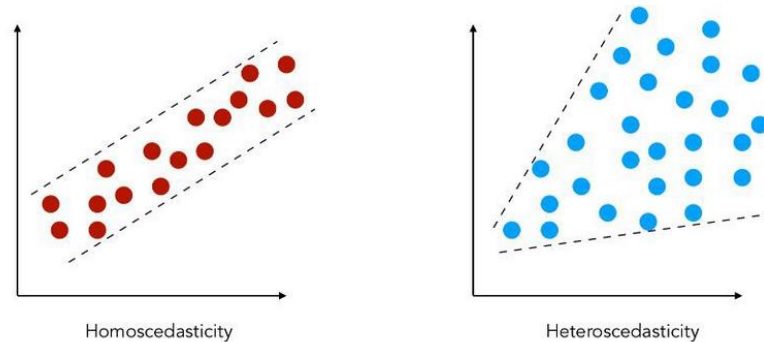


LINEAR REGRESSION ASSUMPTIONS

1. At any given value of X , the population of potential error term has a mean=0.

2. Homoscedasticity: The variance of residual is the same for any value of X .

Visual test - In a plot of residuals against dependent variable or another variable will often produce a fan shape.

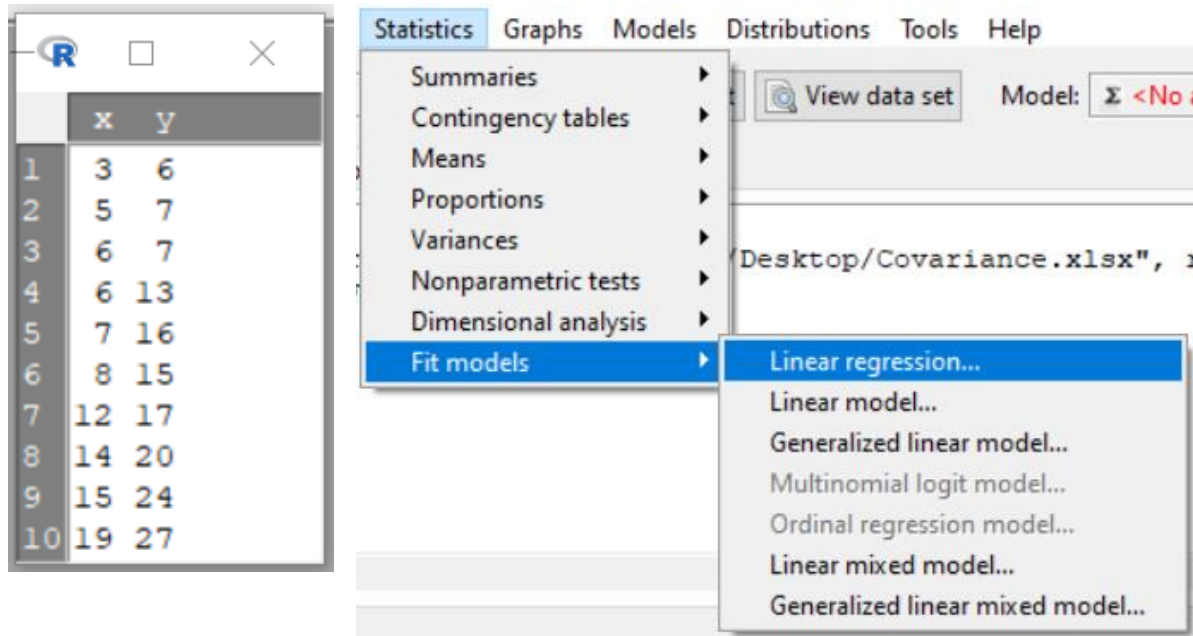


3. Independence: Any error term value is independent of any other error term.

Durbin-Watson test for autocorrelation between residuals. A value of 2.0 indicates there is no autocorrelation detected in the sample

4. Normality: At any given value of x the population of potential error term is normally distributed.

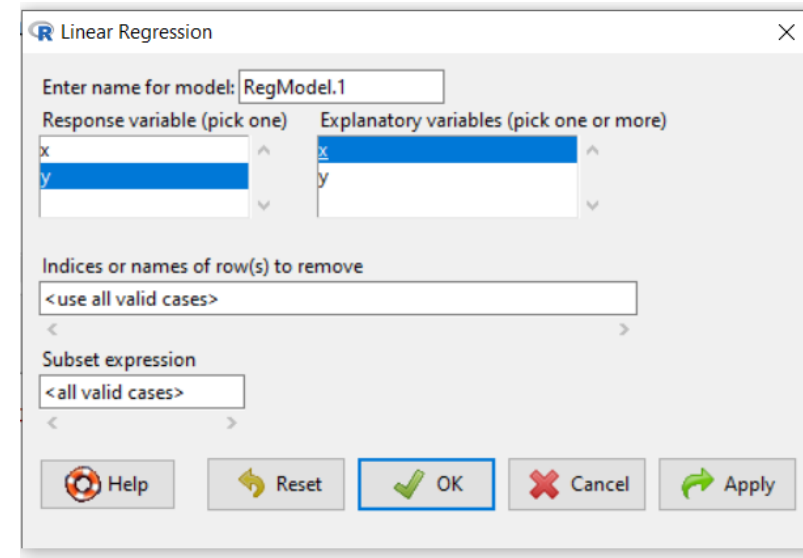
RUNNING SIMPLE LINEAR REGRESSION IN Rcmdr



The screenshot shows the Rcmdr interface. On the left, a data table with columns 'x' and 'y' is displayed. The 'Statistics' menu is open, and 'Fit models' is selected. The 'Linear regression...' option is highlighted in the submenu.

	x	y
1	3	6
2	5	7
3	6	7
4	6	13
5	7	16
6	8	15
7	12	17
8	14	20
9	15	24
10	19	27

```
RegModel.1 <- lm(y~x, data=Dataset)
summary(RegModel.1)
```



The 'Linear Regression' dialog box is shown. The 'Enter name for model' field is set to 'RegModel.1'. The 'Response variable (pick one)' is 'y' and the 'Explanatory variables (pick one or more)' is 'x'. The 'Indices or names of row(s) to remove' is '<use all valid cases>'. The 'Subset expression' is '<all valid cases>'. The 'OK' button is highlighted.

```
Call:
lm(formula = y ~ x, data = Dataset)

Residuals:
    Min       1Q   Median       3Q      Max
-3.610 -1.384 -0.667  1.722  4.078

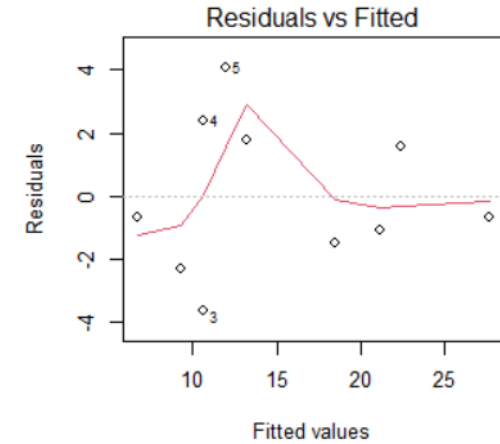
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.7423     1.7280   1.587   0.151
x             1.3113     0.1615   8.120 0.0000392 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.515 on 8 degrees of freedom
Multiple R-squared:  0.8918, Adjusted R-squared:  0.8783
F-statistic: 65.94 on 1 and 8 DF, p-value: 0.00003922
```

SIMPLE LINEAR REGRESSION IN Rcmdr ASSUMPTIONS

1. At any given value of X , the population of potential error term has a mean=0.

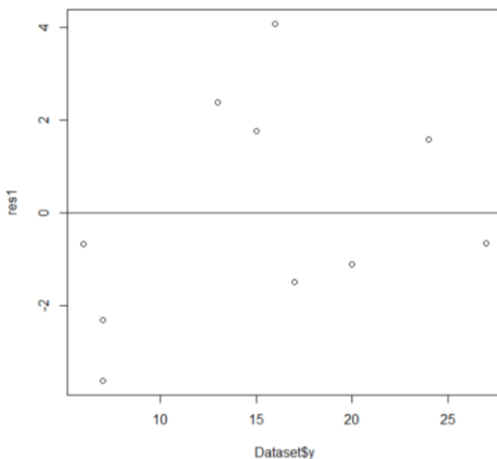
```
plot(RegModel.1, 1)
```



2. **Homoscedasticity**: The variance of residual is the same for any value of X .

Visual test - In a plot of residuals against dependent variable or another variable will often produce a fan shape.

```
res1 <- residuals(RegModel.1)
plot(Dataset$y, res1)
abline(h=0)
```



Another test is the Breusch-Pagan test using the following hypotheses:

H₀: Homoscedasticity is present (the residuals are distributed with equal variance)

H_a: Heteroscedasticity is present (the residuals are not distributed with equal variance)

In Rcmdr we first load the needed library, and after that execute the test.

```
library(lmtest)
```

```
bptest(RegModel1.1)
```

Ideally, we want p-value > 0.05

Out:

studentized Breusch-Pagan test

```
data:  stocks ~ t1y + t10y + hyield + cpi + ppi + oil + indpro + pce
BP = 49.943, df = 8, p-value = 4.191e-08
```

SIMPLE LINEAR REGRESSION IN Rcmdr ASSUMPTIONS

3. Independence of residuals: Any error term value is independent of any other error term. Durbin-Watson test for autocorrelation between residuals. A value of 2.0 indicates there is no autocorrelation detected in the sample. Ideally, we want the p-value > 0.05

Ho (null hypothesis): There is no correlation among the residuals.

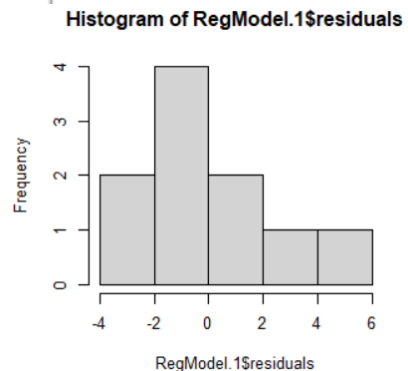
Ha (alternative hypothesis): The residuals are autocorrelated.

```
durbinWatsonTest(RegModel.1)
```

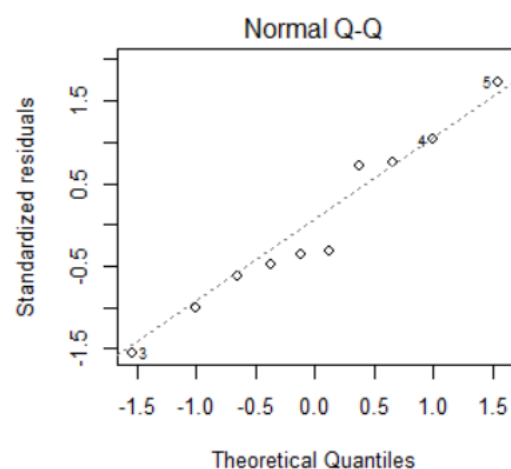
```
lag Autocorrelation D-W Statistic p-value
1      0.2846772      1.413055    0.144
Alternative hypothesis: rho != 0
```

4. Normality of residuals

```
hist(RegModel.1$residuals)
```



```
plot(RegModel.1, 2)
```



The Shapiro-Wilks test for normality is one of three general normality tests designed to detect all departures from normality.

Ho: Data is normally distributed

Ha: Data is not normally distributed

```
normalityTest(LinearModel.1$residuals, test="shapiro.test")
```

Ideally, we want the p-value > 0.05

SIMPLE LINEAR REGRESSION IN Rcmdr RESULTS

Call:

```
lm(formula = y ~ x, data = Dataset)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.610	-1.384	-0.667	1.722	4.078

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	p-value
(Intercept) b_0	2.7423	1.7280	1.587	0.151	
x b_1	1.3113	0.1615	8.120	0.0000392	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.515 on 8 degrees of freedom

Multiple R-squared: 0.8918, Adjusted R-squared: 0.8783

F-statistic: 65.94 on 1 and 8 DF, p-value: 0.00003922

SIMPLE LINEAR REGRESSION IN Rcmdr RESULTS

When we run linear regression, we perform hypotheses testing on the two parameters of the regression line (the intercept and the slope).

$H_0: b_0=0$, $H_a: b_0 \neq 0$, and so on for each predictor(independent variable) X coefficient. We use t-tests because a normal distribution divided by the square root of a chi-square distribution follows a t-distribution. In addition, we do not know the standard deviation or the variance of the population. However, when n is big enough, the t-test value is similar to the z-test value.

If the p-values for each parameter $b_0, b_1, \dots, b_n$ is $< \alpha=0.05$ or another reasonably established α , then we reject the null hypothesis and conclude that there is a linear relationship between the corresponding independent variable and the dependent variable. If we fail to reject the null, there is no linear relationship between the corresponding X and Y. In our case, our equation line is:

$$Y = 2.7423 + 1.3113 * X$$

Notice that the p-value for the intercept says that the intercept is not significantly different from zero, which sometimes is ok. The meaning of the intercept is sometimes problematic. However, it is safest to include the intercept in the model even though it is not significant. Remember that b_0 equals the average of Y when X is zero. If the response function is exactly zero when all the predictors are set to zero or at their reference levels, we won't keep the intercept(there are some economical models in this situation).

Coefficient b_1 is the expected change in the outcome Y per unit change in X. An increase in X by 1 unit is associated with an increase in Y by b_1 units while keeping all the other variables constant.

LINEAR REGRESSION SIMPLE COEFFICIENT OF DETERMINATION (R-squared)

R-square is a measure of usefulness of a linear regression model.

Total variation = Explain Variation + Unexplained Variation

$$\Sigma(y_i - \bar{y})^2 = \Sigma(\hat{y}_i - \bar{y})^2 + \Sigma(y_i - \hat{y})^2$$

SumofSquareTotal = SumofSquareRegression + SumofSquareErrors

$$SST = SSR + SSE$$

$$R\text{-squared} = SSR/SST$$

R-squared is the total variation in the Y variables that is explained by the X variables. It has values between zero and one.

The square root of the R-squared give us a value equal to the coefficient of correlation (r). However, we need to be careful about the interpretation of the results because of some conceptual differences.

SIMPLE LINEAR REGRESSION IN Rcmdr RESULTS

The R-squared value = 0.8918. This means that 89.18% of the variance in the Y variable is explained by the X variable. In general, we want to get an R-squared as much as possible. However, it is not always possible.

The F-test for overall significance has the following two hypotheses:

Ho: the model with no independent variables fits the data as well as your model.

Ha: the model we came up with fits the data better than the intercept-only model.

Our model p-value is :

F-statistic: 65.94 on 1 and 8 DF, p-value: 0.00003922

If none of your independent variables are statistically significant, the overall F-test is also not statistically significant. However, we may have conflicting results.

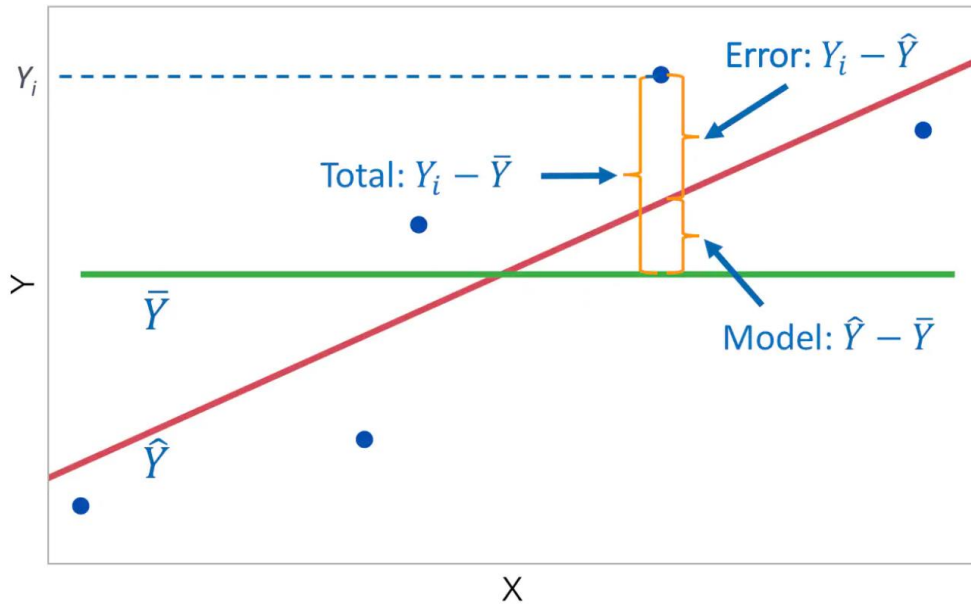
Reason: F-test of overall significance assesses all of the coefficients jointly whereas the t-test for each coefficient examines them individually.

The overall F-test can find that the coefficients are significant jointly while the t-tests can fail to find significance individually.

SIMPLE LINEAR REGRESSION THE F-TEST

The result of our F-test is:

Residual standard error: 2.515 on 8 degrees of freedom
Multiple R-squared: 0.8918, Adjusted R-squared: 0.8783
F-statistic: 65.94 on 1 and 8 DF, p-value: 0.00003922



$$\Sigma(y_i - \bar{y})^2 = \Sigma(\hat{y}_i - \bar{y})^2 + \Sigma(y_i - \hat{y})^2$$

	df	SS	MS	F	Significance F
Regression	1	417.0061856	417.0061856	65.93789225	0.0000
Residual	8	50.59381443	6.324226804		
Total	9	467.6			

	df	SS	MS	F	Significance F
Regression	1	$\Sigma(\hat{y} - \bar{y})^2$	$\Sigma(\hat{y} - \bar{y})^2 / 1 = \text{MSR}$	MSR/MSE	p-value
Residual	(n-2)	$\Sigma(y_i - \hat{y}_i)^2$	$\Sigma(y_i - \hat{y}_i)^2 / (n-2) = \text{MSE}$		
Total	(n-1)	$\Sigma(y_i - \bar{y})^2$	$\Sigma(y_i - \bar{y})^2 / (n-1)$		

SIMPLE LINEAR REGRESSION IN Rcmdr PREDICTION

Let us now predict a value of Y based on a new values of X. Suppose $X_0=21$.

Recall the linear regression equation: $Y = 2.7423 + 1.3113 * X$

To get an estimation of the Y based on the equation called $\hat{y}$ we use the equation from above replacing X with 21. The $\hat{y} = 2.7423 + 1.3113 * 21 = 30.2796$

How confident are we on the newly predicted value? Can we provide a **prediction** interval? Sure, the **prediction** interval is:

$$\hat{y} \pm t_{\alpha/2, (n-2)} * s * \sqrt{1 + DistanceValue} \text{ where}$$

$$DistanceValue = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \text{ and}$$

$$s = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}}$$

LINEAR REGRESSION.

CONFIDENCE vs PREDICTION INTERVAL

We use the following formula to calculate a **confidence interval**:

$$\hat{y}_0 \pm t_{\alpha/2, n-2} * S_{yx} \sqrt{((x_0 - \bar{x})^2 / SS_x + 1/n)}$$

We use the following formula to calculate a **prediction interval**:

$$\hat{y}_0 \pm t_{\alpha/2, n-2} * S_{yx} \sqrt{((x_0 - \bar{x})^2 / SS_x + 1/n + 1)}$$

Thus, a prediction interval will always be wider than a confidence interval.

- $\hat{y}_0$: Estimated mean value of response variable
- $t_{\alpha/2, n-2}$: t-critical value with n-2 degrees of freedom
- S_{yx} : Standard error of response variable
- x_0 : specific value of predictor variable
- $\bar{x}$: mean value of predictor variable
- SS_x : Sum of squares for predictor variable
- n : Total sample size

Suppose we have the following dataset that shows the number of bedrooms and the selling price for 20 houses in a particular neighborhood:

Bedrooms	Price(thousands)
1	120
1	133
1	139
2	185
2	148
2	160
2	192
3	205
3	244
3	213
3	236
3	280
3	275
3	273
4	312
4	311
4	304
5	415
5	396
6	488

```
new <- data.frame(Bedrooms=c(3))
predict(RegModel.1, newdata = new, interval = "confidence")
```

```
> predict(RegModel.1, newdata = new, interval = "confidence")
      fit      lwr      upr
1 251.45 240.087 262.813
```

```
new <- data.frame(Bedrooms=c(3))
predict(RegModel.1, newdata = new, interval = "prediction")
```

```
> predict(RegModel.1, newdata = new, interval = "prediction")
      fit      lwr      upr
1 251.45 199.3783 303.5217
```

MULTIPLE LINEAR REGRESSION

We can explore single and multiple linear regression using the Car dataset from Canvas. The dataset looks like below.

1	Name	Sedan	SUV	Hatchback	Convertible	Wagon	Coupe	Minivan	Drive	Seating	Engine	HP	Length	Width	Height	Wheelbase	Weight	CargoVolume	CityMPG	HighwayMPG	OverallMPG	LowestMSRP	Parent Company HQ
2	Acura ILX	Yes	No	No	No	No	No	No	Front	5	2.4	201	182	71	56	105	3095	12	24	34	28	\$30,000	Japan
3	Acura MDX	No	Yes	No	No	No	No	No	Front	7	3	321	196	77	67	111	4200	34	20	27	21	\$44,300	Japan
4	Acura RDX	No	Yes	No	No	No	No	No	Front	5	2	272	187	74	66	108	4005	33	22	28	22	\$37,600	Japan
5	Acura RLX	Yes	No	No	No	No	No	No	Front	5	3.5	310	198	74	58	112	3930	15	20	29	23	\$54,900	Japan
6	Acura TLX	Yes	No	No	No	No	No	No	Front	5	2.4	206	190	74	57	109	3480	13	30	38	26	\$33,000	Japan
7	Alfa Romeo Giulia	Yes	No	No	No	No	No	No	Rear	5	2	280	183	74	57	111	3695	12	24	33	27	\$38,295	Italy
8	Alfa Romeo Stelvio	No	Yes	No	No	No	No	No	Rear	5	2	280	185	75	65	111	4020	26.5	22	29	24	\$40,295	Italy
9	Audi A3	Yes	No	No	Yes	No	No	No	Front	5	2	184	175	70	56	104	3135	13	27	36	27	\$33,300	Germany
10	Audi A4	Yes	No	No	No	Yes	No	No	Front	5	2	188	186	73	56	111	3630	13	27	34	27	\$39,200	Germany
11	Audi A5	No	No	Yes	Yes	No	Yes	No	AWD	4	2	248	184	73	54	109	3550	12	23	34	27	\$44,200	Germany
12	Audi A6	Yes	No	No	No	No	No	No	AWD	5	3	335	195	74	57	115	4265	14	24	32	25	\$58,900	Germany
13	Audi A8	Yes	No	No	No	No	No	No	AWD	5	3	335	209	77	59	123	4810	13	19	27	21	\$83,800	Germany
14	Audi Q3	No	Yes	No	No	No	No	No	AWD	5	2	228	177	73	62	106	3905	23.7	19	27	22	\$34,700	Germany
15	Audi Q5	No	Yes	No	No	No	No	No	AWD	5	2	248	184	75	65	111	4140	27	22	28	24	\$42,950	Germany
16	Audi Q7	No	Yes	No	No	No	No	No	AWD	5	2	248	200	78	69	118	5080	35.5	19	25	20	\$53,550	Germany
17	Audi Q8	No	Yes	No	No	No	No	No	AWD	5	3	335	197	79	67	118	5000	30.5	17	22	19	\$67,400	Germany
18	Audi TT	No	Yes	No	Yes	No	Yes	No	AWD	4	2	228	165	72	53	99	3160	12	23	31	26	\$44,900	Germany
19	BMW 2 Series	No	No	No	Yes	No	Yes	No	Rear	4	2	248	175	70	56	106	3450	14	24	32	25	\$35,300	Germany
20	BMW 4 Series	No	No	Yes	Yes	No	Yes	No	Rear	4	2	248	183	72	54	111	3470	16	23	34	27	\$44,600	Germany
21	BMW 5 Series	Yes	No	No	No	No	No	No	Rear	5	2	248	195	74	58	117	3950	19	25	33	26	\$53,400	Germany
22	BMW 6 Series	Yes	No	Yes	Yes	No	No	No	Rear	5	3	315	193	75	54	112	4355	13	20	29	20	\$70,300	Germany
23	BMW 7 Series	Yes	No	No	No	No	No	No	Rear	5	2	322	207	75	58	126	4710	18	22	29	21	\$83,650	Germany
24	BMW X1	No	Yes	No	No	No	No	No	Front	5	2	228	175	72	63	105	3725	27	24	33	26	\$34,950	Germany
25	BMW X2	No	Yes	No	No	No	No	No	Front	5	2	228	172	72	60	105	3645	23.5	24	32	25	\$36,400	Germany
26	BMW X3	No	Yes	No	No	No	No	No	Rear	5	2	248	186	74	66	113	4220	32	25	29	24	\$41,000	Germany
27	BMW X4	No	Yes	No	No	No	No	No	AWD	5	2	248	188	76	64	113	4145	19	23	28	25	\$50,450	Germany
28	BMW X5	No	Yes	No	No	No	No	No	Rear	5	3	335	194	79	69	117	4740	36.5	21	26	23	\$60,700	Germany

MULTIPLE LINEAR REGRESSION- INITIAL MODEL

Linear Regression

Enter name for model: RegModel.8

Response variable (pick one): LowestMSRP

Explanatory variables (pick one or more): CargoVolume, CityMPG, Engine, Height, HighwayMPG, HP

Indices or names of row(s) to remove: <use all valid cases>

Subset expression: <all valid cases>

Help Reset OK Cancel Apply

```
lm(formula = LowestMSRP ~ CargoVolume + CityMPG + Engine + Height + HighwayMPG + HP + Length + Seating + Weight + Wheelbase + Width, data = Dataset)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	31587.411	30681.862	1.030	0.304604
CargoVolume	-303.424	87.142	-3.482	0.000623 ***
CityMPG	697.552	315.255	2.213	0.028165 *
Engine	-3777.557	1157.582	-3.263	0.001315 **
Height	-657.352	264.466	-2.486	0.013836 *
HighwayMPG	-720.508	370.360	-1.945	0.053264 .
HP	105.683	17.438	6.060	0.00000000763 ***
Length	19.148	144.394	0.133	0.894648
Seating	-1543.290	1004.508	-1.536	0.126186
Weight	18.403	2.922	6.297	0.00000000221 ***
Wheelbase	-327.423	280.193	-1.169	0.244108
Width	168.853	368.973	0.458	0.647765

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9128 on 182 degrees of freedom

Multiple R-squared: 0.7513, Adjusted R-squared: 0.7362

F-statistic: 49.97 on 11 and 182 DF, p-value: < 2.2e-16

MULTIPLE LINEAR REGRESSION – MODEL CLEANING

In addition to the checks for the case of simple linear regression, we also need to pay attention to multicollinearity. One way to check multicollinearity is to look at the variance inflation factors(VIF).

Multicollinearity means there is correlation between independent variables, and it should not be. We will eliminate those independent variables with high VIF.

```
RegModel.6 <- lm(LowestMSRP~CargoVolume+CityMPG+Engine+Height+HighwayMPG+HP+Length+Seating+Weight+Wheelbase+Width,  
  data=Dataset)  
summary(RegModel.6)  
vif(RegModel.6)
```

```
> vif(RegModel.6)  
CargoVolume    CityMPG      Engine    Height  HighwayMPG      HP    Length    Seating    Weight  
   3.430145    5.633896    2.764385    6.718091    8.417060    4.326493    8.679597    3.749227    13.330850  
Wheelbase      Width  
   9.614106    4.241558
```

VIF limits: if $VIF > 10$ - this a high multicollinearity

if $10 \geq VIF \geq 5$ - this indicate moderate multicollinearity

if $5 \geq VIF \geq 3$ - this indicate minimal multicollinearity

If we have a variable with $VIF > 10$ we should remove that variable from the model. For the other values of the VIF we should proceed with caution. Sometimes we must decide between eliminating a variable and having a minor multicollinearity problem. Eliminating a variable from the model has an impact on all the other variables (p-values, VIFs, etc.)

MULTIPLE LINEAR REGRESSION- FINAL MODEL

Linear Regression

Enter name for model: RegModel.9

Response variable (pick one): LowestMSRP

Explanatory variables (pick one or more): CargoVolume, CityMPG, Engine, Height, HighwayMPG, HP

Indices or names of row(s) to remove: <use all valid cases>

Subset expression: <all valid cases>

Help Reset OK Cancel Apply

```
RegModel.8 <- lm(LowestMSRP~CargoVolume+Engine+Height+HP+Weight+Wheelbase, data=Dataset)
summary(RegModel.8)
vif(RegModel.8)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	52080.95	20480.98	2.543	0.011804	*
CargoVolume	-312.79	85.12	-3.675	0.000311	***
Engine	-4628.40	1080.86	-4.282	2.96e-05	***
Height	-735.67	236.12	-3.116	0.002125	**
HP	114.13	16.51	6.914	7.24e-11	***
Weight	20.37	2.52	8.083	7.68e-14	***
Wheelbase	-516.61	176.63	-2.925	0.003874	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9220 on 187 degrees of freedom
Multiple R-squared: 0.7393, Adjusted R-squared: 0.7309
F-statistic: 88.37 on 6 and 187 DF, p-value: < 2.2e-16

```
> vif(RegModel.8)
```

CargoVolume	Engine	Height	HP	Weight	Wheelbase
3.208360	2.362388	5.249183	3.800593	9.717462	3.745136

MULTIPLE LINEAR REGRESSION- DUMMY VARIABLES

TestScore	IQ	Gender					
93	125	Male					
86	120	Female					
96	115	Male					
81	110	Female					
92	105	Male					
75	100	Female					
84	95	Male					
77	90	Female					
73	85	Male					
74	80	Female					



TestScore	IQ	Gender	DGender
93	125	Male	1
86	120	Female	0
96	115	Male	1
81	110	Female	0
92	105	Male	1
75	100	Female	0
84	95	Male	1
77	90	Female	0
73	85	Male	1
74	80	Female	0



	TestScore	IQ	Gender	DGender
1	93	125	Male	1
2	86	120	Female	0
3	96	115	Male	1
4	81	110	Female	0
5	92	105	Male	1
6	75	100	Female	0
7	84	95	Male	1
8	77	90	Female	0
9	73	85	Male	1
10	74	80	Female	0

Linear Model

Enter name for model: LinearModel.2

Variables (double-click to formula)

DGender
Gender [factor]
IQ
TestScore

Model Formula

Operators (click to formula): + * : / %in% - ^ ()

Splines/Polynomials:
(select variable and click)

B-spline natural spline orthogonal polynomial raw polynomial

df for splines: 5
deg. for polynomials: 2

TestScore ~ IQ + Gender

Model formula help

Indices or names of row(s) to remove
<use all valid cases>

Subset expression
<all valid cases>

Weights
<no variable selected>

Help Reset OK Cancel Apply

```
Call:
lm(formula = TestScore ~ IQ + Gender, data = Dataset)

Residuals:
    Min       1Q   Median       3Q      Max
-6.60  -2.35  -0.10   3.15   4.40

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  38.6000     9.5355   4.048  0.00488 **
IQ           0.4000     0.0935   4.278  0.00366 **
Gender[T.Male]  7.0000     2.6857   2.606  0.03510 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.182 on 7 degrees of freedom
Multiple R-squared:  0.8102, Adjusted R-squared:  0.756
F-statistic: 14.94 on 2 and 7 DF, p-value: 0.002979
```

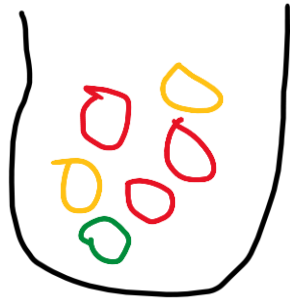
The number dummy variable columns is the number of categorical variables -1.

LOGISTICS REGRESSION

Difference between probability and odds.

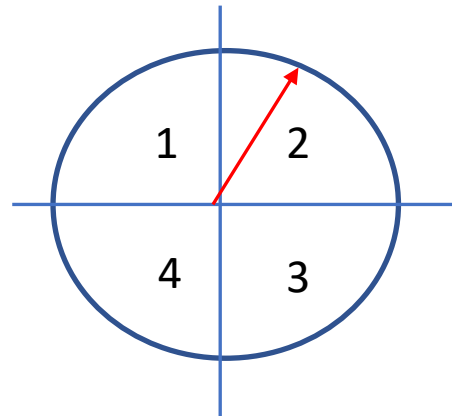
Probability = #of successes / # total number of cases

Odds = #number of success / # number of not successes



$P_{\text{red ball}} = 3/6 = \frac{1}{2} = 0.5$ every probability is $0 \leq P(A) \leq 1$

$\text{Odds}_{\text{red ball}} = 3/3 = 1/1 = 1:1$



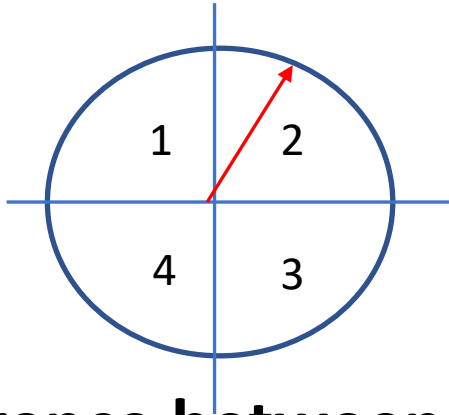
$P_{\text{getting_a_2}} = 1/4 = 0.25$

$\text{Odds}_{\text{getting_a_2}} = 1/3 = 1:3$ are also named **Odds_in_favor**

Odds_against $\text{getting_a_2} = 3/1 = 3:1$

LOGISTICS REGRESSION

Experimental vs theoretical differences



If we run the spin 100 times and get 2 for 30 times.

The experimental probability is $P(2) = 30/100 = 0.3$

However, the theoretical probability is $P(2) = 1/4 = 0.25$

Difference between probability and likelihood:

Probability corresponds to finding the chance of something given a sample distribution of the data.

$$P(\text{height} > 170\text{cm} | \mu = 170, \sigma = 3.5)$$

Likelihood refers to finding the best distribution of the data given a particular value of some feature or some situation in the data.

$$\text{Likelihood}(\mu = 170, \sigma = 3.5 | \text{height} > 170\text{cm})$$

LOGISTICS REGRESSION

Odds provide a measure of the **likelihood** of a particular outcome. They are calculated as the ratio of the number of events that produce that outcome to the number that do not.

Odds (odds of success): It is defined as the chances of success divided by the chances of failure. Say, there is a 90% chance that winning a wager implies that the '*odds are in our favour*' as the winning odds are 90% while the losing odds are just 10%. It is also known defined as odds ratio as it is in the form of a ratio. So the odd ratio of the above discussed example would be:

$$\text{Odds Ratio} = \frac{\text{odds of success}}{\text{odds of failure}} = \frac{90}{10} = 9$$

Formally, the odds ratio of an event A is defined as the probability that A occurs upon the probability that A does not occur (i.e. complement of A). (As shown by equation given below)

$$\text{Odds Ratio} = \frac{P(A)}{1-P(A)} = \frac{P(A)}{P(A')}$$

LOGISTICS REGRESSION

Odds can be demonstrated by examining rolling a six-sided die. The odds of rolling a 6 is 1:5. This is because there is 1 event (rolling a 6) that produces the specified outcome of "rolling a 6", and 5 events that do not (rolling a 1,2,3,4 or 5).

The odds of rolling either a 5 or 6 is 2:4. This is because there are 2 events (rolling a 5 or 6) that produce the specified outcome of "rolling either a 5 or 6", and 4 events that do not (rolling a 1, 2, 3 or 4).

The odds of not rolling a 5 or 6 is the inverse 4:2. This is because there are 4 events that produce the specified outcome of "not rolling a 5 or 6" (rolling a 1, 2, 3 or 4) and two that do not (rolling a 5 or 6).

The probability of an event is different, but related, and can be calculated from the odds, and vice versa. The probability of rolling a 5 or 6 is the fraction of the number of events over total events or $2/(2+4)$, which is $1/3$, 0.33 or 33%.

LOGISTIC REGRESSION

ODDS RATIO

The odds ratio is exactly what it says it is, a ratio of two odds

Fair coin flip

$$P(\text{heads}) = \frac{1}{2} = 0.5$$

$$\text{odds}(\text{heads}) = \frac{0.5}{0.5} = 1 \text{ or } 1:1$$

Loaded coin flip

$$P(\text{heads}) = \frac{7}{10} = 0.7$$

$$\text{odds}(\text{heads}) = \frac{0.7}{0.3} = 2.333$$

$$\text{Odds ratio} = \frac{\text{odds}_1}{\text{odds}_0}$$

$$\text{Odds ratio} = \frac{\frac{p_1}{1-p_1}}{\frac{p_0}{1-p_0}}$$

$$\text{Odds ratio} = \frac{\frac{.7}{.3}}{\frac{.5}{.5}} = \frac{.7}{.3} \times \frac{.5}{.5} = \frac{.35}{.15} = 2.333$$

The odds of getting “heads” on the loaded coin are **2.333x** greater than the fair coin.



LOGISTICS REGRESSION

Problem with odds

The problem with odds ratio is due to the presence of asymmetry created between the odds of winning and the odds of losing. Let us try to understand this with the help of an example. 72 countries participate in the Commonwealth Games.

Considering every country has an equal chance of winning, the odds of the Indian team winning a gold are against, 1 to 71 or $\frac{1}{71} = 0.014$

But assuming an idealistic scenario where the odds of Indian team winning the gold are in favour, 71 to 1 or $\frac{71}{1} = 71$

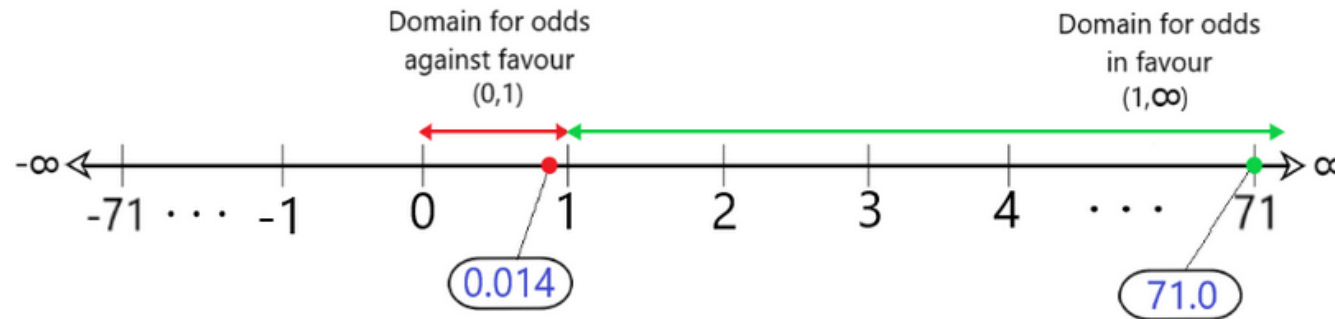


Fig 1: Domain for favourable vs against odds

As observed in Fig 1, when the odds are against our favour, the value always tends to lie between 0 and 1 which is a very small value. However, if the odds are in our favour, the value can lie anywhere in between 1 and infinity, which can be a very big value! To solve this problem, the concept of Log odds came into picture.

LOGISTICS REGRESSION

Log odds: It is the logarithm of the odds ratio. (As shown by the equation given below)

$$\text{Log Odds} = \log\left(\frac{P(A)}{1-P(A)}\right)$$

As per the above-mentioned example,

The log of odds of the Indian team winning a gold are against, 1 to 71 = $\log\left(\frac{1}{71}\right) = -1.85$

The log of odds of Indian team winning the gold are in favour, 71 to 1 = $\log\left(\frac{71}{1}\right) = 1.85$

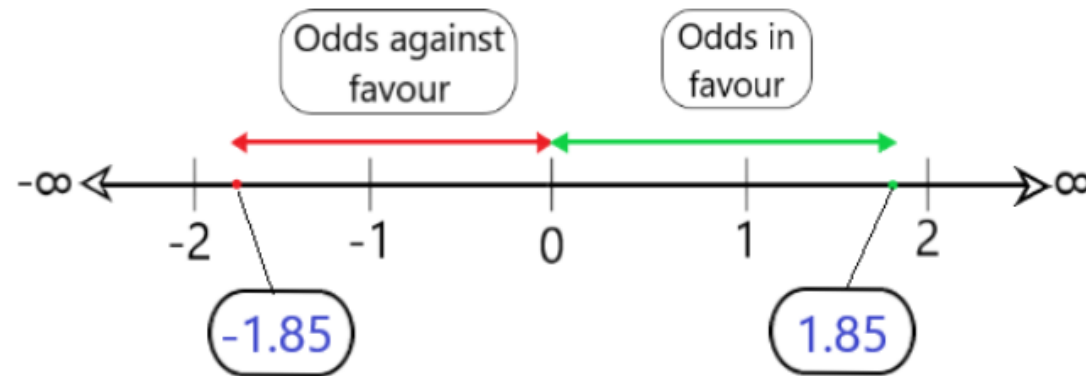


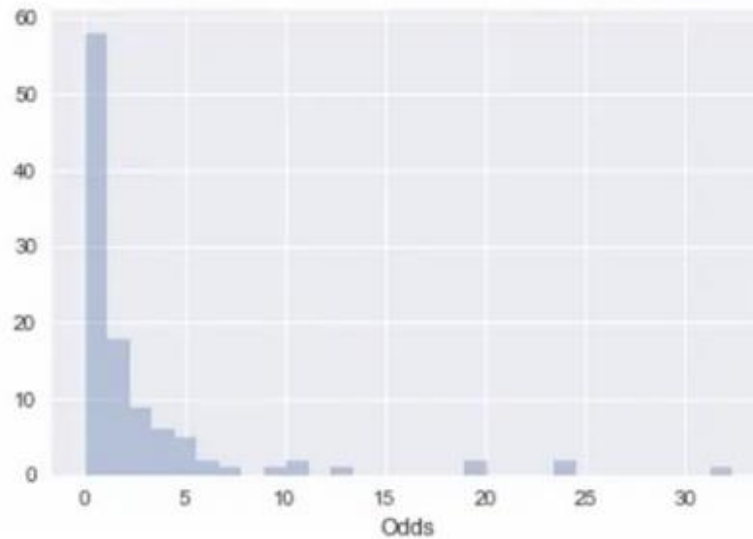
Fig 2: Log Odds

As observed in Fig. 2, taking the log of the odds ratio brings about a certain symmetry in the results, making it easier to interpret and use in various statistics.

INTERESTING NOTE: The log odds of a certain event gives a **normal distribution** when plotted on a histogram! This is what makes log odds so useful.

LOGISTICS REGRESSION

Take odds of random numbers
which add up to 100
Ex: 20/80, 70/30, etc.



Taking $\log(\text{odds})$ of the same

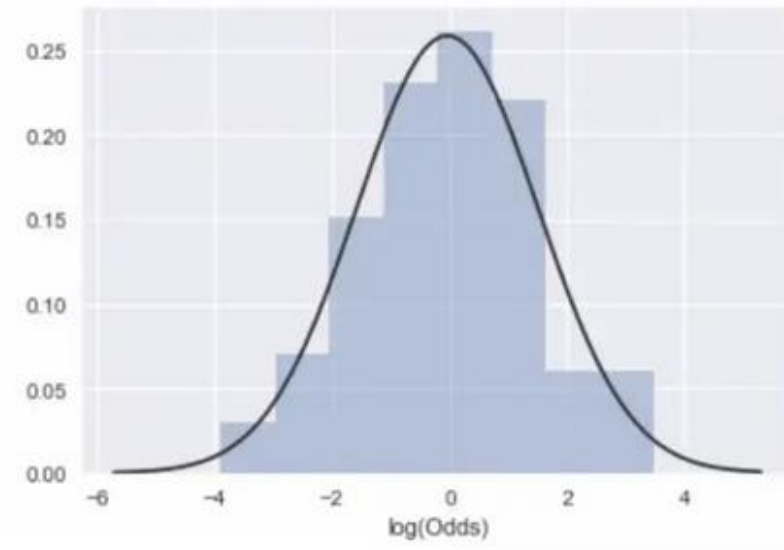


Figure-8: $\log(\text{odds})$ helps getting a Normal distribution

You can see from the plot on the right that how $\log(\text{odds})$ helps us get a nice normal distribution of the same plot on the left. This makes $\log(\text{odds})$ very useful for solving certain problems, basically ones related to finding probabilities in win/lose, true/fraud, fraud/non-fraud, type scenarios.

LOGISTIC REGRESSION

- The odds ratio for a variable in logistic regression represents how the odds change with a 1 unit increase in that variable holding all other variables constant
- For (fictitious) example:
 - Body weight and sleep apnea (two categories: apnea / no apnea)
 - Weight variable had an odds ratio of 1.07
 - This means a one pound increase in weight increases the odds of having sleep apnea by 1.07 (not very high b/c we are looking at 1lb increments)
 - A ten pound increase in weight increases the odds to 1.98, or almost doubles a person's odds of having sleep apnea and a 20 pound increase raises the odds to 3.87 or almost 4x greater (we will learn how calculate that later)
 - This holds true at any point in the weight spectrum



$$y = e^{\beta_1 \cdot \Delta} \quad y = e^{0.014634 \cdot \Delta}$$

$$y = e^{0.014634 \cdot 1} = 1.0147 \text{ or } 1.47\% \text{ increase}$$

$$y = e^{0.014634 \cdot 19} = 1.3205 \text{ or } 32.05\% \text{ increase}$$

$$y = e^{0.014634 \cdot 53} = 2.1719 \text{ or } 117.19\% \text{ increase}$$

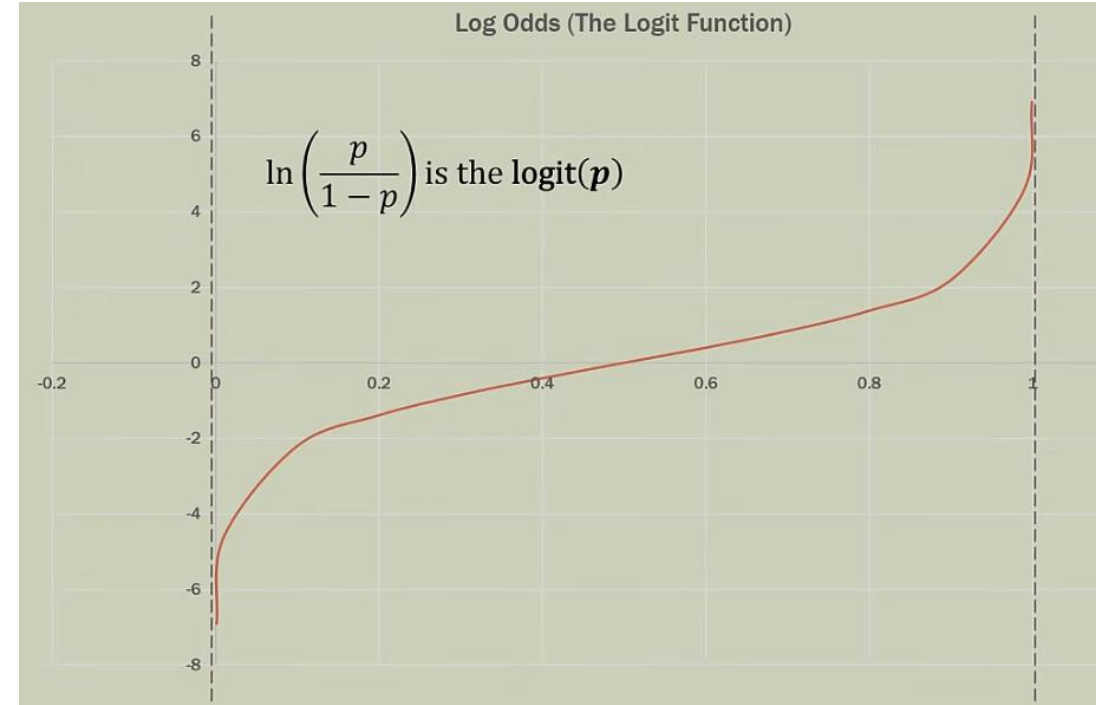
$$y = e^{0.014634 \cdot 74} = 2.9533 \text{ or } 195.33\% \text{ increase}$$

LOGISTIC REGRESSION

In logistic regression the dependent variable follows a Bernoulli distribution (a special case of a Binomial distribution) with only one trial $n=1$. The possible outcomes are 1 for success and 0 for failure with probability of success p and $q=1-p$ the probability of failure.

In logistic regression we **estimate an unknown probability p -hat** for any given combination of the independent variables. For this we need a function that maps the linear combination of the independent variables to the Bernoulli distribution. We can do that using the **Logit function** defined as the natural logarithm of the odds ratio.

$\ln(\text{odds}) \Rightarrow \ln(p/(1-p))$ is the $\text{logit}(p)$ where $\log_e x = \ln(x)$



LOGISTIC REGRESSION

WHAT IS THE INVERSE LOGIT?

In our logit link function graph, 0 to 1 ran along the x-axis but we want the probabilities to be on the y-axis. We can achieve that by taking the inverse of the logit function.

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$

where p is between 0 and 1

$$\text{logit}^{-1}(\alpha) = \frac{1}{1 + e^{-\alpha}} = \frac{e^{\alpha}}{1 + e^{\alpha}}$$

α = some number

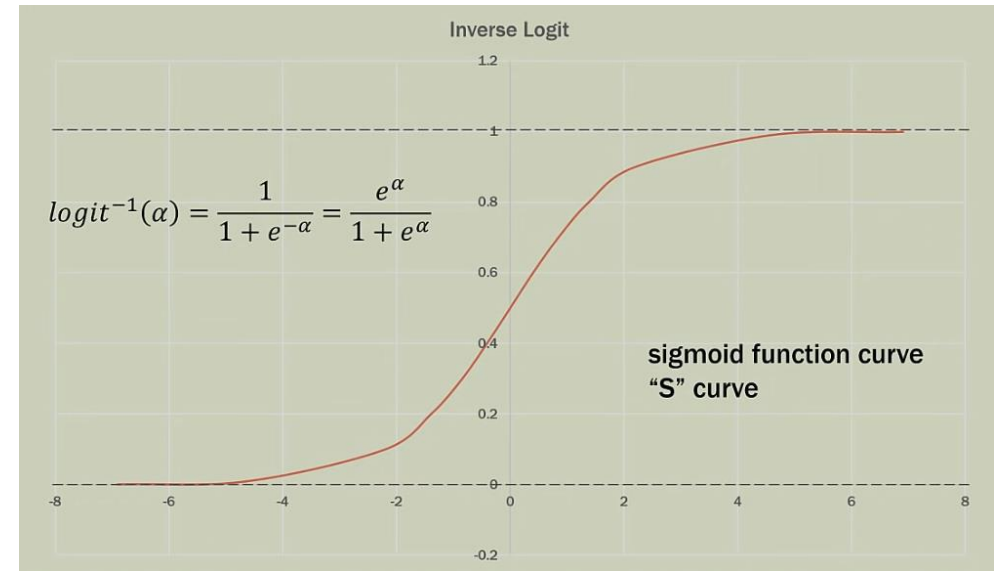
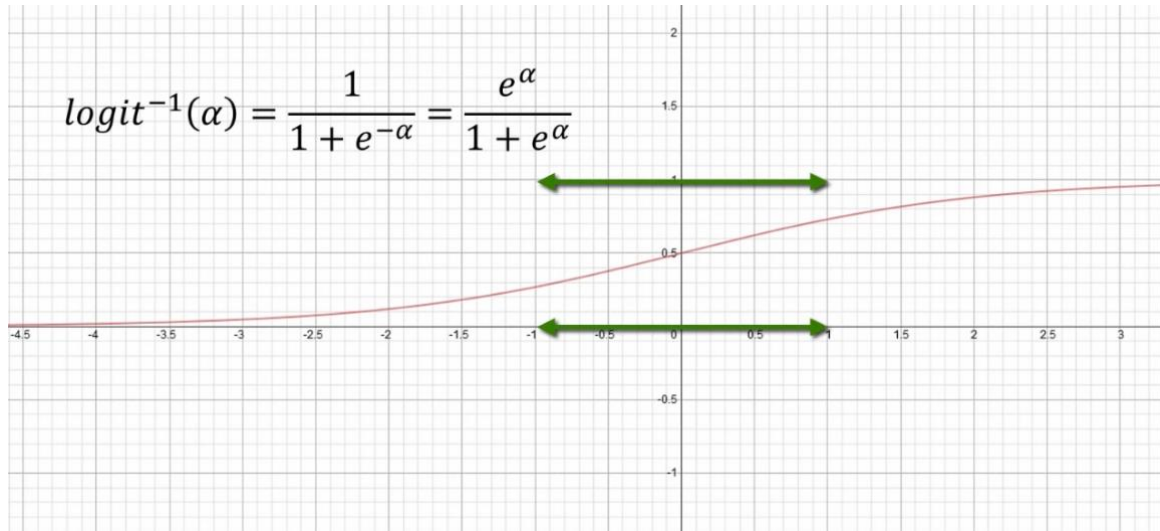
Mean function:

$$\mu_{y|x} = \frac{1}{1 + e^{-\alpha}} = \frac{e^{\alpha}}{1 + e^{\alpha}}$$

In our case, *some number* will be the linear combination of variables and their coefficients. The inverse-logit will return the probability of being a “1” or in the “event occurs” group.



LOGISTIC REGRESSION



The natural logarithm of the odds ratio is equivalent to a *linear* function of the independent variables. The antilog of the logit function allows us to find the estimated regression equation.

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 \quad \text{antilog} \quad \rightarrow \quad \frac{p}{1-p} = e^{\beta_0 + \beta_1 x_1}$$

$$p = e^{\beta_0 + \beta_1 x_1} (1 - p)$$

$$p = e^{\beta_0 + \beta_1 x_1} - e^{\beta_0 + \beta_1 x_1} * p$$

$$p + e^{\beta_0 + \beta_1 x_1} * p = e^{\beta_0 + \beta_1 x_1}$$

$$p(1 + e^{\beta_0 + \beta_1 x_1}) = e^{\beta_0 + \beta_1 x_1}$$

$$\hat{p} = \frac{e^{\beta_0 + \beta_1 x_1}}{1 + e^{\beta_0 + \beta_1 x_1}}$$

**Estimated
Regression
Equation**

LOGISTIC REGRESSION

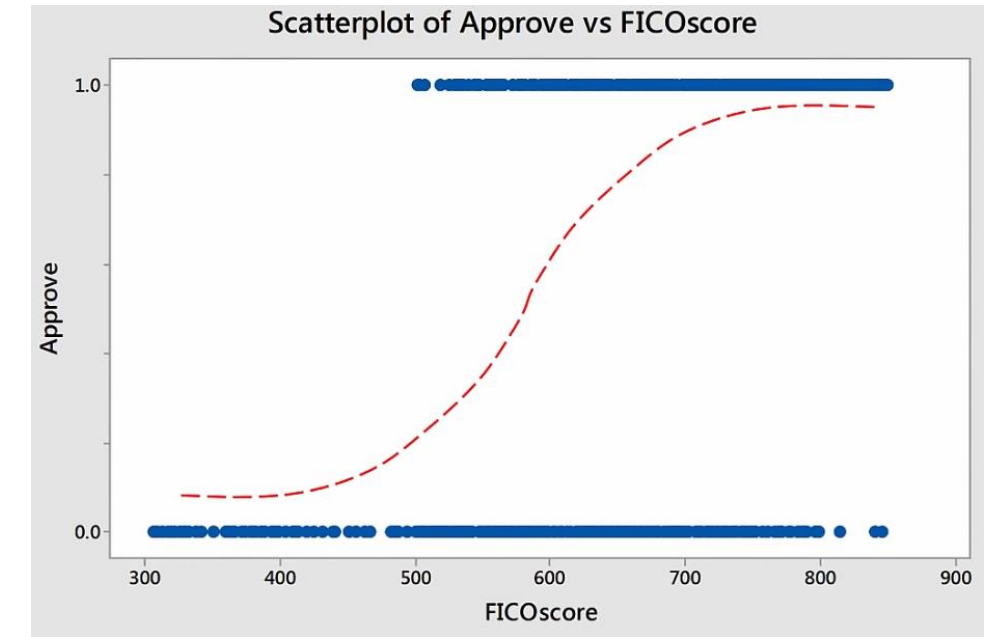
creditScore	approved
655	0
692	0
681	0
663	1
688	1
693	1
699	0
699	1
683	1
698	0
655	1
703	0
704	1
745	1
702	1

$n = 1000$

creditScore is the applicant's credit score

approved is coded "1" for approved and "0" for not approved;
it is a binary, mutually exclusive variable.

* Only 15 of 1000 observations shown



Binary Logistic Regression: Approve versus FICOscore

Deviance Table

Source	DF	Seq Dev	Contribution	Adj Dev	Adj Mean	Chi-Square	P-Value
Regression	1	413.8	31.42%	413.8	413.812	413.81	0.000
FICOscore	1	413.8	31.42%	413.8	413.812	413.81	0.000
Error	998	903.0	68.58%	903.0	0.905		
Total	999	1316.8	100.00%				

Model Summary

Deviance	Deviance	
R-Sq	R-Sq(adj)	AIC
31.42%	31.35%	907.03

This can be interpreted much like the ANOVA table generated in linear regression.

Binary Logistic Regression: Approve versus FICOscore

Coefficients

Term	Coef	SE Coef	95% CI	Z-Value	P-Value	VIF
Constant	-9.346	0.637	(-10.594, -8.097)	-14.67	0.000	
FICOscore	0.014634	0.000940	(0.012791, 0.016476)	15.56	0.000	1.00

Odds Ratios for Continuous Predictors

	Odds Ratio	95% CI
FICOscore	1.0147	(1.0129, 1.0166)

LOGISTIC REGRESSION

Binary Logistic Regression: Approve versus FICOscore

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Binary Logistic Regression: Approve versus FICOscore

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FICOscore	0.014634	0.000940	(0.012791, 0.016476)	15.56	0.000	1.00

$$\hat{p} = \frac{e^{\beta_0 + \beta_1 x_1}}{1 + e^{\beta_0 + \beta_1 x_1}}$$

**Estimated
Regression
Equation**

$$\hat{p} = \frac{e^{-9.346 + 0.014634x_1}}{1 + e^{-9.346 + 0.014634x_1}}$$

**Estimated
Regression
Equation**

where $\hat{p}$ is the estimated probability of being approved and x_1 is a credit score

Odds ratio represent the change in the odds when the FICO score increases by 1 unit.
We would like to have an odds ratio with a confidence interval that DOES NOT contains 1 because a change in the FICO score would result Into a minor increase in the odds of getting approved.

LOGISTIC REGRESSION USING R-commander

	AGE	HOMEOWNR	INCOME	GENDER	MALEMIL
1	NA	<NA>	NA	M	
2	NA	U	5	F	1
3	85	H	1	M	
4	NA	U	1	M	
5	NA	<NA>	NA	<NA>	
6	42	H	6	F	
7	35	H	5	F	4
8	81	H	3	F	
9	72	H	4	M	
10	NA	H	4	M	
11	74	H	2	M	
12	88	U	6	F	
13	59	H	7	M	
14	74	<NA>	5	F	
15	52	<NA>	4	M	
16	NA	<NA>	NA	F	
17	37	<NA>	NA	F	
18	NA	H	7	M	
19	75	H	6	F	
20	27	<NA>	NA	M	
21	59	H	2	F	
22	NA	U	4	F	
23	96	H	6	F	
24	NA	<NA>	NA	F	
25	NA	U	1	M	
26	52	U	1	F	
27	68	H	1	F	
28	87	<NA>	NA	M	
29	NA	U	5	M	
30	48	U	3	M	

Generalized Linear Model

Enter name for model: GLM.4

Variables (double-click to formula)

CARDPROM
FEDGOV
FIRSTT
GENDER [factor]
HOMEOWNR [factor]
IDCODE [factor]

Model Formula

Operators (click to formula): + * : / %in% - ^ ()

Splines/Polynomials:
(select variable and click) B-spline natural spline orthogonal polynomial raw polynomial

df for splines: 5
deg. for polynomials: 2

TARGET_B ~ INCOME + AVGGIFT + LASTT + FIRSTT

Model formula help

Indices or names of row(s) to remove
<use all valid cases>

Subset expression
<all valid cases>

Weights
<no variable selected>

Family (double-click to select)

gaussian
binomial
poisson
Gamma
inverse.gaussian
quasibinomial
quasipoisson

Link function

logit
probit
cloglog

Help Reset OK Cancel Apply

LOGISTIC REGRESSION USING R-commander

```
> summary(GLM.4)

Call:
glm(formula = TARGET_B ~ INCOME + AVGGIFT + LASTT + FIRSTT, family = binomial(logit),
    data = Dataset)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.7577  -1.1453  -0.7223   1.1483   2.5798

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.29000527  0.23750871   1.221   0.22207
INCOME       0.10379307  0.02321802   4.470 0.00000781 ***
AVGGIFT     -0.01666048  0.00604394  -2.757   0.00584 **
LASTT      -0.00160664  0.00033821  -4.750 0.00000203 ***
FIRSTT       0.00017522  0.00003887   4.508 0.00000654 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3180.1  on 2293  degrees of freedom
Residual deviance: 3100.1  on 2289  degrees of freedom
(694 observations deleted due to missingness)
AIC: 3110.1

Number of Fisher Scoring iterations: 4

> exp(coef(GLM.4)) # Exponentiated coefficients ("odds ratios")
(Intercept)      INCOME      AVGGIFT      LASTT      FIRSTT
  1.3364345    1.1093709    0.9834775    0.9983947    1.0001752
```

We interpret this as that each additional **INCOME** dollar multiplies the odds of donating by 1.10, assuming all other variables are held constant. We could also say that each additional **INCOME** dollar taken is associated with a $(1.10 - 1)$ 10% increase in the odds of donating, assuming all other variables are constant.

We interpret this as that each additional AVGGIFT dollar multiplies the odds of donating by 0.98, assuming all other variables are constant. We could also say that each additional INCOME dollar taken is associated with a $(1 - 0.98)$ 2% decrease in the odds of donating, again assuming all other variables are held constant.

The **null deviance** tells us how well the response variable can be predicted by a model with only an intercept term. The **residual deviance** tells us how well the response variable can be predicted by a model with p predictor variables. The lower the value, the better the model is able to predict the value of the response variable.

There is no value for AIC that can be considered “good” or “bad” because we simply use AIC as a way to compare regression models. The model with the lowest AIC offers the best fit. The absolute value of the AIC value is not important.

LOGISTIC REGRESSION USING R-commander

```
> summary(GLM.4)

Call:
glm(formula = TARGET_B ~ INCOME + AVGGIFT + LASTT + FIRSTT, family = binomial(logit),
    data = Dataset)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.7577  -1.1453  -0.7223   1.1483   2.5798

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.29000527  0.23750871   1.221   0.22207
INCOME       0.10379307  0.02321802   4.470 0.00000781 ***
AVGGIFT     -0.01666048  0.00604394  -2.757   0.00584 **
LASTT      -0.00160664  0.00033821  -4.750 0.00000203 ***
FIRSTT       0.00017522  0.00003887   4.508 0.00000654 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3180.1  on 2293  degrees of freedom
Residual deviance: 3100.1  on 2289  degrees of freedom
(694 observations deleted due to missingness)
AIC: 3110.1

Number of Fisher Scoring iterations: 4

> exp(coef(GLM.4)) # Exponentiated coefficients ("odds ratios")
(Intercept)      INCOME      AVGGIFT      LASTT      FIRSTT
  1.3364345    1.1093709    0.9834775    0.9983947    1.0001752
```

McFadden's R-Squared

McFadden's R-Squared = $1 - (\log \text{likelihoodmodel} / \log \text{likelihoodnull})$

log likelihoodmodel: Log likelihood value of current fitted model

log likelihoodnull: Log likelihood value of null model (model with intercept only)

In practice, values over 0.40 indicate that a model fits the data very well.

```
with(summary(GLM.4), 1 - deviance/null.deviance)
[1] 0.02515434
```

Also

#install and load pscl package

```
install.packages('pscl')
```

```
library(pscl)
```

#calculate McFadden's R-squared for model

```
pR2(model)['McFadden']
```

fitting null model for pseudo-r2

McFadden

0.02515434

```
predict (GLM.4, newdata=data.frame(INCOME=10,AVGGIFT=5,LASTT=2,FIRSTT=4), type="response")
```

PRACTICE LOGISTIC REGRESSION

1. Use the healthcare-dataset-stroke-data.xls file
2. Fit a regression model

Call:

```
glm(formula = stroke ~ age + avg_glucose_level + hypertension,  
     family = binomial(logit), data = Dataset)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.0063	-0.3241	-0.1734	-0.0815	3.7895

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-7.578833	0.355584	-21.314	< 2e-16	***
age	0.070585	0.005062	13.944	< 2e-16	***
avg_glucose_level	0.004354	0.001152	3.779	0.000158	***
hypertension	0.384460	0.162331	2.368	0.017866	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

3. Interpret the results

```
Null deviance: 1990.4  on 5109  degrees of freedom
Residual deviance: 1594.4  on 5106  degrees of freedom
AIC: 1602.4
```

```
Number of Fisher Scoring iterations: 7
```

```
> exp(coef(GLM.8))  # Exponentiated coefficients ("odds ratios")
      (Intercept)          age avg_glucose_level      hypertension
      0.0005111576      1.0731359650      1.0043631869      1.4688213302
```

Null deviance shows how well only the intercept can be used to predict the STROKE variable. Residual deviance shows how well our model with all significant variables can be used to predict the STROKE variable. Since the residual deviance is smaller than the null deviance ($1594.4 < 1990.4$), our model (one with all the significant variables) predicts the STROKE variable better than a model with only an intercept. Also, the Akaike information criterion (AIC), a metric used to compare the fit of different regression models, has a lower value (1602.4) when compared with similar values of the previous models. $AIC = 2k - 2\ln(L)$ where k = the number of estimated parameters in the model and L = the maximized value of the likelihood function for the model. We will now focus on interpreting the logistic regression coefficients and odds ratios.

Coefficients:

```
      Estimate Std. Error z value Pr(>|z|)
(Intercept)   -7.578833    0.355584  -21.314 < 2e-16 ***
age            0.070585    0.005062   13.944 < 2e-16 ***
avg_glucose_level 0.004354    0.001152    3.779 0.000158 ***
hypertension    0.384460    0.162331    2.368 0.017866 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
> exp(coef(GLM.8)) # Exponentiated coefficients ("odds ratios")
      (Intercept)      age avg_glucose_level hypertension
0.0005111576      1.0731359650      1.0043631869      1.4688213302
```

a.k.a. Log Odds

or Logit

Intercept

$$\log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X$$

$$P = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

The Intercept: When $X_s = 0$, the probability of having the outcome is $P = e^{\beta_0} / (1 + e^{\beta_0}) = e^{(-7.57)} / (1 + e^{(-7.57)}) = 0.00051$

Age is a continuous variable: The coefficient is 0.07. $e^{\beta} (= e^{0.07} = 1.07)$ tells us how much the odds of the outcome (stroke) will change for each 1-unit change in the predictor(age). An increase of 1 year in age multiplies the odds of stroke by 1.07, or an increase of 1 year in age is associated with an increase of 7% in the odds of stroke.

Avg_glucose_level is a continuous variable: The coefficient is 0.00. $e^{\beta} (= e^{0.00} = 1.00)$ tells us how much the odds of the outcome (stroke) will change for each 1-unit change in the predictor(avg_glucose_level). An increase of 1 unit in avg_glucose_level multiplies the odds of stroke by 1.00, or an increase of 1 unit in avg_glucose_level is associated with an increase of 0% in the odds of stroke.

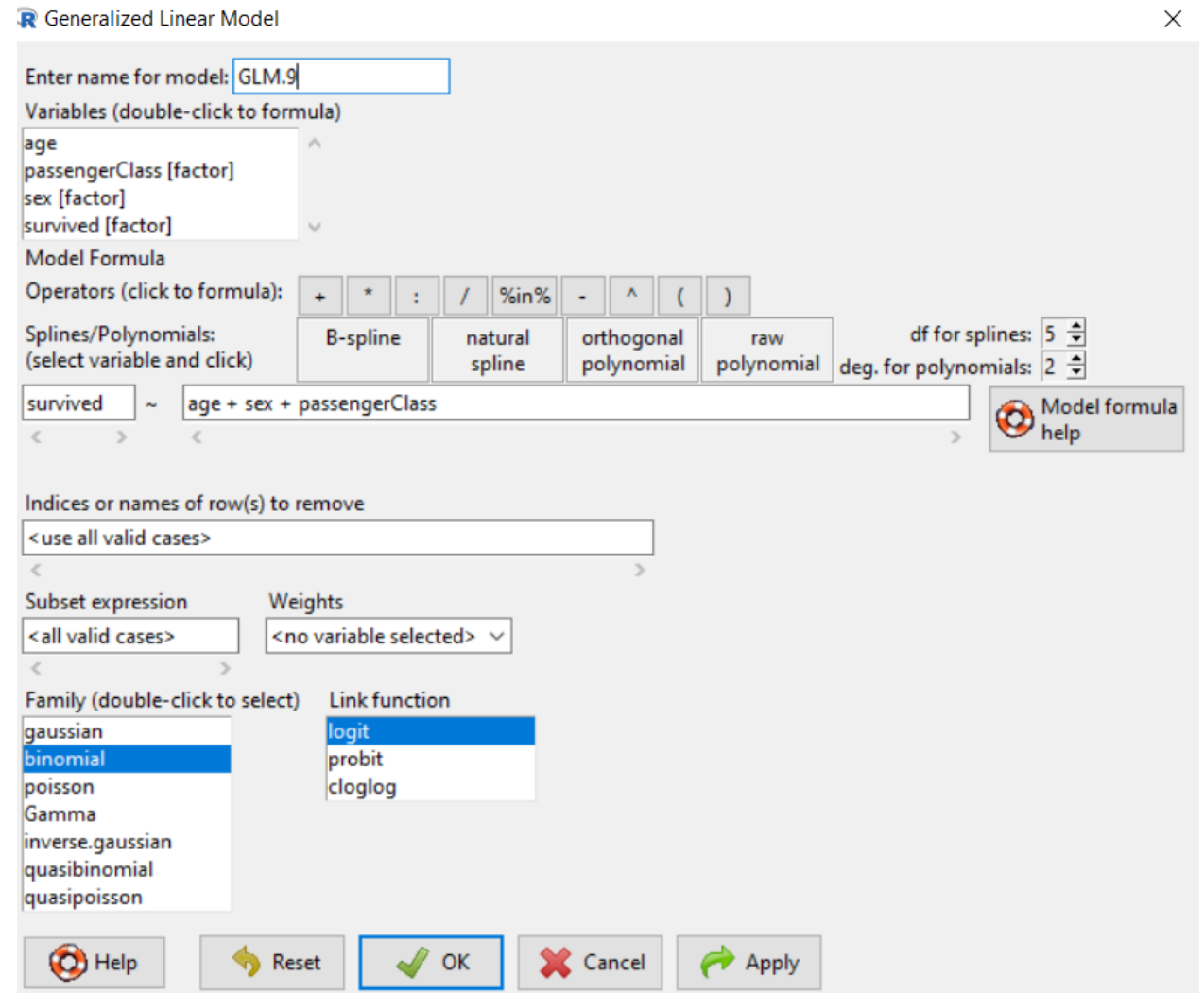
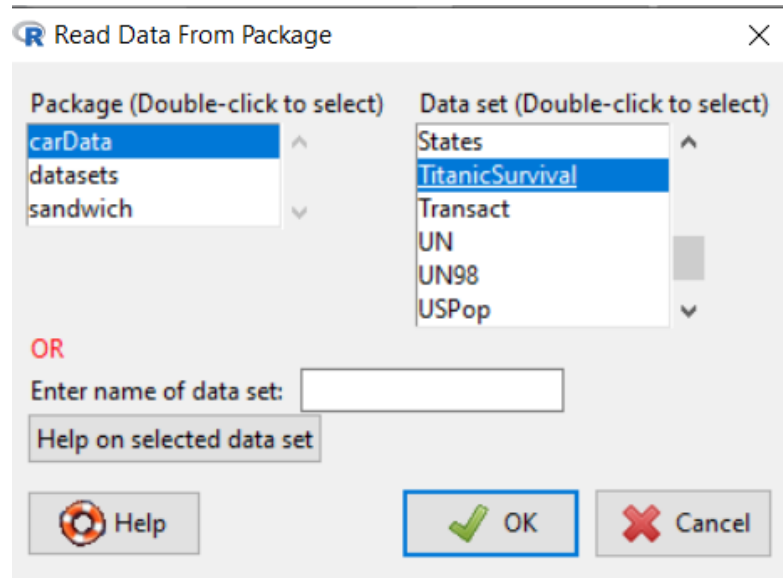
Hypertension is a binary variable: 0 = non-hypertension, 1=hypertension. The coefficient is 0.38. $e^{\beta} (= e^{0.38} = 1.46)$ tells us how much the odds of the outcome (stroke) will change for each 1-unit change in the predictor(hypertension). The hypertension group has 1.46 times the odds of the non-hypertension group of having a stroke. The hypertension group has 46% ($1.46 - 1 = 0.46$) more odds of having a stroke than the non-hypertension group.

PREDICT OUTCOME FROM NEW DATA

```
newdata <- data.frame(hypertension=1,heart_disease=0,age=67.00)
predict(GLM.7, newdata, type="response")
```

PRACTICE LOGISTIC REGRESSION

1. Use one of the R built-in datasets(see screen capture)
2. Fit a logistic regression line for survivor



```
Call:
glm(formula = survived ~ age + sex + passengerClass, family = binomial(logit),
     data = TitanicSurvival)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.6399	-0.6979	-0.4336	0.6688	2.3964

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.522074	0.326702	10.781	< 2e-16	***
age	-0.034393	0.006331	-5.433	0.0000000556	***
sex[T.male]	-2.497845	0.166037	-15.044	< 2e-16	***
passengerClass[T.2nd]	-1.280570	0.225538	-5.678	0.0000000136	***
passengerClass[T.3rd]	-2.289661	0.225802	-10.140	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Null deviance: 1414.62 on 1045 degrees of freedom
Residual deviance: 982.45 on 1041 degrees of freedom
(263 observations deleted due to missingness)
AIC: 992.45

Number of Fisher Scoring iterations: 4

```
> exp(coef(GLM.9)) # Exponentiated coefficients ("odds ratios")
      (Intercept)           age      sex[T.male] passengerClass[T.2nd] passengerClass[T.3rd]
      33.85457044       0.96619149       0.08226211       0.27787894       0.10130084
```

Age – continuous: The coefficient is -0.03. e^{β} ($= e^{-0.03} = 0.96$) tells us how much the odds of the outcome (surviving) will change for each 1-unit change in the predictor(age). An increase of 1 unit in age multiplies the odds of surviving by 0.96, or an increase of 1 unit in age is associated with a decrease of 4% in the odds of surviving.

95% Confidence Interval = $\exp(\beta \pm 1.96 \times \text{Std. Error}) = \exp(-0.03 \pm 1.96 \times 0.006) = [0.9590, 0.9819]$. We are 95% confident that older people had on average 2% to 5% ($1 - 0.95 = 0.05$ and $1 - 0.98 = 0.02$) less odds of surviving than younger people.

Sex[T.male] – binary: The coefficient is -2.49. e^{β} ($= e^{-2.49} = 0.08$) tells us how much the odds of the outcome (surviving) will change for each 1-unit change in the predictor(male). Males had 0.08 ($e^{\beta} = 0.08$) times the survival odds of females. Males had 92% ($1 - 0.08 = 0.92$) less odds of surviving than females.

passengerClass[T.2nd] – category: This variable is interpreted with respect to the reference variable, which in this case is **passengerClass[1st]**. The coefficient is -1.28. e^{β} ($= e^{-1.28} = 0.27$) tells us how much the odds of the outcome (surviving) will change for each 1-unit change in the predictor(passengerClass). **PassengersClass[2nd]** had 0.27 ($e^{\beta} = 0.27$) times the survival odds of **passengerClass[1st]**. **PassengersClass[2nd]** had 73% ($1 - 0.27 = 0.73$) less odds of surviving than **passengerClass[1st]**.

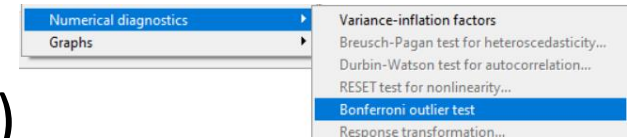
passengerClass[T.3rd] – category: This variable is interpreted with respect to the reference variable, which in this case is **passengerClass[1st]**. The coefficient is -2.28. e^{β} ($= e^{-2.28} = 0.10$) tells us how much the odds of the outcome (surviving) will change for each 1-unit change in the predictor(passengerClass). **PassengersClass[3rd]** had 0.10 ($e^{\beta} = 0.10$) times the survival odds of **passengerClass[1st]**. **PassengersClass[3rd]** had 90% ($1 - 0.10 = 0.90$) less odds of surviving than **passengerClass[1st]**.

Assumptions of logistic regression

Assumption #1: The Response Variable is Binary

Assumption #2: The Observations are Independent - `plot(residuals(GLM10))`

Assumption #3: There is No Multicollinearity Among Explanatory Variables (VIFs). For GVIF square the value and apply the same “rule of thumb”.



Assumption #4: There are No Extreme Outliers (Bonferroni Test)

Assumption #5: There is a Linear Relationship Between Explanatory Variables and the Logit of the Response Variable

This assumption can be tested by including in the model interactions between the continuous predictors and their logs. If such an interaction is significant, then the assumption has been violated.

Term	Coef	Std. Err.	z	Pr > z
age:log (age)	-0.03927	0.03660	-1.073	0.2833

Assumption #6: The Sample Size is Sufficiently Large

As a rule of thumb, you should have a minimum of 10 cases with the least frequent outcome for each explanatory variable. For example, if you have 3 explanatory variables and the expected probability of the least frequent outcome is 0.20, then you should have a sample size of at least $(10*3) / 0.20 = 150$.

LOGISTIC REGRESSION PSEUDO R-SQUARED

Interpretation of R ²	Explanation
Degree of explained variance	R ² =0: the model does not explain the data at all R ² =1: the model does not explain the data perfectly
Improvement over the null model toward the saturated model	R ² =0: the model fits equally well as the null model R ² =1: the model fits equally well as the saturated model
Squared correlation between the observed and predicted values	R ² =0: the model does not predict the dependent variable at all R ² =1: the model predicts the dependent variable perfectly
OLS estimation criterion	Maximizing the R²
Nested model test	Difference in R² is used in F test

Some pseudo-R²s

Pseudo R ²	Formula	Notes
Efron's	$1 - \frac{\sum_{i=1}^N (y_i - \hat{p}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$	Quantifies explained variance, but not improvement over null model.
McFadden's	$1 - \frac{LL_{model}}{LL_{null}}$	Quantifies improvement over null model and explained variance, but not correlation between predicted and observed values.
Cox & Snell's	$1 - \left(\frac{L_{null}}{L_{model}} \right)^{2/N}$	Quantifies improvement over null model, Maximum less than 1. OLS R ² is a special case.
Nagelkerke's	$\frac{R_{Cox \& Snell}^2}{1 - L_{null}^{2/N}}$	A scaled version of Cox & Snell so that maximum is 1
McKelvey & Zavoina's	$\frac{\text{var}(\hat{y}^*)}{\text{var}(\hat{y}^*) + \text{var}(\varepsilon)}$	Uses the latent variable formulation of logistic regression. Used in Mplus
Count	$\frac{\# \text{ Correct}}{N}$	Focused on single prediction. Random prediction has 50% "count R ² "
Coefficient of discrimination (Tjur, 2009)	$\hat{p}_1 - \hat{p}_0$	Focuses on probability prediction.

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